Abstraction Mechanisms for Virtual Avatars

Master's Thesis

submitted by

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Abstract

Virtual reality applications increasingly rely on avatar representations to facilitate social presence and user embodiment. However, photorealistic avatar rendering can trigger the uncanny valley effect, where near-realistic human representations evoke discomfort and reduced trust rather than engagement. Non-photorealistic rendering (NPR) offers a principled alternative, deliberately abstracting avatar appearance while preserving expressiveness and identity.

This thesis presents the design, implementation, and evaluation of a real-time NPR stylisation system for virtual avatars. Eight techniques were developed iteratively: geometry-based inverted hull outlines, screen-space normal edge detection, Sobel edge detection with adaptive skin-to-clothing thresholding, Gaussian pre-filtered Sobel, hierarchical multi-cue edge detection combining depth, normal, and colour discontinuities, isotropic Kuwahara painterly filtering, and two composite techniques combining painterly abstraction with edge rendering. The techniques were implemented and compared across three avatar platforms — a Mixamo humanoid model, the Meta Avatars SDK, and Avaturn — to assess their visual behaviour and robustness under different mesh topologies and rendering pipelines. Avaturn was selected as the primary platform for the user study due to its facial rigging capabilities and higher photorealistic fidelity.

The user study investigates how visual abstraction affects the perceived trustworthiness of an avatar delivering advice. Participants watched short video scenarios in which a speaker — presented either as a real person, a photorealistic Avaturn avatar, or an NPR-stylised avatar — recommended one of two options in a practical decision situation. After each scenario, participants rated how trustworthy they found the speaker's recommendation. The results provide empirical insight into the relationship between avatar abstraction level and social trust, offering practical guidance for avatar design in applications where perceived credibility matters.

Keywords: non-photorealistic rendering, virtual reality, avatar abstraction, trust perception, edge detection, Avaturn, uncanny valley, Kuwahara filter.

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Abbreviations

ANOVA	Analysis of Variance
API	Application Programming Interface
AR	Augmented Reality
CG	Computer-Generated
ESF	Extensible Shader Framework
FPS	Frames Per Second
GPU	Graphics Processing Unit
HCI	Human-Computer Interaction
HLSL	High-Level Shading Language
HMD	Head-Mounted Display
HSV	Hue, Saturation, Value (colour space)
LOD	Level of Detail
NPR	Non-Photorealistic Rendering
PBR	Physically-Based Rendering
RGB	Red, Green, Blue
SDK	Software Development Kit
SSS	Subsurface Scattering
UI	User Interface
URP	Universal Render Pipeline
UV	Texture Coordinate Axes (U and V)
VR	Virtual Reality

1. Introduction

1.1. Motivation

Computer-generated (CG) humanoid characters have become an ordinary part of everyday digital life, especially as generative AI video content is widely encountered on a daily basis. This is no longer limited to passive viewing, as virtual assistants, game characters, and avatars in immersive environments increasingly involve users in sustained interactions with virtual humans [1]. Therefore, engaging with a CG human raises the value of its design, since the character is no longer a background presence but an active participant in the interaction. This leads to several design concerns with consequences across film, games, and other domains where virtual humans appear. One of the most difficult is how realistically or abstractly such a participant should be rendered [2]. This tension between realism and abstraction is the starting point for the work presented in this thesis.

This growing engagement has also given rise to new features of virtual reality (VR), particularly communication through avatars. This form of communication has extended well beyond its origins as a navigational convenience and branched into social presence. Research confirmed this: avatar appearance exerts a measurable influence on social and copresence [3]. This same presence has been linked to interpersonal trust and perceived communication quality in avatar-mediated collaboration [4]. Accordingly, an avatar's design choice influences both trust and social presence.

This design choice can be represented along a fidelity axis. Higher fidelity corresponds to greater human-likeness, where human-likeness refers to the degree to which an avatar visually resembles a real human. Furthermore, realistic appearance and behaviour have been shown to increase users' confidence in an avatar's guidance and their willingness to follow it [5]. However, perceived trust does not scale linearly with likeness across this axis. Mori termed this effect the Uncanny Valley, in which comfort rises with human-likeness only up to a point. It then drops sharply for entities that appear almost but not quite human, before recovering near perfect likeness [6]. Consequently, an avatar that approaches realism without fully achieving it may reduce rather than enhance user confidence. This holds for CG characters generally [2], and it extends specifically to avatar stylisation and trust [7]. These findings motivate the exploration of alternative visual styles that avoid the perceptual pitfalls associated with highly realistic avatars.

Non-Photorealistic Rendering (NPR) is one such approach, allowing an avatar's visual representation to be deliberately positioned along the fidelity axis through controlled abstraction. The open question is not appeal, but whether a deliberately abstracted avatar can still be trusted by the people and reducing realism introduces its own perceptual costs. This question is becoming increasingly relevant as virtual avatars move beyond research settings into widely used consumer applications such as gaming and advertising **rahman2024**. Thus, determining the balance between realism and abstraction is a central challenge in the design of trustworthy virtual avatars.

1.2. Problem Statement

Although NPR provides a potential approach for controlling avatar appearance, applying NPR stylisation to modern photorealistic avatar systems is not straightforward. Production avatar platforms manage rendering through proprietary shader architectures, material systems, and mesh topologies designed for physical fidelity rather than artistic abstraction. This design orientation gives each platform a different set of constraints on what effects can be achieved. These constraints take different forms across production platforms. Some systems expose only a single, restrictive integration point into an otherwise closed lighting pipeline, and

this permits no additional render passes or access to earlier shading stages. Others provide broader pipeline access but introduce different obstacles due to texture.

Prior NPR research has not had to address constraints of this kind. The literature assumes full pipeline access and custom assets under complete authorial control [2], [8]. It remains unclear how NPR techniques behave across production avatar platforms with substantially different integration constraints, or which technique choices remain viable under the most restrictive of them.

This technical challenge is accompanied by an open empirical question. Photorealistic avatar generation has become increasingly accessible, yet the Uncanny Valley problem suggests that realism alone does not guarantee a positive perceptual reception. Two questions remain unanswered for production avatar systems of this kind: whether NPR abstraction can improve how an avatar is perceived in terms of trustworthiness, and how much stylisation is viable before credibility is affected. They are consequential for any domain in which avatar credibility is at stake: social VR, remote consultation, training simulation, and education.

This thesis addresses both challenges by developing and integrating NPR techniques across three avatar platforms with different rendering constraints, and by evaluating the perceptual consequences of that abstraction in a controlled user study of trust.

1.3. Research Questions

The following three research questions guide this work:

RQ1: Can NPR stylisation techniques be implemented and applied across avatar platforms with different rendering architectures, specifically Mixamo, the Meta Avatars SDK, and Avaturn, and what technical constraints does each platform impose on achievable effects?

RQ2: Can NPR stylisation serve as a mechanism for reducing the uncanny valley effect in virtual avatars?

RQ3: Can an NPR avatar be perceived as more trustworthy than a real human speaker?

1.4. Contributions

This thesis makes the following contributions:

- **NPR shader system across three avatar platforms.** An NPR stylisation system was developed and applied across three avatar platforms. The first is a Mixamo humanoid model, used for initial technique development. The second is the Meta Avatars SDK, integrated via the fragment hook evaluated after the lighting stage (`AppSpecificPostManipulation`). The third is Avaturn. It serves as the platform for both the photorealistic baseline and the NPR condition evaluated in the user study. Implementation across the three platforms reveals the constraints specific to each rendering pipeline and demonstrates how NPR techniques can be adapted to different mesh topologies and material systems.
- **Iteratively developed stylisation techniques.** A series of eight stylisation techniques were designed, implemented, and compared through an iterative development process:
  - a geometry-based inverted hull outline
  - toon shading with a quantised diffuse ramp [9]
  - X-Toon extended toon shading with a 2D ramp lookup indexed by depth [10]
  - screen-space normal derivative edge detection
  - Sobel edge detection [11] with adaptive thresholding to distinguish skin from clothing
  - Gaussian prefiltered Sobel [12]

  - hierarchical edge detection combining depth, normal, and colour discontinuities
  - anisotropic Kuwahara painterly filtering [13]

  Each technique addressed limitations observed in its predecessor, and all are selectable at runtime without recompilation. A final technique was selected for the user study based on visual quality assessment across all three avatar platforms.

- **VR parameter tuning interface.** A fully procedural world-space UI panel was implemented in Unity. It enables live exploration of all shader parameters from inside the headset through controller raycasting. This tool was central to the iterative development process and can be reused for future shader experimentation.
- **User study on avatar abstraction and trust.** A user study designed as within subjects was conducted. It used prerecorded video scenarios in which a speaker appears as either a real person, a photorealistic Avaturn avatar, or an avatar rendered with NPR stylisation. The speaker recommends one of two practical options in each scenario. The video format ensures that all three conditions are observed under identical scene and performance contexts. This consistency enables a clean comparison across representation types. The study provides empirical evidence on how visual abstraction affects the perceived trustworthiness of an avatar's advice.

1.5. Thesis Outline

The remainder of this thesis is structured as follows:

Chapter 2 surveys prior work on NPR for 3D characters, avatar rendering and realism in VR, and trust and social presence in virtual environments, positioning this thesis within the existing research landscape.

Chapter 3 introduces the technical and theoretical foundations required for this work, covering edge detection mathematics, the Kuwahara filter, the Meta Avatars SDK rendering pipeline, and relevant concepts of trust and abstraction from human-computer interaction research.

Chapter 4 details the research design, elaborating on the three research questions above, and presents the overall system architecture, the iterative development strategy, and the user study design including stimuli, procedure, and measurement instruments.

Chapter 5 describes the full implementation of the NPR shader system, covering all stylisation techniques in detail, the runtime parameter tuning interface, and the rationale for the final technique selection.

Chapter 6 reports the technical evaluation of each stylisation technique and presents the results of the user study, followed by a discussion of findings and their implications.

Chapter 7 summarises the contributions of this thesis, reflects on its limitations, and identifies directions for future work.

2. Related Work

This chapter surveys the literature most relevant to the central research questions of this thesis. The investigation of Non-Photorealistic Rendering (NPR) as an abstraction mechanism for production avatar systems sits at the intersection of three established areas. The first is the perceptual and affective consequences of avatar realism and the Uncanny Valley phenomenon. The second is trust, credibility, and social presence in virtual agent interactions. The third is the technical design space of NPR rendering for animated human figures. The chapter concludes by identifying the specific gap that this thesis addresses.

2.1. The Uncanny Valley and Avatar Realism

The Uncanny Valley hypothesis was first articulated by Mori, who proposed that human affinity towards an artificial figure increases with its degree of human likeness up to a critical threshold. Subtle imperfections at that threshold cause a sharp drop in comfort [6]. The drop persists until the figure achieves complete human fidelity, after which affinity recovers. Originally framed in the context of physical robots, the hypothesis has since been applied extensively to computer-generated (CG) characters and virtual avatars.

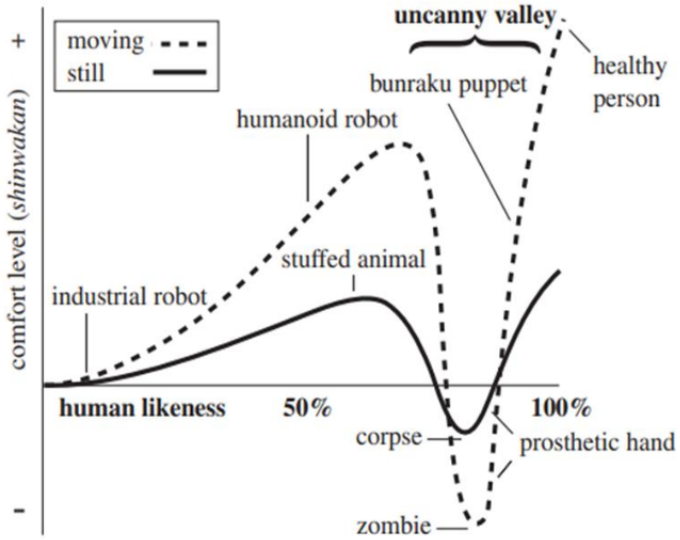


Figure 2.1.: The Uncanny Valley as originally illustrated by Mori [6]. The vertical axis represents comfort level (*shinwakan*). On the contrary, the horizontal axis represents degree of human likeness. The solid curve describes affective response to still figures, while the dashed curve describes figures in motion.

As illustrated in Figure 2.1, Mori plots two curves rather than one. Both follow the same general shape, rising as human likeness increases before dropping sharply in the high-fidelity region and recovering only at full human resemblance. The critical observation is that the curve for moving figures descends far deeper into negative comfort than the curve for still ones. The reference points Mori places inside the valley, a corpse, a zombie, and a prosthetic hand, share a common characteristic: each resembles a human closely across most perceptual dimensions but fails on some dimension that perception detects without effort. For animated digital avatars this implication is direct. A figure at roughly sixty to seventy percent human likeness, the region occupied by a stuffed animal or a humanoid robot, sits on the safe rising slope before the drop. A figure at eighty to ninety-five percent, realistic enough to raise expectations of full human fidelity yet not realistic enough to satisfy them, occupies the most problematic region of the curve, and motion makes that positioning considerably worse.

Furthermore, MacDorman and Ishiguro advanced the Uncanny Valley from a qualitative hypothesis into an empirically tractable research programme. They argued that androids offer a uniquely controllable stimulus for cognitive and social science research. This is because androids can be positioned at any point along the human likeness continuum with experimental control [14]. As part of this methodological contribution, they proposed semantic differential instruments to quantify eeriness and strangeness as measurable affective dimensions. This work established a foundation that transformed the valley from a speculative curve into a quantifiable perceptual response. They further identified the mechanism underlying the valley effect as categorical perception: when an agent occupies an ambiguous position between the human and nonhuman categories, the cognitive mismatch this ambiguity produces triggers a negative affective response. Because androids can be systematically positioned at any point along the human likeness continuum, this categorical account transfers directly to digital avatar research. Any near human representation, including a photorealistic avatar, risks eliciting the same response if its rendering does not resolve the human-nonhuman categorical boundary.

The perceptual consequences of render style choice for virtual humans were investigated systematically by McDonnell et al. [2]. Participants were exposed to video stimuli of the same animated character, rendered across six distinct styles spanning the range from photorealistic to fully stylised. Photorealistic rendering was associated with significantly higher eeriness ratings and lower appeal than stylised conditions, while NPR styles consistently achieved more positive affective responses across both dimensions. McDonnell et al. thereby established that stylised rendering is not merely a technical shortcut but a principled perceptual strategy for sidestepping the Uncanny Valley [2]. This finding constitutes the primary empirical motivation for the NPR approach pursued in this thesis. A companion study by the same group further demonstrated that the perception of emotional actions portrayed by animated characters is largely independent of avatar body style. This independence remained consistent across photorealistic, cartoon, and stylised skeletal representations [15]. Such consistency suggests that NPR abstraction preserves the communicative legibility of avatar behaviour even when surface appearance is substantially stylised.

Modern avatar systems sit squarely in the region of the uncanny valley curve that is most problematic [6]. They achieve surface realism sufficient to trigger high perceptual expectations, but cannot fully meet those expectations in motion timing, facial microexpressions, or gaze behaviour. The practical consequence is that a photorealistic avatar may undermine the trustworthiness it was designed to convey. Whether state of the art production avatars have crossed the valley was investigated by Seymour et al. in a field study at SIGGRAPH. The study compared a near photorealistic digital human to a caricature avatar [16]. Participants rated the photorealistic avatar as more trustworthy and expressed greater affinity towards it. However, the caricature condition did not trigger Uncanny Valley discomfort. It occupied instead a stable, neutral region of the affinity curve. This dissociation suggests that clearly stylised avatars avoid eeriness without sacrificing social legibility, and that trust advantages associated with photorealism are driven by factors beyond the mere avoidance of the valley.

Dill et al. validated the uncanny valley response specifically for CG characters drawn from contemporary commercial films and games. Familiarity ratings followed Mori's predicted curve, with the strongest negative response in the near-human range. Motion amplified this effect for every stimulus tested [17]. This confirmation that the valley extends to digital characters from popular media, not only to physical robots or androids, motivates its consideration across all virtual avatar contexts. A network meta-analysis by Tao et al., pooling 13 independent studies and 2,343 participants, subsequently quantified these effects across three realism levels [18]. Avatars at medium realism produced significantly higher eeriness than those at both abstract and photorealistic levels. By contrast, trustworthiness scaled monotonically with realism and reached its highest values for photorealistic avatars. This dissociation, lower eeriness at both extremes yet higher trustworthiness only at the photorealistic extreme, establishes the opposing pressures that the user study hypotheses are designed to resolve. Kuffner dos Anjos and Pereira further observed that realistic mesh avatars in third-person VR perspective produced pronounced uncanny effects when animation quality was constrained by sensor resolution, while abstract representations avoided the valley entirely in the

same viewing condition [19]. This pattern is consistent with the escape by design principle Mori originally proposed.

2.2. Trust and Credibility in Virtual Agents

Trust in human-agent interaction is a multidimensional construct. The foundational theoretical model for this study is the integrative model proposed by Mayer et al., who defined trustworthiness as comprising three independently evaluable components [20]. Ability is the trustee's domain-specific competence. Benevolence is the degree to which the trustee is genuinely oriented towards the trustor's interests. Integrity is the trustee's adherence to acceptable moral principles. Mayer et al. further distinguished trustworthiness from trust itself. They defined trust as the willingness to be vulnerable to the actions of another party, based on the expectation that the party will act appropriately, irrespective of the ability to monitor or control that party. McKnight and Chervany extended this conceptual structure into a typology separating trusting beliefs from trusting intentions [21]. Trust measurement can therefore span both attitudinal and behavioural dimensions. This distinction is consequential for avatar research: rendering style could plausibly affect the three trustworthiness components differently. NPR abstraction could, for example, reduce perceived competence by signalling an informal or non-professional aesthetic, while leaving perceived integrity intact because the verbal content of the recommendation is unchanged. Collapsing these three components into a single composite score would mask precisely this kind of differential effect. The three-subscale structure is therefore the appropriate operationalisation for a study that investigates rendering style rather than task framing or verbal content as its independent variable.

Alipour et al. conducted a PRISMA-compliant systematic review of 53 empirical studies at the intersection of Uncanny Valley effects and trust [22]. Their novel contribution is a taxonomy of how trust has been operationalised across the literature. The taxonomy distinguishes three approaches. Direct trust measures ask participants to explicitly rate trust or trustworthiness ($n = 24$ studies). A second approach covers trust-related constructs such as competence, credibility, and acceptability ($n = 11$ studies). The third captures trust-related intentions, meaning behavioural indicators such as advice following or willingness to interact ($n = 18$ studies). Among the 41 studies that empirically tested the UVE–trust relationship, 17 reported a direct negative effect, 24 found the relationship conditional on moderating factors including task type, age, and familiarity with the agent, and 10 found no significant effect. A critical observation is that 42 of the 53 included studies employed indirect encounters. This means participants passively viewed static images or video rather than engaging in live interaction. The majority relied exclusively on subjective self-report. Alipour et al. conclude that well-designed future studies should employ dynamic video stimuli and include at least one behavioural indicator alongside self-report ratings [22]. This prescription directly informs the evaluation design of this thesis.

Nowak and Rauh provided an early empirical foundation for the relationship between avatar appearance and credibility in online interaction [23]. In a study where 255 participants evaluated a series of avatars, perceived anthropomorphism and avatar gender expression both predicted credibility judgments. More anthropomorphic avatars were rated as more credible. The study established that avatar design choices constitute meaningful social signals rather than cosmetic variation. Participants, it showed, apply social heuristics to avatar appearance in ways that influence their willingness to engage and trust. Nowak and Biocca extended this framework. They demonstrated that anthropomorphism and perceived agency contribute independently to social presence and copresence, and that very high levels of visual anthropomorphism do not consistently produce higher social presence [3]. The implication is that avatar realism operates under diminishing perceptual returns: beyond a certain threshold of human-likeness, additional anthropomorphic cues do not translate into proportionally stronger social engagement, and may in some configurations reduce it. This pattern is structurally analogous to the uncanny valley, where affinity likewise ceases to scale linearly with human resemblance and inverts before recovering at full fidelity [6].

The direct effect of avatar stylisation on multi-component trust was investigated by Canales et al. [7]. They presented participants with virtual doctors rendered at three stylisation levels: realistic, semi-realistic, and caricatured. Trust was measured across competence, benevolence, and integrity dimensions before and after a short interaction. A behavioural indicator was also collected, via a forced-choice selection of which doctor the participant would consult for a follow-up. Self-reported trust ratings were broadly equivalent across all three style levels. However, the semi-realistic condition attracted the highest rate of behavioural selection, and this suggests that moderate stylisation can match or exceed the behavioural trust of full photorealism. The dissociation between explicit ratings and behavioural preference implies that render style influences implicit social judgment in ways not fully captured by self-report alone.

Convergent evidence was provided by Gao et al. [24]. They compared realistic and stylised virtual agents across trust, social presence, and perceived empathy. Stylised agents achieved trust levels statistically comparable to realistic agents on all measured dimensions, while voice characteristics were found to have a negligible effect on trust outcomes. Gao et al. also identified perceived agent age as a significant moderator: elder-appearing agents were trusted more across both style conditions, independent of render style [24]. This result implies that social characteristics of the avatar, such as perceived demographics, can function as independent trust cues that operate above and beyond the effects of rendering style. Taken together, these results establish that NPR stylisation does not constitute a trust penalty relative to photorealism.

Alimardani et al. isolated the contribution of visual appearance from voice quality in a 2×2 factorial study [25]. They found that mismatches between the two modalities elevated eeriness, yet produced no significant change in the rate at which participants adopted the agent's recommendations. The absence of a trust behavioural effect, despite a clear perceptual one, aligns with the Mayer et al. model: when demonstrated competence is held constant, appearance manipulations do not alter the trust placed in that competence. Additional support comes from Dubosc et al. [26]. They conducted a VR face-to-face interaction study and found that stylised avatars produced significantly lower eeriness and significantly higher trustworthiness ratings than photorealistic avatars, with no interaction between stylisation level and facial expression intensity. This pattern of results suggests that stylisation reduces the uncanny valley response without compromising social credibility. Cihodaru-Ştefanache and Podina's systematic review of 29 empirical studies on embodied conversational agents further identified consistent stylisation across all visual features as the most reliably recommended design strategy for avoiding uncanny valley responses, alongside accurate lip synchronisation and consistency between visual and voice realism levels [27].

Weidner et al. synthesised the effect of rendering style at a much larger scale in a systematic review of 72 studies on avatar and agent visualisation in augmented and virtual reality [1]. Full-body, realistically rendered avatars generally produced stronger presence, embodiment, and task performance than partial-body or more abstract alternatives, but this advantage was task dependent. Highly realistic avatars offered little benefit, and were sometimes a hindrance, in game-like and educational scenarios. This qualification indicates that photorealism's advantages documented across the wider literature are not unconditional, and it motivates testing whether they hold in the video-based advisory scenario examined here.

The Godspeed Questionnaire Series, developed by Bartneck et al., provides validated semantic differential scales for five perceptual dimensions: anthropomorphism, animacy, likeability, perceived intelligence, and perceived safety [28]. The series was originally developed for human-robot interaction. It has since been widely adapted for virtual avatar studies. The likeability subscale functions as an affective trust indicator. This subscale captures warmth and appeal as a proxy for social acceptance. The anthropomorphism subscale is particularly informative in this context because it measures the perceptual mechanism through which rendering style is hypothesised to modulate eeriness. If NPR reduces eeriness by reducing perceived human-likeness, the anthropomorphism subscale would detect that effect directly. If NPR reduces eeriness while leaving perceived human-likeness intact, that is a stronger and more theoretically significant result. Measuring anthropomorphism and eeriness independently makes this mechanism empirically testable.

The eeriness-specific instruments introduced by MacDorman and Ishiguro complement the Godspeed series by targeting the negative affective response to near-human imperfection directly [14]. Ho and MacDorman subsequently refined this operationalisation into a two-subscale instrument that distinguishes perceived eeriness from perceived human-likeness [29]. They validated the instrument across multiple stimulus types, and it now stands as a standalone alternative to the Godspeed eeriness items. Their principal finding is that the two dimensions are orthogonal: the reported correlation between eeriness and human-likeness in their validation data is approximately zero ($r \approx .02$). This orthogonality has a direct methodological consequence for the evaluation design of this thesis. Eeriness and perceived human-likeness could in principle be entangled, if NPR abstraction were to reduce both simultaneously. It would then remain unclear whether eeriness decreased because the stylised avatar is less unsettling, or merely because it reads as less human. Measuring the two dimensions separately makes this question answerable. A reduction in eeriness (D_score) unaccompanied by a corresponding reduction in anthropomorphism (C_score) from C2 to C3 would indicate that stylisation reduces discomfort through a mechanism other than a simple shift in perceived humanness. Such a pattern would carry more theoretical weight than a composite measure could reveal, since it would rule out the simpler explanation. Measuring them as a single composite would foreclose this distinction. The recommendation from Alipour et al. to include a behavioural indicator alongside self-report measures is addressed in the evaluation design of this thesis through a forced-choice preference task [22]. The task captures implicit trust judgment that explicit Likert ratings may not fully reflect. All instruments described in this section are employed in the user study.

2.3. Non-Photorealistic Rendering for Virtual Humans

Non-Photorealistic Rendering encompasses a family of techniques that deliberately depart from physical light simulation. These techniques abstract surface appearance through stylisation, edge detection, quantised shading, or painterly filtering. NPR, applied to virtual human avatars, is not a single technique but a design space with multiple technical dimensions. Each dimension produces a distinctive perceptual register, with different implications for avatar appearance and social reception. Kumar and Poornima provide a comprehensive survey of this design space [30]. They classify methods into five families: stroke-based rendering, colour image analogy, region-based rendering, image filtering abstraction, and direct stylisation.

The foundational contribution to NPR lighting is the cool-to-warm shading model introduced by Gooch et al. [31]. The model replaced the continuous Lambertian diffuse term with a hue shift from cool shadows to warm highlights, and this mimicked the conventions of technical illustration. While the model targets engineering drawings rather than characters, its core insight, that shading can communicate form without physically simulating light, became a reference point for subsequent stylised rendering work.

Silhouette rendering has been approached from two geometrically distinct directions. Geometry-based methods generate outlines using the GPU rasteriser itself. The inverted hull technique described by Lake et al. expands back-facing polygons along vertex normals by a small offset, and this creates a slightly enlarged shell of the mesh [9]. When front-facing geometry is rendered on top, only the protruding rim remains visible as an outline. The approach requires no image-space computation, scales predictably with polygon count, and remains consistent regardless of camera distance, properties that make it well suited to animated characters in real-time applications. Petikam et al. addressed a related limitation in this style of shading [32]. They proposed a shading rig that interpolates between shading states defined at authoring time, so that the tonal boundary remains stable as lighting conditions change rather than sliding with the Lambertian angle. Isenberg et al. provide a systematic survey of silhouette algorithms and conclude that geometry-based methods are robust and distance-independent but structurally incapable of detecting internal surface detail [8]. This structural limitation is particularly consequential for avatar rendering, where facial features, clothing folds, and skin creases all constitute expressive visual cues that lie within, rather than on, the silhouette boundary.

Image-space methods address this limitation directly by operating on the rasterised output rather than on the mesh. This makes them sensitive to any visible discontinuity on screen, regardless of whether it falls on the silhouette or in the interior of the figure. Screen-space normal derivatives, computed via hardware-accelerated `ddx/ddy` intrinsics available on all modern GPUs, detect abrupt changes in surface orientation without additional draw calls or intermediate render targets. This approach captures surface creases and mesh boundaries that geometry expansion cannot reach. The fundamental limitation is that edge thickness is measured in screen pixels rather than world-space units. This causes edges to scale with camera distance. Users in an immersive VR context move relative to other participants. This produces edges that appear overly thick at close range and invisible at distance.

Texture-based edge detection targets the painted surface layer. The Sobel operator, a discrete gradient approximation applied to the diffuse texture via a 3×3 convolution kernel, identifies luminance boundaries corresponding to clothing patterns, facial contours, and painted surface details [11]. It responds to what was painted onto the surface rather than to the surface shape itself. In contrast, geometry-based methods respond to shape rather than surface. Marr and Hildreth demonstrated that low-pass prefiltering is a prerequisite for stable gradient computation [12]. This observation motivates the Gaussian prefiltered variant explored in this work.

Extended toon approaches have explored richer parameter spaces. Barla et al. introduced X-Toon [10]. X-Toon parameterises the shading tone ramp by both view angle and depth. This enables a two-dimensional lookup that encodes artistic intent unavailable in a one-dimensional quantisation step. A separate line of work applies the Kuwahara filter to NPR. This produces region-based painterly abstraction without explicit edge detection. Kyprianidis et al. extended the isotropic Kuwahara with an anisotropic formulation aligned to local image structure. This produces brush-stroke effects oriented along salient contours [13]. The anisotropic extension requires a structure tensor computed from a prior render pass, a constraint directly relevant to the integration target of this thesis. A multiscale variant of this filter, also by Kyprianidis, propagates orientation estimates and filter outputs through a resolution pyramid [33]. This removes artefacts in low-contrast regions while maintaining real-time performance. Praun et al. demonstrated real-time hatching as a complementary branch of this design space [34]. Tonal regions in this branch are conveyed by oriented stroke textures rather than flat fills or edge lines.

The perceptual impact of NPR technique choices on avatar social reception has been studied through a combination of psychophysical and VR experiments. Wisessing et al. demonstrated that toon shading reduces the perceived intensity of facial emotional expression while leaving recognition accuracy intact [35]. Characters rendered in this style therefore convey the correct emotional content, without the perceptual amplification that photorealistic rendering provides. A follow-up study identified lighting parameters as a further modulator of eeriness [36]. Brighter key lighting and lighter shadow regions significantly reduced eeriness and increased appeal for both photorealistic and stylised characters. The absolute reduction was larger for photorealistic rendering, since it starts from a higher eeriness baseline. Together these findings provide a psychophysical mechanism for why NPR stylisation may reduce eeriness in an avatar advisory context. Zibrek et al. conducted a within-participants VR study in which the same animated character was presented in photorealistic and non-photorealistic render conditions [37]. Participants evaluated emotional recognition, social presence, and eeriness across both conditions. Photorealism did not significantly improve emotional recognition or social presence. In some conditions the NPR condition was rated more positively. This challenges the assumption that rendering quality scales linearly with social utility. Zibrek and McDonnell extended this investigation to embodied VR [38]. Photorealism significantly increased place illusion, the sense of physically inhabiting the virtual space, but had a weaker and less consistent effect on social presence. This dissociation suggests that the social judgment of an avatar, including trust, is less dependent on photorealism than on animation naturalness and behavioural coherence.

Avatar representation level across three discrete conditions, abstract, semi-realistic, and realistic, was examined by Latoschik et al. in an immersive social VR context [39]. The semi-realistic condition produced social experiences comparable to the realistic condition on several dimensions, while the abstract condition

diverged more substantially. This supports the view that a threshold of representational fidelity exists, above which additional photorealism yields diminishing perceptual returns. NPR avatars may already satisfy this threshold.

What connects nearly all of this body of work is a shared implicit assumption: the developer controls the rendering pipeline. Techniques are designed and validated against custom meshes, open rendering frameworks, or research engines where shader code, render passes, and material systems can be freely modified. The most directly related prior work on the avatar side, Canales et al., applied stylisation to custom assets in an open pipeline and compared coarse-grained style levels rather than specific technical approaches [7]. This assumption of full pipeline access does not hold for production avatar SDKs, where the rendering stack is managed by the SDK provider and exposes only constrained integration points.

2.4. Research Gap

The reviewed literature establishes three relevant findings. First, NPR stylisation is a principled strategy for avoiding the Uncanny Valley, with clear empirical support from McDonnell et al. and Zibrek et al. [2], [37]. Second, stylised avatars do not carry a trust penalty relative to photorealistic avatars in controlled evaluations, as demonstrated by Canales et al. and Gao et al. [7], [24]. Third, all prior work on both NPR rendering and avatar trust perception has assumed open pipeline access. This work relied on custom assets and research engines rather than production SDK systems.

Three specific gaps remain unaddressed. First, no published work has applied fine-grained NPR stylisation within a closed, compute-skinned, LOD-managed production avatar SDK, where the developer cannot modify the core lighting pipeline, add intermediate render passes, or alter material management. The NPR techniques surveyed above were all developed and validated under full rendering stack access. Whether the same techniques can be effectively integrated under production SDK constraints, and whether their perceptual properties hold in that context, remains an open question. This gap is addressed by the technical contribution of this thesis (RQ1 and RQ2).

Second, while prior work compared coarse stylisation levels, the literature has not examined the technical design space within NPR under production SDK constraints. Which technique families remain viable when render-pass access and material management are controlled by an SDK, and which best preserves avatar legibility while achieving perceptual abstraction, has not been evaluated. This gap is addressed by the iterative technique development and selection described in Chapters 4 to 6 (RQ2).

Third, no prior study has measured the trust and eeriness consequences of NPR stylisation applied to a photorealistic production avatar under a controlled evaluation protocol that uses validated multi-component instruments and a behavioural measure. Canales et al. [7] is the closest prior work, but compared coarse style categories rather than a specific NPR shader pipeline, used a between-subjects design, and did not measure eeriness or anthropomorphism as separate dimensions. This gap is addressed by the user study of this thesis (RQ3). The study tests three hypotheses: that NPR reduces eeriness relative to photorealism (H1), that NPR does not reduce self-reported trust (H2), and that eeriness negatively predicts trust across all three conditions (H3).

This thesis addresses all three gaps. It integrates a suite of NPR techniques into the Meta Avatar SDK under real production constraints, evaluates the visual and perceptual trade-offs across technique variants, and measures the trust and eeriness implications of NPR abstraction in a within-subjects, video-based user study. The study presents three conditions: a recording of a real person serving as the trust ceiling baseline (C1), an Avaturn avatar rendered at full photorealistic quality (C2), and the same Avaturn avatar rendered with the selected NPR shader pipeline (C3). Holding the avatar platform constant between C2 and C3 isolates the perceptual effect of NPR stylisation from platform variation. The evaluation instrument combines the Godspeed likeability and anthropomorphism subscales [28], eeriness scales grounded in MacDorman and Ishiguro [14] and refined by Ho and MacDorman [29], and a multi-component trust battery structured

around the ability, benevolence, and integrity dimensions of the Mayer et al. integrative trust model [20], operationalised within the trusting-beliefs typology of McKnight and Chervany [21]. A forced-choice behavioural preference task supplements the self-report battery, following the paradigm of Canales et al. [7] and the methodological prescription of Alipour et al. [22].

3. Background

This chapter introduces the technical and theoretical foundations required for the work described in subsequent chapters. It covers the concept of virtual avatars and visual representation, edge detection mathematics, the Kuwahara painterly filter, and the human factors concepts of trust and social presence relevant to the user study.

3.1. Mathematical Notation

Throughout this thesis, scalars are denoted by italic letters x, vectors by bold lowercase letters $\mathbf{x}$, and matrices by bold uppercase letters $\mathbf{X}$. The i-th element of vector $\mathbf{x}$ is written x_i. Convolution of a signal I with a kernel K is written $K * I$. In shader pseudocode, `ddx(f)` and `ddy(f)` refer to the hardware-computed screen-space partial derivatives of a value `f` with respect to the horizontal and vertical screen axes, respectively.

3.2. Edge Detection

An edge in an image is conventionally defined as a location where intensity changes rapidly. Detecting such locations involves estimating the spatial gradient of the image intensity function, then applying a threshold to identify pixels where the gradient magnitude exceeds some criterion.

3.2.1. Gradient-Based Detection

Given an image $I(x, y)$, the gradient is the vector $\nabla I = (\partial I/\partial x,\ \partial I/\partial y)$. Its magnitude

$$M = \|\nabla I\| = \sqrt{\left(\frac{\partial I}{\partial x}\right)^2 + \left(\frac{\partial I}{\partial y}\right)^2} \tag{3.1}$$

provides a scalar measure of edge strength at each pixel. In practice, the partial derivatives are approximated by discrete convolution with finite-difference kernels.

The image intensity function $I(x, y)$ used above is not the raw colour channels of a rendered frame, since RGB gradients would respond independently to hue differences that carry no perceptible brightness change. Edge detection in this thesis therefore operates on a single luma channel derived from a perceptually weighted sum of the red, green, and blue components:

$$L = 0.2126\,R + 0.7152\,G + 0.0722\,B \tag{3.2}$$

following the coefficients defined in ITU-R Recommendation BT.709 for high-definition video [40]. The weighting reflects the differing sensitivity of the human visual system to each primary. Green contributes most strongly since photopic vision peaks near the middle of the visible spectrum, whereas blue contributes least. This single luma channel, rather than the raw colour channels, is then differentiated. Edge strength consequently aligns with perceived brightness contrast rather than an arbitrary numerical difference between colour channels.

The Sobel operator [11] estimates the gradient using two 3×3 kernels:

$$G_x = \begin{bmatrix} -1 & 0 & +1 \\ -2 & 0 & +2 \\ -1 & 0 & +1 \end{bmatrix}, \qquad G_y = \begin{bmatrix} +1 & +2 & +1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{bmatrix}. \tag{3.3}$$

G_x responds to horizontal intensity transitions (vertical edges). G_y responds to vertical transitions. The gradient magnitude is $M = \sqrt{G_x^2 + G_y^2}$, computed pointwise after convolving the image with each kernel separately. The ±2 weights along the central row and column give the Sobel operator a small amount of spatial smoothing relative to the simpler Prewitt kernel. This improves noise robustness marginally.

The Roberts Cross operator uses a 2×2 diagonal kernel pattern instead:

$$R_1 = \begin{bmatrix} +1 & 0 \\ 0 & -1 \end{bmatrix}, \qquad R_2 = \begin{bmatrix} 0 & +1 \\ -1 & 0 \end{bmatrix}. \tag{3.4}$$

The edge magnitude is $|R_1| + |R_2|$. This approximates the gradient diagonal without a centre sample. Roberts Cross is faster and more local than Sobel but more sensitive to noise. This thesis uses it specifically for colour edge detection within the hierarchical technique, where its two-sample footprint keeps the per-fragment sample count low.

3.2.2. Screen-Space Normal Derivatives

Modern GPU pipelines expose per-fragment screen-space derivatives of any interpolated value via the `ddx` and `ddy` intrinsics. These are computed using the finite difference between adjacent 2×2 pixel quads. Applying this to the world-space surface normal $\mathbf{N}$ gives

$$M_N = \sqrt{\|\texttt{ddx}(\mathbf{N})\|^2 + \|\texttt{ddy}(\mathbf{N})\|^2}, \tag{3.5}$$

which responds to abrupt changes in surface orientation: creases, mesh boundaries, and regions of high curvature. No additional texture samples are required, and the computation executes at negligible cost relative to the surrounding fragment shader. The significant practical limitation is that screen-space derivatives are measured in pixel units. A fixed-world-space edge, such as the crease between a jacket lapel and its collar, projects to a varying number of pixels depending on camera distance. This causes edge thickness to scale with the viewpoint rather than with the surface.

3.2.3. Gaussian Pre-Filtering

Differentiation amplifies high-frequency components. Marr and Hildreth [12] established that stable edge detection requires low-pass smoothing before gradient computation. Because convolution and differentiation are both linear operations, the smoothed gradient can be written as

$$\nabla(G_\sigma * I) = (\nabla G_\sigma) * I, \tag{3.6}$$

where G_σ is a Gaussian kernel with standard deviation σ. Canny's influential edge detector [41] built on this principle: smooth with a Gaussian, compute the gradient of the smoothed image, and apply non-maximum suppression to obtain thin, stable edge maps. A true two-pass separable convolution is not available in a single-pass fragment shader. The Gaussian can instead be approximated by sampling the source texture at a fixed set of offsets with precomputed weights. This approximation accepts that the smoothed values are recomputed independently at each sample position.

3.3. The Kuwahara Filter

The Kuwahara filter [13] is a structure-preserving smoothing operator that produces piecewise-uniform regions with sharp inter-region boundaries. This visual effect resembles painterly brushwork. The filter divides the neighbourhood around each pixel into k overlapping subregions. It then computes the mean and

variance of pixel values within each subregion, and sets the output to the mean of whichever subregion has the lowest variance. Formally, for subregions $\{Q_i\}_{i=1}^{k}$ and their respective means μ_i and variances σ_i^2:

$$F_{\text{Kuwahara}}(x, y) = \mu_j, \qquad j = \arg\min_i \sigma_i^2. \tag{3.7}$$

The effect is that pixels on uniform surfaces are assigned the mean colour of their most locally uniform neighbourhood, while pixels near strong boundaries pick the subregion that falls entirely on one side. Edges are therefore sharpened rather than blurred, even as internal regions are smoothed towards a uniform painterly fill.

The classical isotropic formulation uses four quadrant-shaped subregions of equal size. This produces a rotationally symmetric filter that does not prefer any orientation. An anisotropic extension by Kyprianidis et al. [13] aligns the filter kernel with local image structure, by computing a structure tensor from a prior render pass. This produces brush-stroke effects oriented along salient contours. The anisotropic version produces superior results visually, but it requires an intermediate render target that is incompatible with a single-pass fragment hook.

3.4. Virtual Avatars and Visual Representation

An avatar is a digital body that represents a user within a shared virtual space. The term covers a wide range of forms, from abstract geometric markers to fully articulated, photorealistic human figures, and the choice of representation carries consequences that extend well beyond visual preference.

Avatar representations can be placed along a continuum of anthropomorphism and visual fidelity. Abstract or stylised avatars, at one end, retain the structural properties of a humanoid figure without committing to the surface detail of a real person. Photorealistic avatars, at the other end, attempt to reproduce the user's physical appearance with enough accuracy to approach the visual fidelity of a photograph. Where an avatar sits along this continuum has measurable consequences for how users are perceived, how much social presence is felt in the interaction, and how the avatar is trusted [3], [39].

Avatar appearance also shapes the behaviour of the person embodying it. Yee and Bailenson [42] demonstrated through a series of experiments that users assigned more attractive or physically taller avatars adopted behaviours consistent with those characteristics: they disclosed more personal information, negotiated more assertively, and maintained shorter interpersonal distances. This tendency, termed the Proteus effect, illustrates that visual self-representation in virtual environments is not a passive interface element but an active influence on how users act and present themselves.

Contemporary avatar platforms have made high quality avatar creation accessible without requiring manual artistic work. Services such as Avaturn reconstruct a personalised, fully rigged avatar from a single uploaded photograph, via learned three-dimensional reconstruction. This produces a mesh with facial blendshape animation support that integrates directly into standard game engine pipelines. Such accessibility makes individualized photorealistic representation practical for a wide range of applications. It also sharpens the perceptual stakes: an avatar that looks like a real person will inevitably be compared to one, and any shortfall in that comparison has consequences both for immersion and for how the user is received by others.

3.5. Trust and Social Presence in Avatar Interaction

Trust in the context of human-avatar interaction is typically understood as a social judgment: the degree to which an observer is willing to rely on the entity represented by the avatar. It is related to, but distinct from, social presence. Social presence refers to the subjective sense of sharing a space with another being. Both constructs are influenced by avatar appearance, but through different mechanisms. Social presence depends strongly on the perceived anthropomorphism of the avatar and the responsiveness of its motion [3]. Trust

is additionally sensitive to perceived competence, predictability, and familiarity, including the absence of uncanny valley discomfort [6].

The dominant theoretical account of interpersonal trust in organisational and HCI contexts is the Ability-Benevolence-Integrity (ABI) model of Mayer et al. [20]. The model identifies three independent components. *Ability* refers to the trustee's perceived competence within a relevant domain: a person, or an avatar, may be trusted to give technically sound advice without being trusted to act in the observer's interest. *Benevolence* refers to the extent to which the trustee is perceived to care about the observer's wellbeing beyond its own motivations. *Integrity* refers to whether the trustee is perceived to adhere to acceptable principles, including honesty and avoidance of deception. The three components are theorised to be separable: a stylistic change to an avatar's appearance could, for example, reduce perceived competence by signalling an informal or non-professional aesthetic while leaving perceived integrity intact, since integrity judgements may rely more heavily on verbal content than on visual form. Measuring the three components separately rather than as a single composite is therefore necessary to detect these differential effects.

Measurement instruments for the ABI dimensions in human-computer interaction contexts were operationalised by McKnight and Chervany [21]. Their trusting-beliefs typology provides item-level questionnaire language for each component. The instrument used in this thesis adapts McKnight and Chervany's items into the advisor-evaluation context described in Chapter 4.

A complementary set of measurement instruments focuses on perceptual responses to avatar appearance. The Godspeed questionnaire developed by Bartneck et al. [28] provides semantic differential items for five subscales: anthropomorphism, animacy, likeability, perceived intelligence, and perceived safety. The questionnaire was originally validated on robotic systems, and it has since been applied extensively to avatar stimuli. The anthropomorphism and likeability subscales are relevant here: the anthropomorphism subscale measures how human-like a stimulus is perceived to be, and the likeability subscale provides an affective assessment complementary to the cognitive trust subscales.

Eeriness is the affective component most directly associated with uncanny valley responses. It is not well-captured by the Godspeed items. Ho and MacDorman [29] developed a separate eeriness instrument consisting of six semantic differential items targeting strangeness, unsettlingness, and perceptual discomfort. Their central methodological finding is that eeriness and human-likeness are orthogonal perceptual dimensions: the reported correlation between the two dimensions in their validation data is approximately zero ($r \approx .02$). This orthogonality is relevant to the measurement design of this thesis, because it means that a reduction in eeriness from a photorealistic avatar to an NPR avatar cannot be explained by a simple shift in perceived humanness alone. If eeriness falls while anthropomorphism remains stable, some other perceptual mechanism is responsible.

A systematic review by Alipour et al. [22] examined the relationship between uncanny valley discomfort and trust across 42 empirical studies. Their review found a consistent negative association: stimuli that produce higher eeriness ratings tend to receive lower trust ratings, and this association is independent of whether trust is measured with cognitive or affective instruments. Alipour et al. also found that the association is strongest when stimuli are presented as dynamic video clips rather than static images. This finding motivates the use of video stimuli in this thesis.

A consistent finding across this literature is that the relationship between visual fidelity and perceived trustworthiness is not linear. Near-photorealistic representations can produce lower trust scores than clearly stylised ones, consistent with the uncanny valley hypothesis. This non-linearity motivates the experimental design of this thesis. The design tests whether NPR abstraction can reduce eeriness while leaving trust intact, rather than assuming that any reduction in realism will necessarily reduce trust.

4. Methodology

This chapter describes the research questions guiding this work, the three avatar platforms used as development and evaluation targets, the NPR stylisation techniques developed across those platforms, and the design of the user study that evaluates their perceptual consequences.

4.1. Research Questions

Three research questions structure the contributions of this thesis.

RQ1: Can NPR stylisation techniques be implemented and applied across avatar platforms with different rendering architectures, specifically Mixamo, the Meta Avatars SDK, and Avaturn, and what technical constraints does each platform impose on achievable effects?

This question is primarily technical. It asks first whether a viable integration path exists for distinct platforms that differ substantially in mesh topology, shader accessibility, and pipeline structure. It then asks what each integration path permits and what it prevents, since the architectural constraints of each platform determine which NPR techniques are realisable and how they appear in practice.

RQ2: Can NPR stylisation serve as a mechanism for reducing the uncanny valley effect in virtual avatars?

This question asks whether NPR stylisation can relocate a digital character away from the near-human fidelity region of the uncanny valley by reducing perceived photorealism while preserving sufficient feature legibility for social judgment. It drives the iterative technique development documented in Section 4.4: each version introduces a different abstraction approach, and the overall goal is to identify a configuration that achieves a stable, readable stylised appearance without the residual realism that triggers uncanny responses. The user study then tests whether the selected composite realises this perceptually.

RQ3: Can an NPR avatar be perceived as more trustworthy than a real human speaker?

This is the central empirical question. Uncanny valley theory predicts that avatars at near-human fidelity attract stronger negative affect than representations at lower fidelity [6], and this raises the theoretical possibility that an NPR avatar could surpass the trust level of a photorealistic one. RQ3 extends this reasoning: it asks whether stylisation can bring perceived trustworthiness close enough to match or exceed that of a real human speaker. This baseline therefore provides the trust ceiling against which both avatar conditions are compared, and the formal hypotheses tested are stated in Section 4.8.1.

4.2. User Study Overview

To address RQ2 and RQ3, a within-subjects study was conducted in which participants viewed video stimuli of an avatar delivering a spoken recommendation across three everyday decision scenarios. Three conditions were presented to every participant: a real human speaker (C1), a photorealistic avatar (C2), and the same avatar rendered with an NPR composite (C3). Holding the avatar platform constant between C2 and C3 isolates the perceptual effect of the stylisation from any variation in avatar identity or animation quality.

After each video, participants rated the advisor on perceived eeriness, likeability, and trust across three dimensions: competence, benevolence, and integrity. At the close of the session, participants selected which advisor they would most trust to follow, providing a behavioural measure of trust alongside the self-report scales. The full study design, questionnaire structure, and analysis plan are described in Chapter 6.

4.3. Avatar Platforms and Rendering Constraints

The NPR techniques developed for this thesis are evaluated across three distinct avatar platforms, each offering a specific combination of capabilities and integration constraints relevant to NPR development. Mixamo offers the most flexible environment for initial shader implementation due to its unrestricted access to URP shaders, but its exported characters lack facial rigging. In contrast, the Meta Avatars SDK provides native Unity integration and built-in body tracking, and this supports evaluation of NPR behaviour on animated avatars in a VR environment. However, it restricts shader customisation to a proprietary injection interface, and NPR shader development is hence the most constrained on this platform. Finally, Avaturn combines unrestricted URP access with high fidelity facial reconstruction and full blendshape animation, though it requires an additional import step for its glTF assets.

Mixamo.

Figure 4.1.: Mixamo: baseline avatar for NPR shader development.

The Mixamo platform provides an online library of rigged humanoid characters and skinned animations that can be exported to Unity as standard FBX assets [43]. Since Mixamo avatars are standard skinned meshes without proprietary shader restrictions, all NPR techniques can be implemented directly within the URP rendering pipeline. Consequently, Mixamo is the least constrained of the three platforms, and it is therefore the most appropriate environment for the initial exploration and development of each proposed approach before adaptation to the more constrained platforms. Nevertheless, Mixamo character exports do not include facial rigging, and the avatar's face remains static throughout all renders generated on this platform. As a result, Mixamo is unsuitable for the user study, although the Mixamo character (Figure 4.1) served as the primary prototyping target during the initial development phase, where unrestricted shader access facilitated rapid prototyping and iterative refinement.

Meta Avatars SDK.

Avatar rendering, face tracking, and body tracking are handled internally by the Meta Avatars SDK, Meta's avatar framework for the Meta Quest 3 [44]. Since one objective of this thesis is to evaluate NPR avatars in VR on this device, the Meta Avatars SDK was examined for the shader customisation capabilities it provides for NPR integration (Figure 4.2). Its built-in face and body tracking eliminates the need for external motion capture systems, and the proposed NPR shaders can thus be evaluated on naturally animated avatars in a VR environment. However, custom NPR shaders cannot be integrated through standard URP material slots due to the SDK's internal rendering architecture. Instead, NPR rendering logic must be injected through hook functions exposed by the SDK at defined points in the shader pipeline. This constrains both the range

Figure 4.2.: Meta avatar: baseline avatar for NPR integration within the Meta Avatars SDK.

of achievable NPR effects and the structural requirements of the shader code. The technical details of this integration are described in Chapter 5.

Avaturn.

Figure 4.3.: Avaturn: baseline avatar for the user study.

Avaturn is an avatar creation service that reconstructs a personalised rigged avatar with facial blendshape targets from three uploaded photographs (Figure 4.3) [45]. The service exports its output as `.glb` files, which implement the glTF 2.0 standard and bundle geometry, PBR materials, and skeletal rigging within a single self-contained file [46]. Since Unity does not natively support this format at runtime, an additional import package is required to load the avatar, as described in Chapter 5. Once loaded, it functions as a standard skinned mesh with PBR materials and unrestricted access to the URP rendering pipeline, and the proposed NPR shaders can be implemented directly. Consequently, Avaturn served as the primary platform for the final user study, since it is the only platform among the three that combines unrestricted shader access with personalised facial reconstruction and full blendshape animation. This complemented the prototyping workflow established with Mixamo and the VR evaluation performed using the Meta Avatars SDK.

Table 4.1.: Summary comparison of the three avatar platforms across dimensions relevant to NPR integration.

	Mixamo	**Meta Avatars SDK**	**Avaturn**
Shader access	Full URP, unrestricted	Hook injection only (SDK post lighting)	Full URP, unrestricted
Transparency	Explicit alpha cutout	SDK managed	Explicit alpha cutout
Facial animation	None	SDK managed face tracking	Blendshape animation
Key challenge	No facial rigging	Shader manipulation	Initial Unity import
Strength	Full shader freedom	Native Quest 3 integration	High fidelity facial quality

4.4. NPR Technique Design

NPR techniques in computer graphics encompass a wide range of abstraction levels [8]. Choosing a level that is too low may place an avatar within the uncanny valley, whereas excessive stylisation can reduce visual detail and compromise expressive readability [47], [48]. Thus, selecting the appropriate degree of abstraction requires calibrating the balance between realism and visual expressiveness.

This calibration was performed through an iterative design process in which several NPR techniques were developed and evaluated. Each version addressed a specific limitation identified in the preceding version, and the final selection was validated through a formal user study. Main techniques are labelled V1 through V6. Extended variants carry a decimal suffix, as in V2.2 and V4.2, indicating a directed modification of the preceding version.

Development proceeded in sequence across the three platforms described in Section 4.3. The SDK's shader hook architecture imposed implementation constraints that shaped several architectural decisions, and the final transfer to Avaturn served as a cross-platform verification of each technique's visual output. The following sections describe the motivation, design rationale, and visual characteristics of each approach. Full implementation details are provided in Chapter 5.

4.4.1. Shared Rendering Structure

All proposed NPR techniques share a common two-pass rendering structure. The first pass generates the silhouette outline using the inverted hull method, which is applied consistently across all three platforms. The second pass computes the NPR surface effect for each technique within the forward rendering pass. Separating outline generation from surface shading allows the outline width, colour, and depth behaviour to be adjusted independently while ensuring that the outline geometry is rendered at a consistent depth regardless of the transparency mode of the surface material.

4.4.2. V1 — Inverted Hull Outline

Object silhouettes are a defining visual feature of nonphotorealistic rendering [8]. Accordingly, a robust outline generation technique was required as the foundation for all NPR techniques developed in this thesis. The inverted hull method, also known as the shell method, was selected because it produces stable geometric silhouettes without requiring screen-space processing [8]. It generates an outline shell by enlarging back-facing polygons and relies on GPU face culling to produce the outline, eliminating the need for edge detection or additional rendering passes. The method operates across two dedicated stages:

1. **Outline Pass.** The mesh is rendered with front-face culling enabled, displacing each vertex outward along its world-space surface normal by a configurable distance w:

$$\mathbf{P}' = \mathbf{P} + \mathbf{N} \cdot w \tag{4.1}$$

where **P** is the original vertex position and **N** is the world-space surface normal. Only the back-facing geometry is rasterised, forming an enlarged shell around the mesh.

2. **Forward-Lit Pass.** The mesh is subsequently rendered with standard back-face culling. The front-facing surfaces overwrite the interior of the previously rendered shell, leaving only the protruding rim visible as a silhouette border with controllable thickness.

This approach is well suited to real-time NPR avatar rendering because it requires no offline mesh preprocessing and executes entirely within the GPU vertex transformation stage [8]. As a result, its computational cost scales predictably across assets of varying complexity. However, the expanded shell and the original surface may produce similar depth values, which can introduce z-fighting artefacts along the silhouette boundary. To prevent this, a depth bias is applied to shift the outline geometry slightly towards the camera in clip space.

Despite its stability and low rendering cost, the inverted hull method has several limitations when applied to avatar rendering. Figure 4.4 shows the results across all three platforms. Surfaces with degenerate or coincident normals, such as flat planes, produce no visible outline because displacing vertices along identical surface normals does not generate an expanded shell. In addition, the method does not register interior surface boundaries. Consequently, creases between body regions, clothing seams, and facial contours receive no edge marking, as the method captures only the external silhouette of the mesh [8]. This limitation is shared by geometry-based outline methods and motivates the interior edge detection techniques introduced in V3.

(a) Mixamo: inverted hull silhouette outline.

(b) Meta SDK: inverted hull silhouette outline.

(c) Avaturn: inverted hull silhouette outline.

Figure 4.4.: Inverted hull geometry outline applied to all three avatar platforms. The slight floor penetration visible at the Meta Avatar's feet in (b) reflects a tracking boundary offset during capture and is not a property of the outline technique.

Alpha Cutout for Transparency

Avatars on the Mixamo and Avaturn platforms contain alpha-masked elements such as eyelashes and hair that are represented as flat cards. Their visible shape is defined by discarding fragments below an opacity threshold. Without alpha testing in the outline pass, the expanded hull would be generated from the full geometric extent of each card. This produces solid rectangular borders rather than outlines that follow the visible hair silhouette, as shown in the left column of Figure 4.5. Alpha testing is therefore applied using the GPU's built-in fragment clip operation to discard fragments whose texture alpha value falls below a

configurable cutoff threshold α_{cutoff}:

$$\text{if } \alpha_{\text{texture}} < \alpha_{\text{cutoff}} \quad \text{then discard} \tag{4.2}$$

This test executes in both the outline pass and the forward-lit pass to maintain silhouette consistency. Figure 4.5 shows the resulting outline with the threshold applied. Unlike alpha blending, this binary approach does not require depth sorting and introduces negligible additional fragment processing cost.

Conversely, the Meta Avatar platform handles hair and eyelash rendering through its internal pipeline. Transparent fragments are therefore already discarded internally by the SDK, meaning that manual alpha testing is not required during outline generation. The same two-pass rendering structure is hence maintained on this platform, although the underlying outline masking approach differs from the explicit alpha cutout used on Mixamo and Avaturn. The full technical details are discussed in Section 5.4.1.

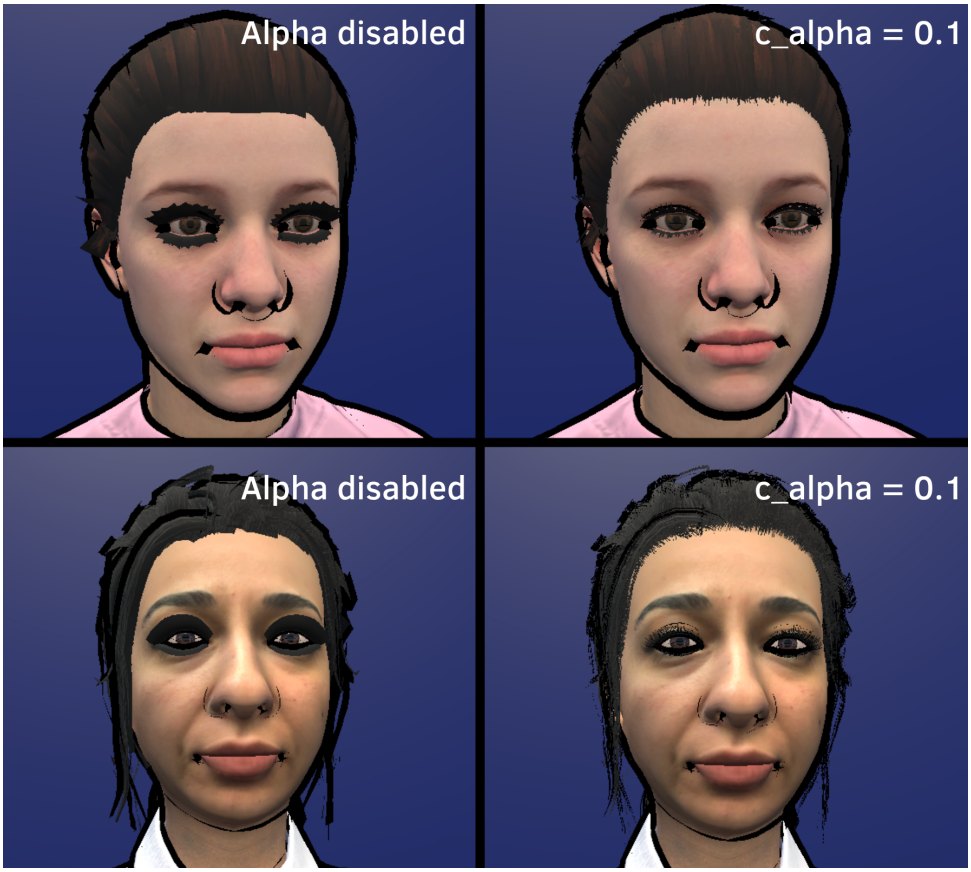


Figure 4.5.: Alpha cutout on the inverted hull outline pass (Mixamo, top; Avaturn, bottom). Left: alpha testing disabled. Right: alpha cutout enabled at $\alpha_{\text{cutoff}} = 0.1$.

4.4.3. V2 — Toon Shading

Toon shading, also known as cel shading, is one of the most widely adopted techniques in NPR, where stylisation is achieved by simplifying the tonal representation of a surface rather than reproducing physically accurate illumination [9]. Discretising the illumination response into uniform tonal regions produces the characteristic flat appearance associated with hand-drawn animation. Correspondingly, the technique is well suited to interactive character rendering [9]. Accordingly, toon shading was selected as the V2 baseline because it provides a controlled starting point from which each subsequent technique introduces a targeted increase in stylisation. However, implementing this stylisation across the three avatar platforms requires adaptations that account for differences in each platform's rendering architecture.

On Mixamo and Avaturn, the continuous illumination response is converted into discrete tonal regions using the following quantisation function:

$$t = \frac{\lfloor s \cdot n \rfloor}{n} \tag{4.3}$$

where s represents the surface lighting response and n defines the number of shading bands. This operation replaces continuous illumination gradients with discrete tonal transitions, producing the characteristic stepped appearance of toon shading [9]. Figure 4.7 shows the resulting appearance at the default band configuration used in this thesis, with the specific value and its selection criteria given in Section 5.4.2. The resulting tonal structure consists of highlight, midtone, shadow, and deep shadow regions.

Several refinements were introduced to improve the visual quality and temporal stability of the baseline toon shading model. Tonal region transitions were softened at each band boundary. This prevents normal jitter near a boundary from snapping a whole surface patch unpredictably between adjacent bands. Additional ambient illumination was introduced to prevent heavily shadowed regions from becoming visually disconnected from the avatar. A rim highlight was also added to improve visual separation between the avatar and its background where flattened shading reduces depth cues (Figure 4.6). Finally, scene cast shadows were deliberately excluded because they introduce an additional visual variable that could interfere with evaluating the effect of tonal quantisation in isolation.

Unlike Mixamo and Avaturn, where intermediate lighting values are directly accessible, the Meta Avatars SDK exposes only the final composited appearance after its internal rendering process. Consequently, quantisation is applied to the available colour output rather than to the surface lighting response itself. Despite the different implementation, both approaches achieve the same intended stylisation by replacing smooth illumination gradients with discrete tonal regions.

Across all three platforms, V2 successfully establishes a simplified graphic appearance through tonal quantisation. However, the technique has two important limitations. First, the distribution of tonal regions remains closely coupled to the lighting response. This leaves little independent artistic control over the tonal transitions. Second, V2 does not introduce additional edge information for interior features such as facial contours, clothing folds, or structural boundaries, as a technique driven purely by shading. These constraints motivate the subsequent development stages. V2.2 extends the tonal representation through X-Toon shading to provide greater control over the tonal distribution, while V3 introduces interior edge detection to enhance the visibility of local avatar features beyond what tonal quantisation alone can convey.

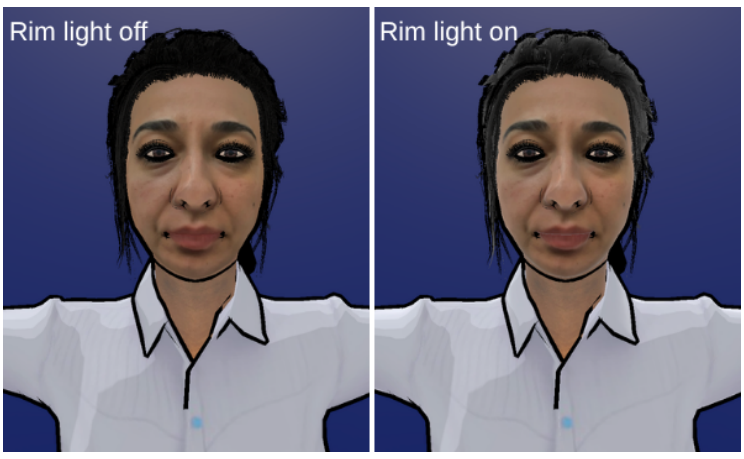


Figure 4.6.: Rim highlight effect in V2 toon shading.

(a) Mixamo: four discrete shading bands with ambient and rim.

(b) Meta SDK: four discrete shading bands applied to the fully composited colour produced by the SDK renderer.

(c) Avaturn: four discrete shading bands with ambient and rim.

Figure 4.7.: V2 (Toon Shading) across all three avatar platforms.

4.4.4. V2.2 — X-Toon Shading

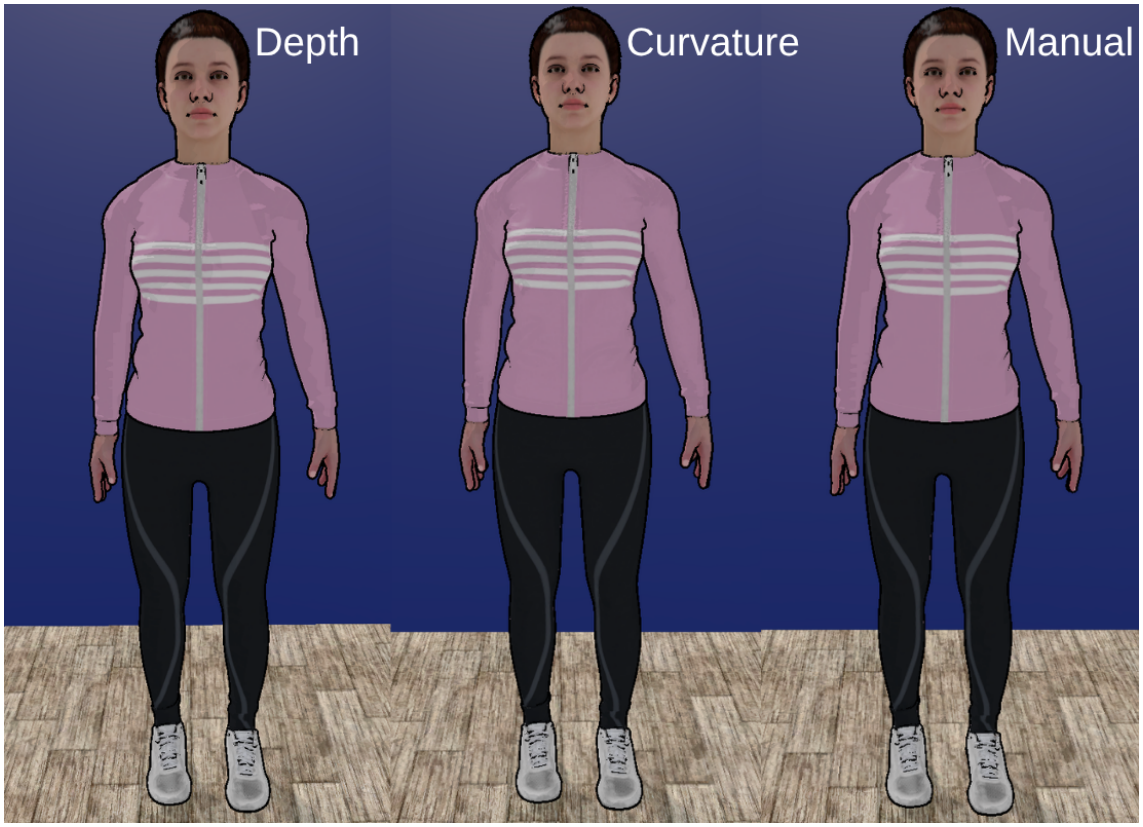
X-Toon shading extends conventional toon shading by replacing the 1D toon ramp with a 2D ramp texture, where the horizontal axis represents the lighting response and the vertical axis controls the abstraction level [10]. This second axis allows the surface appearance to be adjusted independently of the lighting conditions, and this provides finer stylisation control. A higher abstraction level reduces local tonal contrast and softens transitions between shading regions towards a flatter, more uniform appearance. Figure 4.8 illustrates the ramp texture used in this work.

Figure 4.8.: 2D ramp texture used for X-Toon shading.

In addition to tonal abstraction, Barla et al. described a shape detail extension that modifies the surface normal field by blending the original mesh normals with normals derived from a geometrically simplified shape [10]. This blending was incorporated into the Mixamo and Avaturn implementations. It resulted in reducing local surface variation and producing more coherent stylised shading.

Barla et al. further described an attribute mapping mechanism that controls the abstraction level through three driving modes: depth, curvature and manually controlled [10]. All three were evaluated in this work. The user study uses a fixed camera distance much closer to the avatar than the general platform comparison setup, and at that distance the depth and curvature modes described above produce almost no variation, as Section 4.5 explains in detail. In comparison with V2, X-Toon produced a softer and more controllable stylised appearance by allowing the stylisation level to be adjusted independently of the illumination response, and was consequently retained as the toon shading component in all later technique combinations.

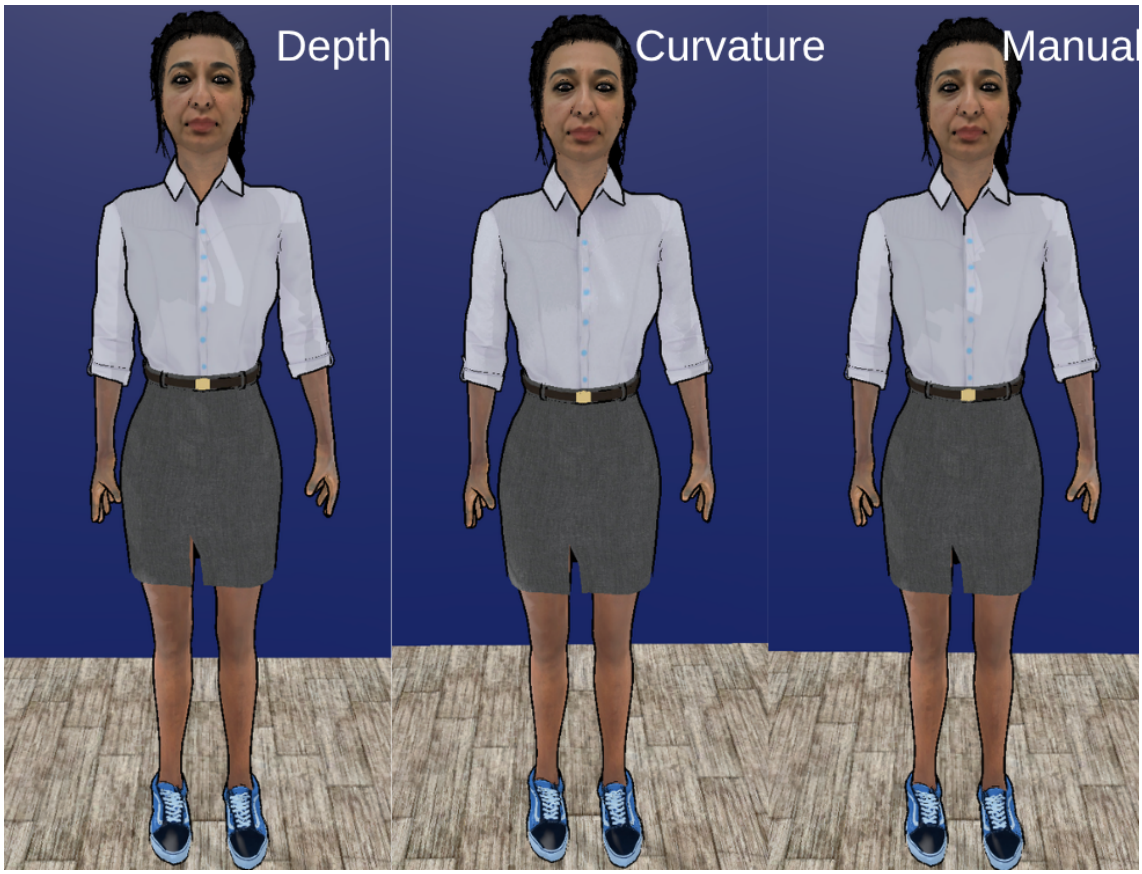
Despite this additional control, X-Toon remains driven only by shading and therefore does not introduce edge information for interior features. Facial contours, clothing folds, and texturally encoded details thus remain visually unmarked unless they produce sufficient variation in the tonal response. This limitation is inherited from V2 and motivates the edge-based techniques introduced in V3.



(a) Mixamo: depth, curvature, and manual abstraction modes.

(b) Meta SDK: depth, curvature, and manual abstraction modes.

Figure 4.9.: V2.2 (X-Toon) across the three avatar platforms. From left to right: depth-driven, curvature-driven, and manually controlled abstraction.



(c) Avaturn: depth, curvature, and manual abstraction modes.

4.4.5. V3 — Screen Space Normal Edge Detection

Silhouette extraction techniques can be classified into image space, object space, and hybrid approaches [8]. Unlike the object-space outline method introduced in V1, image-space techniques analyse per-pixel information generated during rendering to identify visually significant boundaries. One commonly used cue is the surface normal, whose orientation changes abruptly at creases, folds, and other geometric features. In prior work, per-pixel surface normals stored in G-buffer channels have been used as a cue for edge detection [49]. Discontinuities across these same channels detect silhouette, border, and crease edges in real-time NPR rendering, and crease edges in particular are recovered from the normal buffer [50]. It therefore gives image-space techniques access to the interior geometric structure that silhouette methods alone leave undetected.

V3 builds on this image-space principle and introduces the first interior edge signal into the proposed rendering pipeline. It detects changes in the interpolated world-space surface normal across neighbouring screen pixels, and the resulting edge marks trace geometric contours in the interior of the mesh where tonal variation alone would not identify boundaries. This detection is realised through GPU screen-space derivative functions that evaluate the difference between adjacent fragments directly within the fragment pass. Consequently, they require neither a stored normal texture nor an additional rendering pass.

The GPU derivative operators compute the horizontal and vertical gradients of the world-space surface normal $\mathbf{N}$, and the magnitude of the resulting gradient, E_N, is defined as:

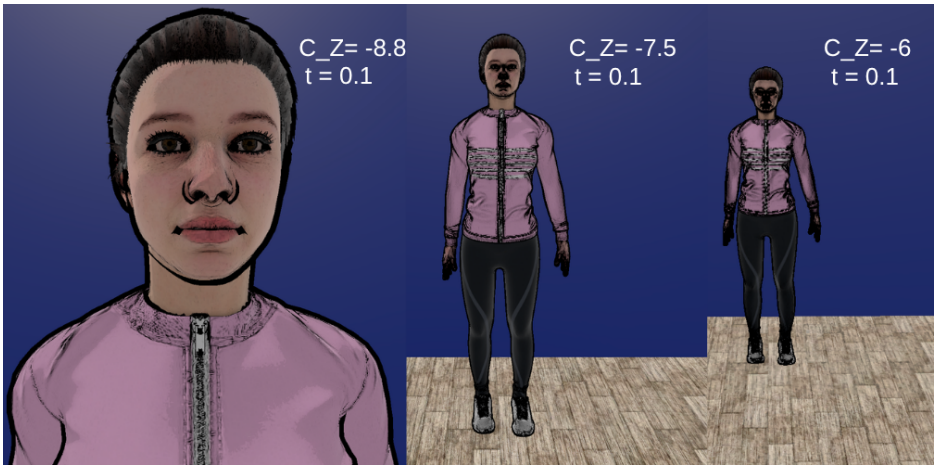
$$E_N = \sqrt{\left\|\frac{\partial \mathbf{N}}{\partial x}\right\|^2 + \left\|\frac{\partial \mathbf{N}}{\partial y}\right\|^2} \tag{4.4}$$

The gradient magnitude is high where the surface orientation changes abruptly, such as along geometric creases and clothing folds, and remains close to zero over smooth surfaces. A smoothstep function then converts this raw magnitude into a continuous edge response, and this prevents abrupt temporal changes as animated surfaces cross the detection threshold. The resulting edge mask is blended with the X-Toon shaded surface using the same compositing formulation adopted throughout the remaining edge detection techniques.

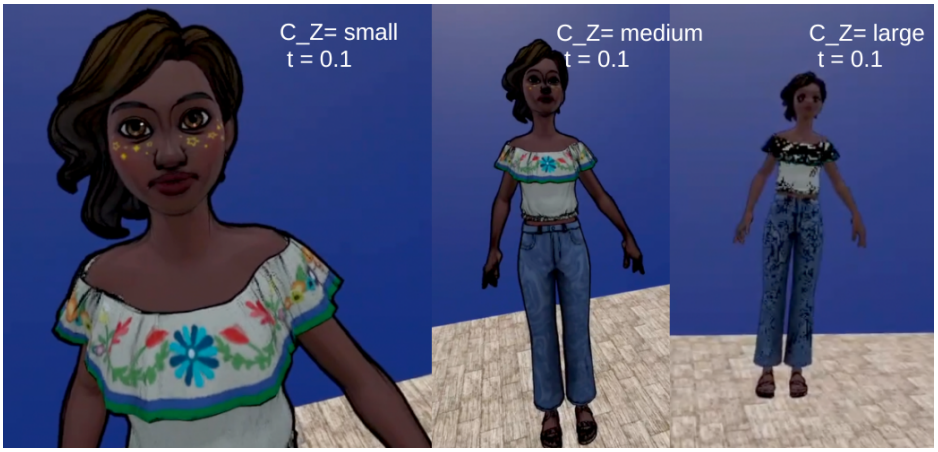
Across all three platforms, V3 complements the silhouette outline established in V1 by introducing interior contours derived from surface geometry. Figure 4.10 illustrates this across the three avatar platforms, where the edge output varies with both the normal information each platform exposes and the viewing distance. Mixamo and Avaturn both compute derivatives from shading normals. This produces a rich edge network that improves the readability of clothing folds, seams, and other regions with pronounced surface curvature. The Meta Avatars SDK, by contrast, exposes only interpolated geometric normals through the shader hook. This restricted input results in considerably sparser interior contours, most noticeably on clothing surfaces where finer surface curvature would otherwise be detected.

Despite these improvements, the technique has two important limitations. First, because the normal gradient is evaluated in screen space, its magnitude depends on image resolution rather than object scale. As the viewing distance increases, geometric features span fewer screen pixels. The interior edges consequently appear progressively thicker. Second, the method responds only to changes in surface orientation, so texture boundaries and colour transitions that do not coincide with a normal discontinuity remain undetected.

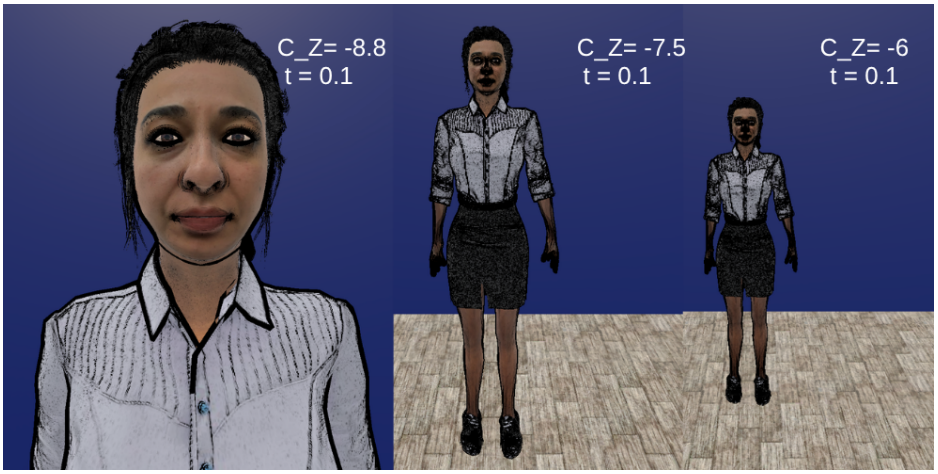
These limitations prevented V3 from being selected for the final rendering pipeline. Edge thickness scales with the viewing distance, and this produces inconsistent results whenever a viewer moves relative to the avatar. A fixed camera distance avoids this specific problem. The user study recordings use exactly this fixed distance (Section 5.6). However, the reliance on surface orientation remains regardless of distance. Subsequently, this dependence leaves many perceptually important facial and material boundaries unmarked, since they are defined primarily by texture rather than geometry. Addressing this second limitation required an edge detector sensitive to colour boundaries rather than surface orientation. This need motivates the Sobel-based approach introduced in V4.



(a) Mixamo: normal edge detection at increasing viewing distances.



(b) Meta Avatars SDK: normal edge detection at increasing viewing distances.



(c) Avaturn: normal edge detection at increasing viewing distances.

Figure 4.10.: V3 (Screen Space Normal Edge Detection) across the three avatar platforms. C_Z: camera-to-avatar viewing distance; t: detection threshold.

4.4.6. V4 — Sobel Edge Detection

The Sobel operator is a gradient-based edge detector and one of the most commonly applied methods in digital image processing [11]. V4 applies the Sobel operator directly to the diffuse albedo texture rather than the surface normal field. This targets colour boundaries in the surface appearance, including clothing patterns, facial details, and painted surface structures, while avoiding the edge thickness growth with viewing distance observed in V3.

The gradient magnitude M is computed from the two Sobel kernels introduced in Section 3.2.1 (Equation (3.3)). Pixels for which M exceeds a configurable threshold are classified as edge pixels. They are then composited over the X-Toon shaded surface to produce the final stylised appearance.

Across the three avatar platforms, however, this output quality depends strongly on the available texture source. On Mixamo, the relatively clean painted albedo enables the detection of clothing boundaries and texture transitions with limited interference from unrelated intensity variations. In contrast, Avaturn uses photographic textures that contain illumination information captured during the reconstruction process. Consequently, the Sobel operator responds not only to material boundaries but also to baked lighting variations. As shown in Figure 4.11, this effect is particularly visible around the nose bridge, where the transition between highlight and shadow produces a strong response despite not representing a meaningful texture boundary. On the Meta Avatars SDK, the Sobel operator was applied to the available composited rendering output rather than the original diffuse texture. Although this introduces responses tied to scene lighting, the rendered texture contains less high-frequency photographic variation. Edge structures are thus cleaner and more consistent than on Avaturn, particularly along clothing boundaries.

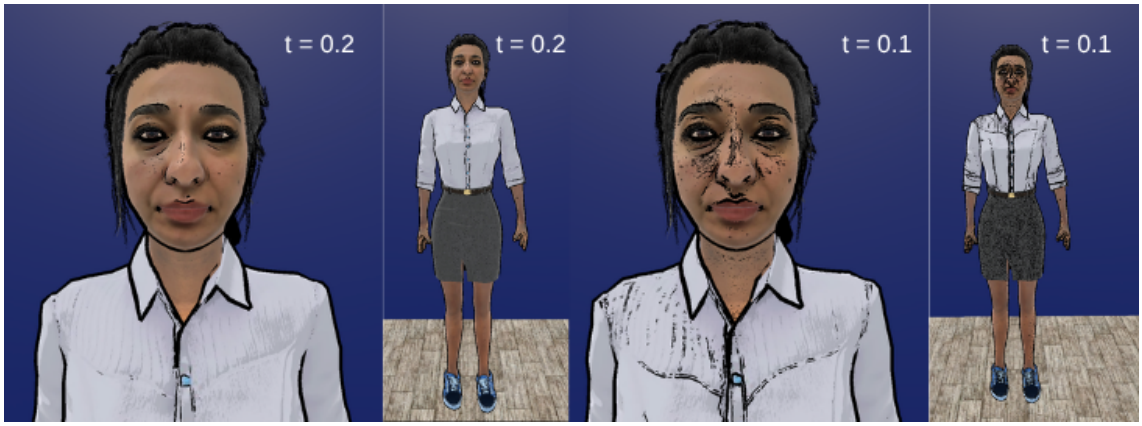
These input differences directly affect the threshold calibration required on each platform. Mixamo produced almost no unwanted edge responses at $t = 0.2$, whereas Avaturn yielded markedly more artefacts at the same threshold. This arose from its photographic texture content. Increasing the threshold on Avaturn suppressed these responses, but it also removed meaningful facial contour lines and eliminated segment boundaries present at the lower setting, as Figure 4.12 shows. The Meta Avatars SDK requires lower threshold values because its composited output produces weaker gradients at material boundaries. This same property reduces sensitivity to fine texture variations and results in comparatively cleaner edge patterns. These observations confirmed that the Sobel response depends on the available appearance representation rather than solely on underlying geometry. The threshold must therefore be calibrated to each platform's signal characteristics rather than treated as a fixed operator property.

Compared with V3, Sobel edge detection successfully captures texture-defined boundaries that remain invisible to normal-based methods, including clothing patterns, facial details, and painted surface structures. The operator responds to all local intensity variations, however. Consequently, fine texture details, photographic illumination, and compression artefacts introduce unwanted responses alongside genuine structural boundaries. A further source of false detections arises from UV seams, which are cuts where the mesh surface is split apart so it can be unwrapped into a flat 2D texture. Neighbouring pixels on opposite sides of such a cut can correspond to distant locations in texture space, and the Sobel operator therefore interprets these discontinuities as strong edges. Both categories of false detection motivate V4.2, where Gaussian smoothing is introduced before gradient computation to reduce sensitivity to high-frequency variation.

(a) Mixamo: Sobel edge detection at two threshold values.

(b) Meta Avatars SDK: Sobel edge detection at two threshold values.

Figure 4.11.: V4 (Sobel Edge Detection) across the three avatar platforms. t: Sobel threshold.



(c) Avaturn: Sobel edge detection at the baseline threshold ($t = 0.2$).

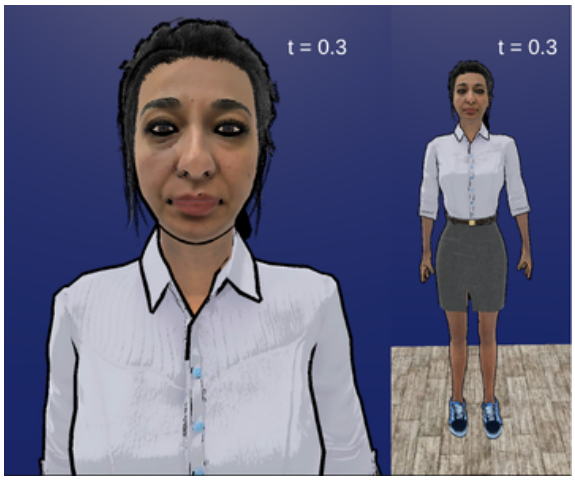


Figure 4.12.: Avaturn: Sobel edge detection at an increased threshold. Noise is reduced compared with Figure 4.11c, but meaningful facial contour lines are also eliminated.

4.4.7. V4.2 — Gaussian Prefiltered Sobel

The Sobel operator computes image gradients through partial derivatives. As this differentiation amplifies noise alongside genuine intensity changes, a preliminary noise suppression step is necessary to stabilise the edge response. Gradient operators therefore benefit from Gaussian prefiltering, since the Gaussian function is smooth in both spatial and frequency domains. This reduces variation at fine spatial scales without the ringing artefacts produced by box-averaging filters [11], [12]. Canny showed that the optimal 1D step edge detector under additive noise conditions takes the form of the first derivative of a Gaussian kernel, which is mathematically equivalent to Gaussian smoothing followed by differentiation [41]. On this basis, V4.2 applies a discrete Gaussian filter before the Sobel operator. This approximates the theoretically grounded detection strategy described above.

A discrete 3×3 Gaussian kernel is applied to the texture before the Sobel operator computes the image gradients:

$$G = \frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix} \tag{4.5}$$

Gaussian prefiltering produced cleaner edge maps than V4 across all three platforms, though the degree of improvement differed markedly by texture source. On Mixamo, Gaussian smoothing reduced edge responses caused by compression artefacts and subtle texture variation while preserving the major clothing boundaries and facial features. On the Meta Avatars SDK, prefiltering similarly attenuated high-frequency artefacts in the composited rendering output. Edge structures were consequently cleaner, particularly on clothing regions. Avaturn remained the most challenging platform because its photographic albedo contained baked illumination that introduced strong intensity gradients unrelated to surface structure. Although Gaussian smoothing attenuated some of these responses, many illumination gradients remained sufficiently strong to influence the Sobel output.

Overall, Gaussian prefiltering produced cleaner edge maps compared with V4 by reducing unwanted texture responses. However, this improvement came at the cost of weaker structural contours, particularly along features with limited contrast. Because Gaussian smoothing attenuated both noise and genuine edges, reducing false detections suppressed some meaningful structural information. Consequently, a compromise between contour preservation and noise reduction remained unavoidable.

Among the evaluated platforms, Avaturn exhibited the weakest performance under both V4 and V4.2. The baked illumination present in the photographic texture prevented reliable separation of structural boundaries from lighting variations. Gaussian prefiltering was expected to resolve this by suppressing noise on the skin texture enough to extract facial segments such as the lips and eyebrows. It reduced some of this noise, but not enough to recover these segments reliably. This shortfall motivated the texture preprocessing pipeline described in Section 4.6 instead, where the source texture was edited to reduce gradients introduced by illumination before edge extraction. Once this preprocessing was applied, the basic Sobel operator already produced reliable structural edge detection, without the added cost of Gaussian prefiltering. Section 4.5 reports this outcome in detail.

Although Gaussian prefiltering improves the robustness of texture-based edge detection, its effectiveness remains dependent on the quality of the input texture. The following techniques therefore investigate post-processing stylisation approaches that operate on the rendered image rather than directly on the texture. This reduces the influence of albedo quality on edge detection and yields greater control over the final stylised appearance.

(a) Mixamo: Gaussian prefiltered Sobel edge detection.

(b) Meta Avatars SDK: Gaussian prefiltered Sobel edge detection.

(c) Avaturn: Gaussian prefiltered Sobel edge detection.

Figure 4.13.: V4.2 (Gaussian Prefiltered Sobel) across the three avatar platforms. t: Sobel threshold.

4.4.8. V5 — Hierarchical Edge Detection with Multiple Cues

Hierarchical edge detection combines multiple complementary boundary cues into a single, unified representation rather than relying on any one signal alone [51]. V5 adopts this approach as a post-processing architecture that operates on the rendered image rather than on the source texture, and this addresses the dependence on albedo quality identified in V4.2. Depth discontinuities recover object silhouettes, surface normal discontinuities reveal geometric structure, and colour gradients computed using the Roberts Cross operator detect texture boundaries that have no corresponding geometric signal [11], [52]. Each layer carries an independent weight, and the responses are blended through adjustable parameters so that the contribution of each cue can be controlled interactively.

Although the same hierarchical formulation was evaluated across all three avatar platforms, differences in the rendering architecture affected which edge cues could be computed. On Mixamo and Avaturn, V5 operated as a post-processing effect that sampled the depth, normal, and colour buffers provided by the URP. By contrast, the Meta Avatars SDK restricts custom rendering logic to a per-fragment pass without access to global screen-space depth and normal buffers. Consequently, the depth cue could not reliably separate the

avatar from the background, so its weight was set to zero and the composite relied on the normal and colour cues instead. Figure 4.14 shows the background bleeding that motivated this choice.

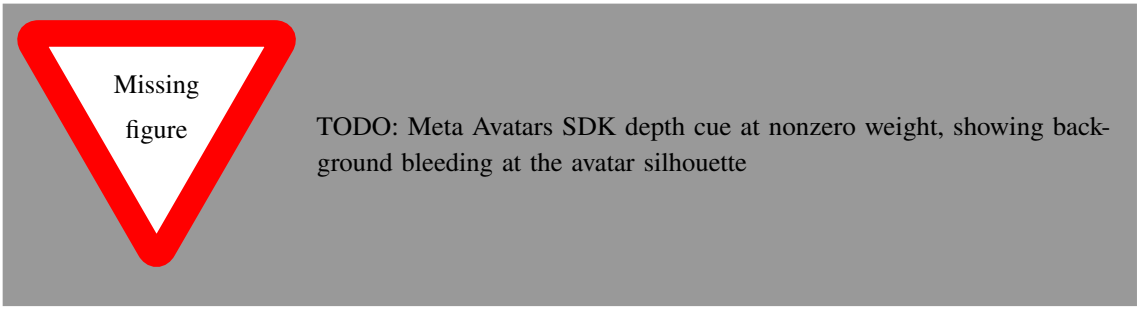


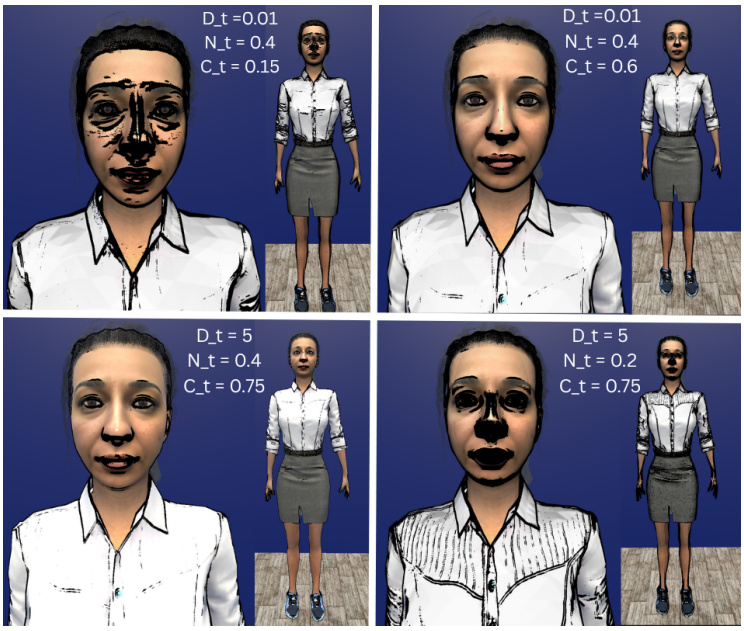
Figure 4.14.: Meta Avatars SDK: depth cue enabled at nonzero weight. Without a dedicated avatar mask, the depth proxy cannot separate the avatar from the background, motivating the zero depth weight used in the final composite.

Figure 4.15 compares the results across the three avatar platforms. V5 detects a broader range of structural boundaries by combining multiple edge detection techniques on the Mixamo and Avaturn rendered frames. The depth threshold has little visual influence, since the inverted hull outline already provides the silhouette. The normal and colour thresholds instead control the density of interior contours. These contours trace geometric curvature, appearing along the underside of the nose and the outer lip edges, but do not produce clear boundaries between adjacent facial segments. Nonetheless, the colour gradient layer occasionally responds to illumination gradients in the rendered image, particularly around the nose bridge where shading transitions are interpreted as texture edges. On the Meta Avatars SDK, the absence of screen-space depth prevents silhouette detection, although combining the remaining cues still produces richer interior detail than the previous techniques. The composited SDK output contains baked lighting gradients absent from the raw albedo, making the texture layer more prone to false responses on this platform. The colour layer therefore replaces the Roberts Cross computation with an XDoG operator on this platform. This reduced these false responses, as Figure 4.15(b) shows.

Compared with the previous techniques, V5 provides the most complete boundary representation by integrating silhouette, geometric, and texture information within a single framework. Nevertheless, because each layer contributes independently to the final edge map, the overall edge density increases and requires weight adjustment to avoid visually cluttered results. Moreover, the contribution of each layer remains spatially uniform across the rendered image, regardless of the local image content. This motivated the exploration of region-based stylisation in V6, where the surface appearance is reorganised into coherent colour regions rather than extended with additional edge marks.

(a) Mixamo: hierarchical edge detection at varying C_t (top row) and N_t (bottom row).

(b) Meta Avatars SDK: hierarchical edge detection at varying threshold combinations with XDoG pre-blur enabled and disabled.



(c) Avaturn: hierarchical edge detection at varying C_t (top row) and N_t (bottom row).

Figure 4.15.: V5 (Hierarchical Edge Detection with Multiple Cues) across the three avatar platforms. N_t: normal threshold; C_t: colour threshold. D_t is not varied as a display parameter because the inverted hull outline already provides the silhouette boundary on Mixamo and Avaturn regardless of the depth threshold value, and the Meta Avatars SDK per-fragment pass has no access to the screen-space depth buffer.

4.4.9. V6 — Kuwahara Painterly Filter

The Kuwahara filter is a nonlinear smoothing operator that replaces each pixel with the mean colour of whichever overlapping neighbourhood subregion exhibits the lowest internal variance [53]. Subregions straddling a tonal boundary exhibit higher variance and are not selected, so uniform regions collapse into broad painterly fills while boundaries remain sharp. In its original formulation, however, the neighbourhood is partitioned into four fixed square subregions, and the output is consequently insensitive to local feature orientation: brushwork is isotropic regardless of surface geometry. Kyprianidis et al. resolved this by computing a smoothed structure tensor and deriving an elliptical sampling window whose major axis aligns with the dominant direction of local image structure [13]. The resulting elliptical kernel causes marks to follow local surface curvature, rather than imposing the fixed rectangular grid of the isotropic variant.

V6 applies this filter to the rendered avatar surface to introduce a form of abstraction qualitatively different from the contour techniques of V3 through V5. Where those techniques add stylised marks over a photorealistic surface, the Kuwahara filter reorganises the surface appearance itself into coherent colour regions, and the result is analogous to a painted or illustrated representation. A post-processing architecture is adopted on Mixamo and Avaturn because the anisotropic variant requires multiple rendering passes. The Meta Avatars SDK single-pass constraint does not permit this. On the SDK platform, Gaussian smoothing of the structure tensor requires a separate render pass and cannot be performed within the hook. The tensor is instead estimated from screen-space derivatives at individual pixels, producing noisy orientation estimates.

Figure 4.16 compares the results across the three avatar platforms. On Mixamo and Avaturn, the full anisotropic implementation produced coherent painterly regions with well-oriented brushstrokes that follow the contours of clothing and facial features. On the Meta Avatars SDK, the noisy tensor caused the inverse-variance weighting to favour the most uniform colour patch in each neighbourhood rather than following feature orientation. Coherent painterly regions were consequently absent, and local colour saturation was amplified instead.

Achieving a clearly painterly effect, however, comes at a cost: at a kernel radius large enough for the painted regions to be perceptually distinct, fine facial features such as eyebrows, lip boundaries, and iris details tend to be subsumed into the surrounding skin region. Reducing the kernel radius preserves these details but diminishes the painted quality of the result.

This trade-off between abstraction strength and feature legibility suggested that the Kuwahara filter is most effective when combined with a contour detection layer that restores structural boundaries.

(a) Mixamo: standard render (left) and Kuwahara painterly filter (right).

(b) Meta Avatars SDK: standard render (left) and Kuwahara painterly filter (right).

(c) Avaturn: standard render (left) and Kuwahara painterly filter (right).

Figure 4.16.: V6 (Kuwahara Painterly Filter) across the three avatar platforms.

4.5. Platform and NPR Technique Selection for the User Study

Three properties were required for a valid perceptual trust comparison: unrestricted URP shader access to apply the NPR techniques, full blendshape animation for animated facial expression in the video stimuli, and photorealistic facial reconstruction to support the comparison with the real human speaker in the user study. On this basis, Avaturn was selected as the platform, since it is the only candidate among those evaluated in Section 4.3 that satisfies all three requirements. The Avaturn avatar used for both conditions C2 and C3 was reconstructed from photographs of the same individual who served as the real human speaker in C1, and rendering style is the primary variable that distinguishes the three conditions. Further preprocessing was also required to support reliable edge detection, and the procedure is described in Section 4.6.

The NPR composite for condition C3 was assembled from five candidate components: an inverted hull outline, Sobel edge detection, and XToon tonal shading, alongside two post-processing passes, a Kuwahara painterly filter and a hierarchical depth edge layer. Parameter settings for all five were determined through a parameter adjustment study, and full details are in Section 4.7. It also established whether the two post-processing passes should be included in the final composite.

Among the techniques evaluated in Section 4.4, three were set aside before the composite was assembled for the parameter adjustment study. Basic toon shading (V2) was replaced by X-Toon (V2.2), since the 2D ramp provides independent control over the abstraction level that procedural quantisation cannot offer. However, this control depends on how the abstraction level is driven. The depth and curvature modes described in Section 4.4.4 respond to camera distance and surface curvature respectively. Since the camera stays fixed and close to the avatar throughout the study recordings, this leaves little room for either quantity to vary across the visible surface. Manual mode avoids this problem entirely. It sets the abstraction value directly rather than computing it from camera distance or surface geometry. Manual mode was therefore the natural choice as the driving mode for the C3 composite. The parameter adjustment study then fixed this value, referred to there as the detail bias (Section 4.7). Screen-space normal edge detection (V3) was excluded primarily because it cannot recover facial features defined by colour or texture contrast rather than geometric curvature. Smooth skin produces minimal normal discontinuities, and the eyebrow boundary, lip contour, nostril edge, and eyelid margin consequently remain invisible to the technique. These are precisely the landmarks that determine facial legibility in a conversational advisor context. V3 detects clothing folds and seams reasonably well through normal map discontinuities, but the Sobel operator subsumes this capability entirely: V4 captures the same clothing contours and additionally identifies edges at boundaries defined by colour contrast, a category that V3 cannot detect. Gaussian prefiltered Sobel (V4.2) was also excluded. Its additional sampling cost no longer produced a detectable improvement over the basic Sobel operator once the head albedo was preprocessed to reduce baked illumination gradients (see Section 4.6). The hierarchical pass was the only candidate to introduce a depth cue, but its two remaining layers replicate detection strategies already set aside earlier in this section: the surface normal layer fails on smooth facial skin for the same reasons that excluded V3, and the colour gradient layer adds no information beyond what the Sobel operator already provides. Both passes were then evaluated in the parameter adjustment study. Participants ranked the unmodified configuration as significantly more trustworthy than every tested variant of each, and both were excluded. Full details of the visual assessment and parameter adjustment study results are described in Chapter 6.

4.6. Avaturn Avatar Preprocessing

To reduce the edge detection artefacts caused by the reconstructed albedo texture identified in Section 4.4.7, the head albedo texture was preprocessed in Blender before the avatar was re-exported as a GLB file. Four texture operations were applied to address two distinct problems in the head albedo: baked illumination variation that introduced spurious edge responses at boundaries unrelated to surface structure, and the absence of a strong tonal cue at the lip boundary.

To address the first problem, a Gaussian blur attenuated baked ambient occlusion, subsurface scattering, and pore detail from the photogrammetric albedo, and suppressed the high-frequency intensity variation before Sobel detection. A skin tone unification pass then blended skin pixels towards a common brightness and saturation baseline. This collapsed the baked luminance falloff into a consistent tonal surface. The brow and eye socket regions were excluded from this step through UV masks derived from geometry vertex positions, since the chromatic contrast at those features was necessary for the Sobel operator to locate eyebrow and eyelid edges accurately.

To resolve the second problem, saturation and brightness were enhanced across the mouth area, and a saturated, uniform colour zone was produced against the flattened skin. A lipline was then placed at the lip boundary. It provided a sharp tonal transition that Sobel detection and the X-Toon ramp in Unity could reliably identify. This preprocessed avatar served as the basis for all video stimuli in the user study.

Figure 4.17 shows the effect of the preprocessing pipeline and two NPR techniques applied to the modified avatar.

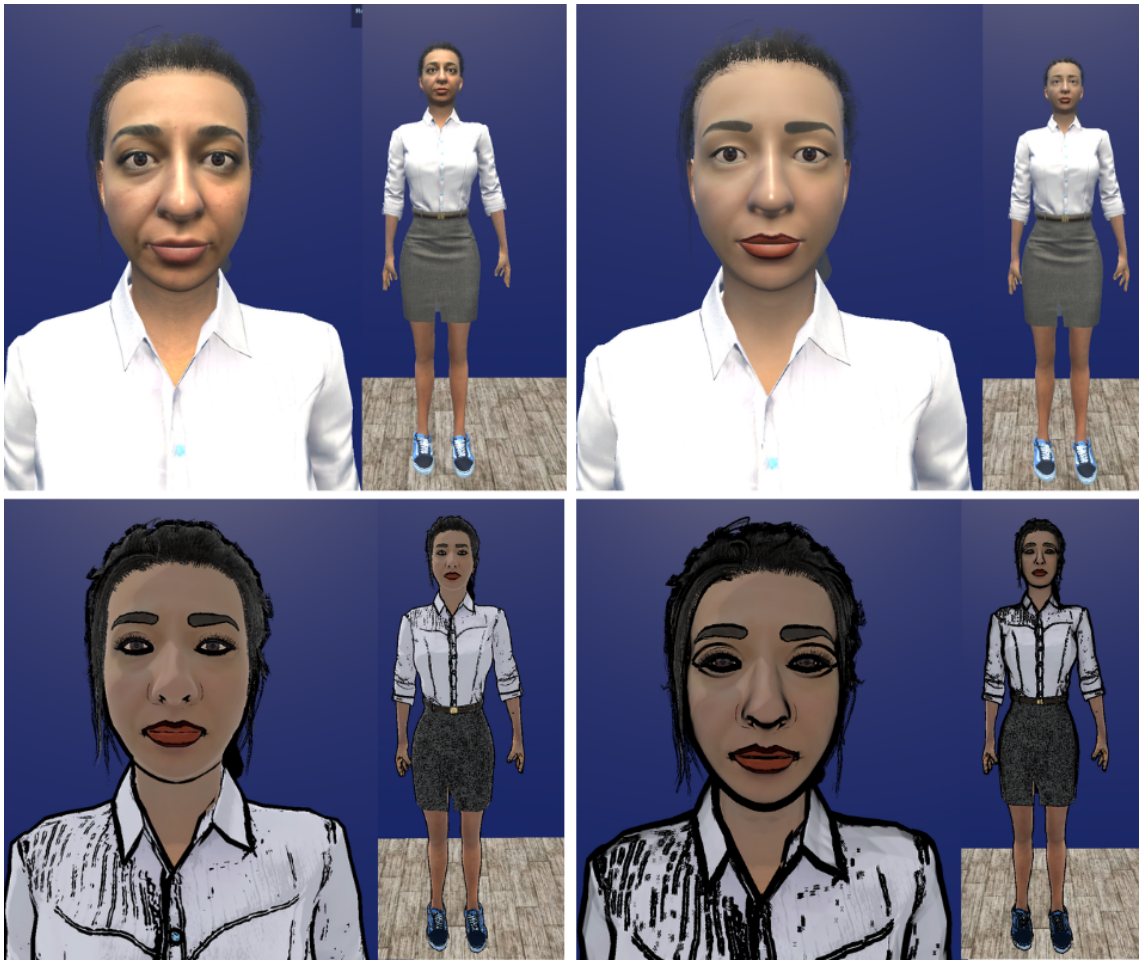


Figure 4.17.: Avaturn avatar rendering progression. From left to right: (1) default Avaturn with no NPR applied; (2) preprocessed head texture with no shader applied; (3) Gaussian prefiltered Sobel on the body combined with the customised Gaussian Sobel shader on the head and hierarchical edge detection post-processing; (4) hierarchical edge detection post-processing as the primary effect, combined with the Gaussian Sobel pass. Images (3) and (4) show intermediate development configurations and do not represent the final NPR composite used in the user study.

4.7. Parameter Adjustment Study

Within the composite described in Section 4.5, the outline establishes the silhouette, Sobel gradients identify interior contours at colour boundaries, and tonal shading controls colour quantisation. The two candidate post-processing passes each targeted a property the other three components could not provide: the Kuwahara filter would smooth texture regions into coherent painted areas, and the hierarchical layer would add depth-based boundary detection at geometry discontinuities where colour contrast is insufficient for Sobel.

A parameter adjustment study established both the parameter settings for all five components and whether the two post-processing passes should be included in the final composite. Participants ordered rendered images of the avatar by perceived trustworthiness as an advisor, and the result of each round fixed one parameter group for all subsequent rounds. The study covered the parameter groups with the greatest expected influence on perceived edge density, tonal balance, or abstraction strength; parameters with comparatively minor expected visual impact, including rim light presence, Kuwahara kernel size, Gaussian blur strength, sector count, and normal edge strength, were instead set through direct visual inspection during development, to keep the study at a manageable length for participants.

One parameter group was varied while all remaining settings were held at the level fixed by the preceding test in each round. Table 4.2 lists all stages and the values examined at each.

Table 4.2.: Parameter groups and values tested in each round of the parameter adjustment study. R4A and R5A carry forward the R1E configuration as the no-treatment baseline for their respective rounds.

Round	Parameter group	Values tested (A, B, C, . . . in order)
R1	Inner line threshold	0.01, 0.02, 0.03, 0.04, 0.05
R2	Outline width and depth offset	width: 0.003, 0.005, 0.007; offset: on / off
R3	Shadow strength and detail bias	strength: 0.0, 0.5, 1.0; bias: 0.0, 0.5, 1.0
R4	Kuwahara hardness and sharpness	none (= R1E); (18, 1); (9, 9); (1, 18)
R5	Hierarchical depth threshold	none (= R1E), 0.01, 0.02, 0.03, 0.04, 0.05

45 pilot participants took part in the parameter adjustment study. Participants were required to assign a unique rank to every image in the round. Any ballot with a missing rank or a tied rank was excluded, and the number of valid responses consequently varies slightly by round, from 40 to 44 of the 45 submitted. Images were converted to Borda-count points in each round, where the image ranked last received 1 point and the image ranked first received points equal to the number of options, and the totals were summed across participants. An omnibus Friedman test established whether the ranking within a round differed from chance, and a pairwise Wilcoxon signed-rank test compared the configuration ranked first against its closest competitor [54]. The same nonparametric approach is applied to the advisor ranking task in the main study (Section 4.8.4). Table 4.3 reports the outcome of each round.

Table 4.3.: Statistical test results for the parameter adjustment study by round.

Round	Best configuration	Friedman	Runner-up test
R1	threshold 0.03	$\chi^2_F(4) = 36.48$, $p < .001$	$p = .31$ vs. 0.05 (not distinguished)
R2	width 0.003, offset on	$\chi^2_F(5) = 126.19$, $p < .001$	$p = .82$ vs. width 0.005, offset on (not distinguished)
R3	shadow 0.5, bias 1.0	$\chi^2_F(4) = 3.18$, $p = .53$	no configuration differed significantly
R4	none (no Kuwahara)	$\chi^2_F(3) = 38.97$, $p < .001$	$p = .003$ vs. best Kuwahara variant (significant)
R5	none (no hierarchical layer)	$\chi^2_F(5) = 117.20$, $p < .001$	$p < .001$ vs. best threshold (significant)

Two outcomes determine the final composite. In R1, threshold 0.05 received the most first-place votes (20 of 43 participants), yet threshold 0.03 accumulated the higher cumulative Borda score (160 vs. 149

points), since it was ranked more consistently near the top across all participants. The pairwise Wilcoxon signed-rank test found no significant difference between the two ($p = .31$), and the omnibus Friedman result confirmed a clear directional trend: the two lowest thresholds were significantly disfavoured, and the top two configurations were accordingly treated as equivalent. In R2, the narrower outline width (0.003, depth offset on) accumulated the highest Borda total but was not statistically distinguished from the standard width (0.005, depth offset on, $p = .82$), and was therefore retained since no significant preference justified using the wider option. In R3, no significant differences emerged across any configuration ($p = .53$), and the Borda winner was shadow strength 0.5, detail bias 1.0, with no configuration statistically distinguishable from another. In both R4 and R5, however, the baseline without any post-processing was significantly preferred over every tested variant of the Kuwahara filter and the hierarchical depth layer respectively. Because trustworthiness is the primary outcome measure of the main study, both passes were excluded from the C3 composite on this basis, consistent with Section 4.5. The per-round ranking distributions for all configurations are provided in Chapter B.

For R1 and R3, where the pairwise tests found no statistically distinguishable winner, the overall impression question served as a secondary tiebreaker. R1E (threshold 0.05) received the most nominations overall with seven, and R3B (shadow strength 0.0, detail bias 0.5) followed with six. Both were consequently selected for their respective rounds. Table 4.4 lists the complete parameter set established by the parameter adjustment study.

Table 4.4.: Final C3 composite parameter set established by the parameter adjustment study.

Parameter	Value	Source
Outline width	0.003	R2 Borda winner
Depth offset	On	R2 Borda winner
Inner line threshold	0.05	R1 overall impression
Shadow strength	0.0	R3 overall impression
Detail bias	0.5	R3 overall impression
Kuwahara filter	Off	R4 excluded (significant)
Hierarchical depth layer	Off	R5 excluded (significant)

Figure 4.18 shows every rendered variant presented to participants across all five rounds, including the Kuwahara and hierarchical variants that were ultimately not retained.

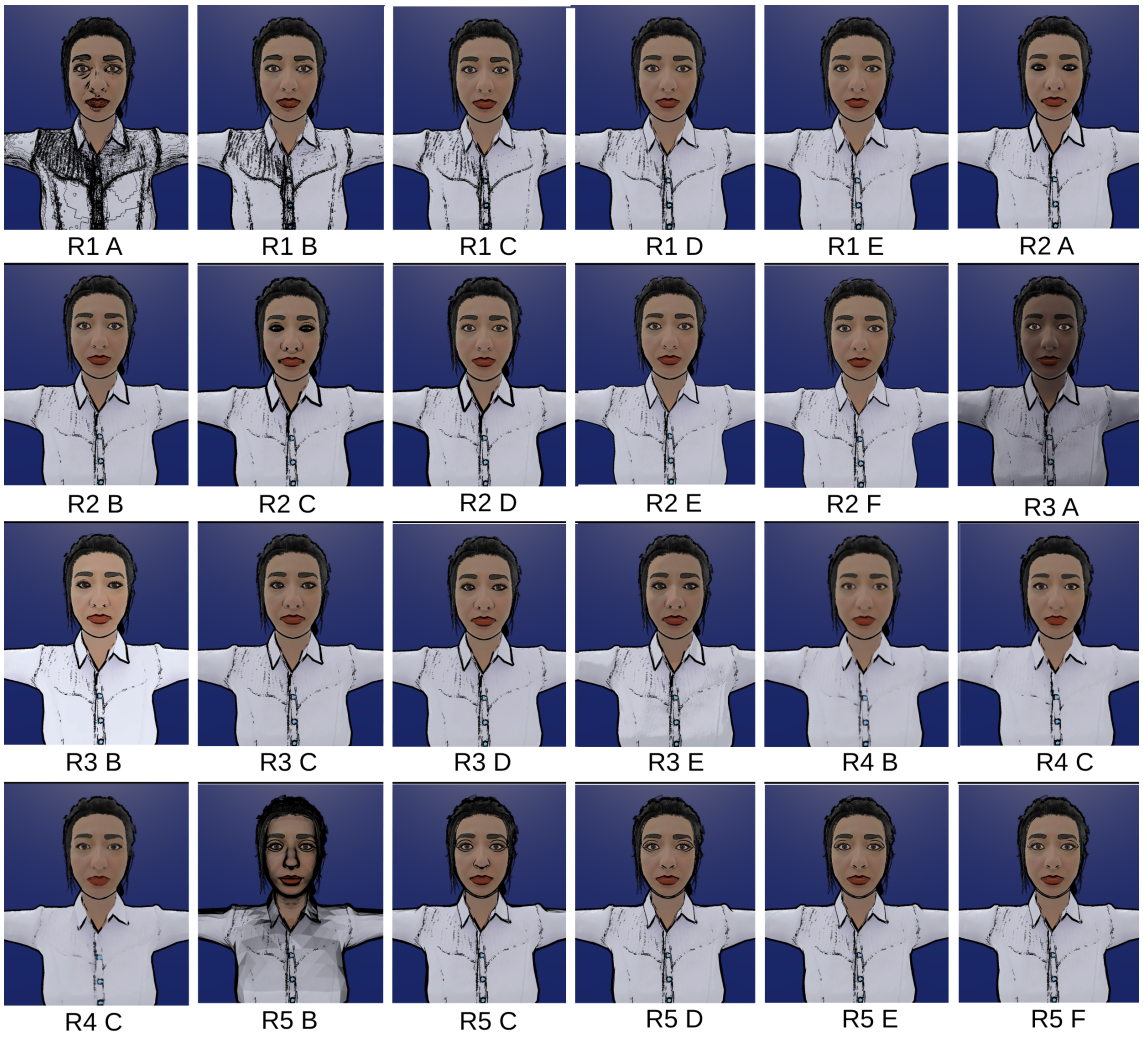


Figure 4.18.: All rendered variants presented during the parameter adjustment study. Labels follow the convention R*n*X, where *n* is the round number and X is the variant letter in the order listed in Table 4.2. R4A and R5A are identical to R1E and are not shown as separate images.

4.8. User Study Design

4.8.1. Objective and Hypotheses

The user study addresses RQ2 and RQ3 by measuring how the visual representation of a speaker affects perceived eeriness and trustworthiness in a concrete decision situation. Three hypotheses are tested.

H1: The NPR avatar (C3) will produce lower eeriness ratings than the photorealistic avatar (C2).

Photorealistic avatars occupy the uncanny valley region where negative affect peaks, while stylised abstraction moves the representation towards the stable ascending slope at lower fidelity [6]. H1 tests whether this shift produces a measurable reduction in eeriness between C2 and C3.

H2: The NPR avatar (C3) will not produce significantly lower perceived trust than the photorealistic avatar (C2).

Prior work found that stylisation level does not reduce perceived trust when the advisor role and content are held constant [7], [24]. H2 tests whether this pattern holds when rendering style is the primary variable that changes between conditions.

H3: Eeriness will negatively predict trust ratings across all conditions.

Eeriness functions as a specific negative affective state that consistently suppresses positive social evaluations, including trust [22], [29]. H3 tests whether this relationship is detectable across all three conditions within each participant.

4.8.2. Conditions

The study compares three presentation conditions:

C1 — Real Person: A video recording of the same human speaker whose photograph was used to generate the Avaturn avatar, so that face identity, voice, and demographic appearance are held constant across all three conditions. C1 serves as the trust ceiling baseline against which both avatar conditions are compared.

C2 — Photorealistic Avatar: The same speaker scenario rendered using a photorealistic Avaturn avatar with standard PBR shading. Facial expressions are driven by blendshape animation captured in the same recording session as C1, so that expression timing and facial gestures are identical across all three conditions.

C3 — NPR Avatar: The Avaturn avatar rendered with the NPR technique selected through the parameter adjustment study on the basis of its balance between abstraction and expressiveness.

Using the same Avaturn geometry for both C2 and C3 holds avatar identity, facial structure, and animation constant across the two avatar conditions, and render style is consequently the primary independent variable between them. Canales et al. applied the same principle in their stylisation study, holding speaker identity and scripted content constant across conditions while varying only the avatar rendering style [7]. The theoretically critical comparison is C2 versus C3. Since animation is held constant between these two conditions, any observed difference in trust or eeriness ratings is attributable primarily to the change in rendering style, with one material-level exception addressed in Section 6.1.4.

4.8.3. Scenarios

Three video scenarios present naturalistic situations in which each speaker delivers a recommendation that participants must decide whether to follow. Each scenario has two plausible options and the speaker recommends one with explicit reasoning. Several design principles govern the scenario construction.

Ecological validity. The three scenarios are drawn from everyday situations that participants are likely to encounter in their social and professional lives: a career decision between two job offers, route planning in an unfamiliar city, and selection of a learning resource for career development. Grounding the scenarios in contexts drawn from daily life ensures that trust judgments reflect responses to the advisor rather than the effort of interpreting an unfamiliar situation.

Vignette anchoring. Before each video, participants read a short context paragraph that anchors them in a specific, concrete situation with defined personal circumstances. Without this anchoring, trust ratings would reflect individual differences in life experience rather than responses to the speaker. The vignette eliminates this source of variance across participants, and every participant enters each video from the same defined role.

Recommendation alternation. The three scenarios are designed so that the recommended option alternates across videos: Scenario 1 recommends Option A, Scenario 2 recommends Option B, and Scenario 3 recommends Option A. This prevents participants from detecting a systematic preference for either option label across the study.

Presentation order alternation. Within each script, the recommended option is not always discussed last. Discussing the recommended option last in every video would introduce a recency effect, where participants

associate recommendation with position rather than with the speaker's reasoning. Scenario 1 and Scenario 3 discuss the recommended option first. Scenario 2 discusses the recommended option last. All scripts conclude with an explicit recommendation statement regardless of discussion order.

Script argumentative structure. Each spoken recommendation follows a fixed five-step structure, applied uniformly across all three scenarios: (1) state the recommendation explicitly; (2) acknowledge the primary drawback of the recommended option; (3) introduce the key advantage of the recommended option with an explicit contrastive marker; (4) name the central weakness of the alternative option; and (5) close with a concise restatement. This structure ensures the advisor engages visibly with trade-offs before concluding, which holds rhetorical strategy constant across scenarios. Any difference in trust ratings between conditions therefore reflects the visual presentation of the speaker rather than variation in argumentation pattern between scenarios.

The three scenarios and their vignette texts are as follows.

Scenario 1 — Career Choice. The participant is placed in the role of a recent graduate choosing between two job offers, with financial pressure and location held constant as confounds. The speaker recommends Job A, judging career flexibility to outweigh the higher salary offered by Job B. This scenario primarily activates competence trust.

Scenario 2 — Travel Route. The participant must reach a hotel for an urgent meeting, choosing between a direct 50 minute route that arrives ten minutes before the meeting and a faster 20 minute route requiring one transfer that arrives forty minutes early. The speaker recommends the faster option with the transfer, arguing that its 40 minute buffer provides enough recovery time to outweigh the transfer risk, whereas the direct route leaves an insufficient margin. This scenario activates competence trust through reasoning about risk and benefit, and benevolence trust through prioritising the participant's ability to arrive on time.

Scenario 3 — Learning Resource. The participant chooses between a paid online course with a named certificate and a free tutorial series covering the same material. The speaker recommends the paid course on the grounds that a recruiter cannot verify a free tutorial series, making the recommendation rest on credential value rather than content superiority. This scenario primarily activates integrity trust.

Scenario order is held constant across all participants: Career in Video 1, Travel in Video 2, and Learning in Video 3. The three conditions are counterbalanced across those fixed positions using a 3×3 Latin square (Table 4.5), with one study version per row. Each appears exactly once per presentation slot, and this controls for any systematic advantage conferred by video order. Sequential carryover between specific condition pairs is not addressed by this design and is noted as a limitation in Chapter 7. In the questionnaire, the three video slots are named Video 1, Video 2, and Video 3 so that participants cannot identify the rendering style. Each version's condition mapping is stored separately and applied during data analysis. The complete questionnaire, including all vignette texts as presented to participants, is reproduced in Chapter A.

Table 4.5.: Latin square counterbalancing across three study versions. Each condition appears exactly once per version, with scenario positions held fixed.

Version	Video 1 (Career)	Video 2 (Travel)	Video 3 (Learning)
A	C2 — Photorealistic Avatar	C1 — Real Person	C3 — NPR Avatar
B	C3 — NPR Avatar	C2 — Photorealistic Avatar	C1 — Real Person
C	C1 — Real Person	C3 — NPR Avatar	C2 — Photorealistic Avatar

4.8.4. Procedure and Measures

Procedure

The study is conducted entirely online via Google Forms. Each participant completes a consent and demographic questionnaire, then watches three video blocks in the order determined by their assigned study version. Each block follows four sequential steps:

1. Read the vignette establishing the scenario.
2. Record an initial option preference before watching (this pre-video baseline is used to compute preference shift as a secondary exploratory measure).
3. Watch the 23 second recommendation video.
4. Complete the per-block questionnaire.

After all three blocks, participants complete a post-study section. The entire session takes approximately 20 to 25 minutes. The questionnaire underwent iterative shortening across three successive rounds of piloting, in response to concerns about participant burden and completion time. The form described in this section is the final one used for data collection.

Per-Block Questionnaire

The per-block questionnaire is administered after each video. Parts B, C, and D use a five point semantic differential scale. Parts E through H use a seven point Likert scale.

Part A — Comprehension check. Two gating items verify that participants watched the video and identified the correct recommendation. Participants who fail either item are excluded from analysis for that block.

Parts B, C, D — Semantic differential (5 point). Part B (likeability) and Part C (anthropomorphism) are administered before Part D (eeriness) so that participants form a general social impression before attending to uncanny affect. A reversed order would prime observers towards negative perceptions and risk inflating eeriness scores before the trust battery. Part B uses the five-item Godspeed likeability subscale. Part C uses the five-item Godspeed anthropomorphism subscale, also from Bartneck et al., measuring how human-like participants perceive the advisor [28]. Part D captures eeriness through eight items from the validated instrument of Ho and MacDorman. Their validation showed that the Godspeed subscales are too strongly intercorrelated to measure distinct constructs, and that no Godspeed subscale independently captures this specific affective state [29]. Each polar pair is preceded by a short definition to support participants unfamiliar with the terms. This instrument treats eeriness as a distinct affective response from likeability, and H3 can consequently be examined independently.

Parts E, F, G — Trust battery (7 point Likert). Nine items operationalise the Ability-Benevolence-Integrity model of Mayer et al. within the trusting beliefs framework of McKnight and Chervany [20], [21]. Parts E, F, and G cover Competence, Benevolence, and Integrity respectively, with three items each. These three components are measured independently rather than as a composite, and this allows differential effects of rendering style on each trust dimension to be detected [21]. The fourth construct in the McKnight and Chervany typology, predictability, is excluded given that it tends to merge with competence and integrity when participants encounter the advisor for the first time [21].

Part H — Response to recommendation. Part H records agreement with the recommendation and the participant's option choice after watching. This post-video choice is compared with the pre-video baseline to compute a preference shift score.

An embedded attention check item is included in each of the three blocks, with a different target response per block to prevent pattern detection.

Post-Study Section

After all three video blocks, participants complete a post-study section. Three separate ranking questions each ask participants to order all three advisors: from most to least wanted for consultation (P1), from most to least trustworthy overall, and from most to least unsettling overall. P1 is the primary behavioural measure, adapted from the post-study advisor selection paradigm of Canales et al. [7]. It captures implicit trust judgment that explicit Likert ratings may not fully reflect, and this addresses the methodological gap identified by Alipour et al. [22]. The trustworthiness and unsettlingness rankings provide behavioural analogues to the corresponding Likert measures, Parts E through G and Part D respectively, and implicit and explicit judgments can consequently be compared directly for both dimensions. A rank order is used for all three questions rather than a forced binary choice, since both avatar conditions are expected to score below C1. Limiting participants to two options would concentrate responses on C1 and leave insufficient variance to distinguish C2 from C3.

A visual style manipulation check (Part I) asks participants to rate the perceived visual stylisation of each advisor simultaneously. Its placement after all video blocks prevents awareness of rendering differences from priming trust ratings during data collection. The post-study section closes with two open-ended questions, each administered once rather than per block to limit response fatigue: the reflection question (Part J) asks participants to describe the most important factor in their trust judgment, and a final open question asks what they believe the study was investigating, and this serves as a check for demand characteristics.

The demographics section, completed before the first video block, also includes an item measuring dispositional trust (seven point Likert: *In general, I tend to trust people I meet for the first time*) as an individual-level covariate. Mayer et al. identify this disposition as a stable individual difference that must be controlled when comparing trust ratings across conditions [20].

Analysis Plan

The primary analysis is a one-way repeated-measures MANOVA with condition (C1, C2, C3) as the within-subjects factor and all six subscale composite scores (B_score, C_score, D_score, E_score, F_score, G_score) as dependent variables. If the omnibus MANOVA is significant (Pillai's trace, $p < .05$), univariate follow-up ANOVAs are conducted per subscale, and pairwise comparisons with Bonferroni correction ($\alpha_{\text{corrected}} = .05/3 = .017$) are applied thereafter. Effect sizes are reported as partial η^2. The theoretically critical pairwise comparison is C2 versus C3. H2 predicts equivalence rather than a difference, and this leads to a non-significant pairwise result providing insufficient evidence on its own. A TOST (two one-sided tests) equivalence test with a pre-specified bound of Cohen's $d = 0.3$ is therefore applied to the C2 versus C3 trust composite to yield positive evidence of negligible difference. For H3, person-centred multilevel regression is used, with eeriness as the within-participant predictor of trust. This approach isolates the relationship between eeriness and trust from variance across conditions in both measures. Three post-study ranking questions, covering P1, trustworthiness, and unsettlingness, are each analysed with a Friedman test comparing mean advisor ranks across conditions. Pairwise Wilcoxon signed-rank tests with Bonferroni correction ($\alpha_{\text{corrected}} = .017$) are then applied to each pair, and C2 versus C3 is the critical pairwise comparison in each case [54].

Sample Size

A power analysis was conducted for a three-condition within-subjects ANOVA. It targeted a medium effect size of partial $\eta_p^2 = 0.06$, $\alpha = .05$, and power = 0.90. The minimum required sample is $n = 38$ participants. The recruitment target is set to $n = 48$, with sixteen participants per study version across three versions. This number accommodates anticipated attrition and exclusions from the comprehension check, and it preserves balance across all three counterbalancing patterns.

5. Implementation

This chapter documents the technical realisation of the NPR avatar system. It covers the development environment, the integration path for each avatar platform, the shared rendering structure, and the HLSL implementation details of each stylisation technique developed to date. The runtime tooling built to support interactive parameter exploration and evaluation capture is described at the end of the chapter. The animation and recording pipeline used to produce the video stimuli for the user study is presented afterward.

5.1. Development Environment

The system was developed in Unity 6 (version 6000.0.60f1) using the Universal Render Pipeline (URP). The target hardware is the Meta Quest 3 standalone VR headset. The Meta Avatars SDK version 40.0.1 was used for the avatar platform managed by the SDK. All shader code is written in HLSL and targets the URP forward rendering path.

5.2. Avatar Platform Integration

5.2.1. Mixamo

Mixamo characters were exported as FBX assets and imported into Unity as standard skinned mesh renderers. As introduced in Section 4.3, this SDK-free structure allows every URP material to be replaced directly with a custom NPR shader.

5.2.2. Meta Avatars SDK

The Meta Avatars SDK generates its shader source from a set of packaged templates managed through the `OvrShaderManager` component. This generated source is not intended to be edited directly. Custom logic is instead injected through the hook function that the SDK exposes at a defined point in the shader pipeline.

The hook used in this thesis was `AppSpecificPostManipulation`, which executes per fragment after all SDK PBR lighting (subsurface scattering, eye glints, hair shading) has been computed. It receives the accumulated lit colour and returns a modified colour. This grants full control over the final fragment output without disrupting any SDK rendering logic.

5.2.3. Avaturn

Avaturn follows a different integration path from the Meta SDK's hook-based model. The additional import step required for its `.glb` export, introduced in Section 4.3, is handled by the GLTFast package, which produces a standard Unity GameObject hierarchy with skinned mesh renderers and PBR materials assigned from the glTF material definitions, including albedo, normal, and metallic-roughness textures. Custom NPR materials are therefore assigned to the resulting skinned meshes using the same approach as Mixamo: direct material replacement after load, and blendshape animation targets exported by Avaturn are preserved through this process.

5.2.4. Common Scene Environment for Avatar Platform Comparison

Jody, Avaturn, and the Meta Avatars SDK avatar each occupy a separate scene, since every platform requires its own configuration. The same room asset is reused across all three: an enclosure with a tiled floor, brick walls, and one dark blue accent wall, plus a controlled skybox and no emissive surfaces. Camera position and framing are matched as closely as possible so that comparison renders share an equivalent viewpoint. Jody and Avaturn additionally share a directional light. It is elevated $-30°$ and rotated $-30°$ horizontally relative to the avatar forward vector. This places it above and towards the front left of the avatar. Its intensity is fixed at 1.0 lux with a neutral white colour. Moreover, the Meta Avatars SDK scene carries additional lighting beyond this configuration in order to improve visibility of the applied technique during the comparison. Ambient intensity and the GI contribution, baked from the shared skybox, nonetheless align across all three platforms.

5.3. Shared Rendering Structure

Each NPR technique on the Mixamo and Avaturn platforms is a standalone multi-pass shader, and technique switching is performed by replacing the active material set. By contrast, all techniques on the Meta SDK platform share a single shader that dispatches to the corresponding technique based on an active keyword. Runtime switching then requires only toggling this keyword without recompilation.

On Mixamo and Avaturn, each avatar mesh is further split across several submesh materials, four on Mixamo and six on Avaturn, so that technique parameters can be tuned separately for regions such as skin, hair, and eyelashes. Their vertex shaders also carry a vertex fetch macro shared with the Meta Avatars SDK codebase. This macro has no effect on Mixamo and Avaturn, since standard Unity vertex attributes are already available directly, and it is kept only for consistency across all three platforms' shaders.

The automatic LOD management system on the Meta SDK platform replaces avatar materials as camera distance changes, and this reverts any custom shader assignment made at load time. A dedicated runtime script resolves this: it performs an initial material replacement across the different detail levels once the avatar finishes loading, then polls the active materials on each renderer at a configurable interval. If the script detects a material assigned by the SDK, a LOD transition has reverted the custom assignment. It then reapplies the NPR material and copies the base colour and normal texture references from the newly active SDK material into its properties.

5.4. NPR Technique Implementation

All shader parameters described in the following subsections, as well as the choice of which NPR technique is active, can be adjusted either in the Unity Inspector during offline development or through a custom UI panel inside the running application. Changes made either way take effect immediately without shader recompilation.

5.4.1. V1 — Inverted Hull Outline Pass

The outline pass implements the inverted hull silhouette by displacing each vertex outward along its world-space normal by a configurable width property, identical in structure across all three platforms:

```
float3 worldPos = P + N * width; // displace along the surface normal
```

Listing 5.1: Outline vertex displacement (unified across platforms).

In the Mixamo and Avaturn shaders the pass culls front faces and writes depth using a standard depth test. The Meta SDK shader applies the same front-face culling. However, it disables depth writing and uses a looser depth test instead to avoid conflicting with the SDK's depth compositing. The fragment outputs a flat outline colour. A configurable depth bias shifts the clip-space Z coordinate towards the camera to prevent z-fighting where the expanded shell meets the original surface.

Alpha Cutout Handling

Alpha cutout is controlled by a toggle flag. It enables both the outline pass and the forward-lit pass to run the following before any displacement or shading is applied:

```
half alpha = SAMPLE_TEXTURE2D(_MainTex, sampler_MainTex, uv).a; // base alpha
clip(alpha - _AlphaCutoff); // discard below threshold
```

Listing 5.2: Alpha cutout (outline and forward-lit passes).

This implements the cutoff test motivated in Chapter 4. The value `_AlphaCutoff` = 0.1 was selected visually as the lowest setting that reliably removes fully transparent regions without clipping the visible hair silhouette, shown in Figure 4.5 in Chapter 4. The Meta Avatars SDK does not require this flag. Its fragment pipeline resolves hair and eyelash transparency before the hook is invoked, and any fragment the SDK would discard is therefore already clipped by the time the outline colour is applied.

5.4.2. V2 — Toon Shading

This technique is realised as two shader implementations across the three avatar platforms. Mixamo and Avaturn share the same toon quantisation code, while the Meta Avatars SDK requires a separate implementation. The outline pass carries over unchanged from Section 5.4.1 across all three platforms. The Meta variant omits the explicit alpha test, since this is already handled by the SDK (Section 5.4.1).

Mixamo and Avaturn

The toon pass quantises the Lambertian dot product $\mathbf{N} \cdot \mathbf{L}$ per step using a per-band smoothstep, shared without modification between the two platforms:

```
float steps  = max(1.0, _ToonSteps); // number of shading bands
float scaled = NdotL * steps; // map into band index
float band   = floor(scaled); // current discrete band
float frac   = scaled - band; // offset within the band
float blend  = smoothstep(1.0 - _ToonSmoothness, 1.0, frac); // edge blend
float toon   = saturate((band + blend) / steps); // combined 0-1 value
toon = lerp(1.0 - _ShadowStrength, 1.0, toon); // remap to shadow range
```

Listing 5.3: V2 toon quantisation (Mixamo and Avaturn, unified).

The default value of `_ToonSteps` is four. This decision has been made because visual assessment confirmed that it provides an effective balance between shape legibility and graphic simplification, shown in Figure 4.7 in Chapter 4. Moreover, each band boundary receives an independent narrow blend of width `_ToonSmoothness`, so normal jitter near any seam produces a localised crossfade rather than snapping the whole surface patch between bands. An additive ambient term and a rim highlight are then added to the quantised diffuse.

Meta Avatars SDK

This implementation runs inside the hook and receives the fully composited PBR colour produced by the SDK's renderer. The quantisation therefore operates on RGB directly and preserves hue while stepping luminance:

```
float  bands      = max(2.0, _ToonColorBands); // number of colour bands
float3 posterized = floor(color.rgb * bands + 0.5) / bands; // round to band
color.rgb = lerp(color.rgb, posterized, _ToonPosterizeStrength); // blend

float lum  = dot(color.rgb, float3(0.2126, 0.7152, 0.0722)); // luminance
color.rgb  = lerp(float3(lum, lum, lum), color.rgb, _ToonSaturation); // desat
```

Listing 5.4: V2 posterisation (Meta SDK).

A plain floor operation always rounds down, and this introduces a darkening bias. Rounding to the nearest band avoids this bias and distributes the quantisation error symmetrically. Saturation adjustment runs after posterisation to prevent clamping artefacts. Since the input already contains Meta's subsurface scattering, anisotropic hair, and eye glints, posterisation stylises an already physically correct image rather than reconstructing lighting from scratch.

Table 5.1.: V2 implementation comparison across the three avatar platforms.

	Mixamo / Avaturn	**Meta SDK**
Quantisation method	Per-step smoothstep, NdotL	Round-to-nearest, composited RGB
Rim light	Additive, computed	Pre-composited by Meta PBR
Outline depth write	`ZWrite On`	`ZWrite Off`
Shadow source	None (`GetMainLight` without shadow coords)	Pre-composited by Meta PBR

Visual results for V2 across all three platforms are documented in Figure 4.7 in Chapter 4.

5.4.3. V2.2 — X-Toon Shading

The same split into two implementations as V2 (Section 5.4.2) carries over: Mixamo and Avaturn share the forward-lit pass, while the Meta Avatars SDK uses a separate implementation. The outline pass remains unchanged from V2 across all three platforms. The forward-lit pass replaces the procedural quantisation of V2 with a 2D ramp texture lookup, shown in Figure 4.8. Moreover, this adds an abstraction compression stage that is driven by a choice of three modes: depth, curvature, or manual.

Mixamo and Avaturn

The forward-lit pass begins by optionally modifying the surface normal via Normal Field Abstraction, which is described in Section 4.4.4. The abstract normal map is a blurred version of the surface normal map, and it is sampled when this option is active. It is then transformed from tangent space to world space through the TBN matrix. The result is linearly interpolated with the geometry normal:

```
// sample and unpack the abstract normal map
float3 abstractN  = UnpackNormal(SAMPLE_TEXTURE2D(
                        _AbstractNormalMap, sampler_AbstractNormalMap, uv));
float3x3 TBN      = float3x3(tangentWS, bitangentWS, normalWS); // TBN basis
float3 abstractWS = normalize(mul(abstractN, TBN)); // to world space
normalWS          = normalize(lerp(normalWS, abstractWS, _NormalSmoothing)); //
    blend towards simplified normal
```

Listing 5.5: Normal Field Abstraction (Mixamo and Avaturn).

The U coordinate for the ramp starts from the same Lambertian dot product used in V2 toon shading (Section 5.4.2). Shadow attenuation is now applied to this term as well, unlike in V2. The result is then remapped and interpolated towards the ramp midpoint to control light sensitivity.

The V coordinate (abstraction level) is computed from the fragment's world-space camera distance in depth mode, from the screen-space normal derivative magnitude in curvature mode, or from a constant in manual mode, selected at compile time via `shader_feature_local` keywords. The ramp is then sampled at `(rampU, rampV)`. An abstraction compression stage then flattens the lighting bands as the abstraction level rises:

```
float abstractU = lerp(rampU, 0.5, rampV * 0.6); // compress towards midpoint
float dynSmoothing = lerp(_RampSmoothing, _RampSmoothing + 0.35, rampV); //
    widen blend as abstraction rises
float shadowMask = smoothstep(0.5 - dynSmoothing, 0.5 + dynSmoothing, abstractU
    ); // soft lit/shadow threshold
float3 toonAlbedo = albedo * rampColor; // tint by ramp
float3 shadowedAlbedo = lerp(toonAlbedo * _ShadowColor.rgb, toonAlbedo,
    shadowMask); // blend towards shadow colour
float3 finalColor = lerp(albedo, shadowedAlbedo, _ShadowStrength); // apply
    overall shadow strength
```

Listing 5.6: Abstraction compression and shadow mask (Mixamo and Avaturn).

As *V* increases, the U compression pulls rampU towards the midpoint while the smoothstep width widens, and this softens band boundaries rather than sharpening them. This same mask then blends the ramp-tinted albedo towards a configurable shadow colour. An overall shadow strength parameter controls how much of the effect applies over the plain albedo. A Blinn-Phong specular highlight is added after the shadow stage. A configurable smoothness parameter controls the transition width, and values near zero produce a sharp specular highlight.

Meta Avatars SDK

The Meta implementation runs inside the hook. Unlike the Mixamo and Avaturn shaders, this hook does not own the vertex stage. World-space position is instead forwarded from the SDK vertex output as an explicit fifth argument to the effect function. This makes the same U axis computation as the standalone platforms possible. Thus, `GetMainLight` is called with a per-fragment shadow coordinate derived from that world-space position. The shadow-attenuated NdotL is then computed using the same approach as on the standalone platforms:

```
float4 shadowCoord = TransformWorldToShadowCoord(positionWS); // per-fragment
    shadow coordinate
Light mainLight = GetMainLight(shadowCoord); // main light
float3 lightDir = normalize(mainLight.direction); // light dir
float NdotL = dot(normalWS, lightDir) * mainLight.shadowAttenuation; // shadow-
    attenuated Lambertian term
float rampU = lerp(0.5, saturate(NdotL * 0.5 + 0.5), _XToonLightSensitivity);
    // remap towards ramp midpoint
```

Listing 5.7: U axis (shadow-attenuated NdotL), Meta SDK.

The surface tangent frame is not available in the hook, and this leaves Normal Field Abstraction via an abstract normal map unsupported. Only the geometric smoothing path is available on this platform. The abstraction compression and dynamic smoothing stages match those on the standalone platforms. The same `lightDir` and shadow attenuation from the U axis computation are reused for the specular term.

Table 5.2.: V2.2 implementation comparison across the three avatar platforms.

	Mixamo / Avaturn	**Meta SDK**
U axis source	NdotL, shadow-attenuated	NdotL, shadow-attenuated (per-fragment `GetMainLight`)
V axis computation	`shader_feature_local` keywords (Depth / Curvature / Manual)	`multi_compile` keywords (same three modes, runtime-switchable)
Normal Abstraction	TBN abstract normal map or geometric smoothing	Geometric smoothing only (no TBN in hook)
Shadow source	Per-vertex `GetShadowCoord`, interpolated	Per-fragment `TransformWorldToShadowCoord`

Figure 4.9 in Chapter 4 shows the V2.2 output across all three platforms.

5.4.4. V3 — Screen-Space Normal Edge Detection

The V2.2 X-Toon ramp shading is retained as the surface shading base. The gradient magnitude E_N, introduced in Section 4.4.5 (Equation (4.4)), maps to an edge weight through

$$E = \text{smoothstep}(\tau - \sigma,\ \tau + \sigma,\ E_N) \cdot \alpha_N \tag{5.1}$$

where τ is the threshold centre, σ is the threshold half-width, and α_N is the edge weight scale. Both `ddx` and `ddy` operate on screen-space pixel derivatives rather than object-space geometry, and the resulting edge width therefore scales with viewing distance rather than object scale, as discussed in Section 4.4.5. These three symbols are exposed as shader properties on all three platforms, and can be adjusted at runtime through the parameter tuning interface described in Section 5.5.1:

- `_NormalEdgeThreshold` (τ)
- `_NormalEdgeSmoothness` (σ)
- `_NormalEdgeStrength` (α_N)

This edge mask is composited over the shaded surface by linear interpolation towards a configurable line colour.

Mixamo and Avaturn

The normal must be decoded from the PBR normal map before the derivative is computed. The vertex shader builds the tangent frame from the mesh tangent attribute, and the fragment shader then unpacks the normal map and transforms it into world space:

```
float3 geomNWS = normalize(nWS); // interpolated geometry normal
float3 nWS = geomNWS; // fallback if normal map disabled
// decode the tangent-space normal map and transform it into world space
float3 normalTS = UnpackNormalScale(
                      SAMPLE_TEXTURE2D(_BumpMap, sampler_BumpMap, uv),
    _BumpScale);
float3x3 TBN = float3x3(tangentWS, bitangentWS, geomNWS); // tangent frame
```

```
nWS = normalize(mul(normalTS, TBN)); // rotate normal into world space
```

Listing 5.8: Normal decode via TBN (Mixamo and Avaturn).

When the normal map is disabled, `nWS` falls back instead to the bare interpolated geometry normal. The resulting `nWS` then feeds into the same gradient magnitude and threshold computation shown in Listing 5.9 for the Meta SDK platform, with `nWS` in place of `worldNormal`.

Meta Avatars SDK

On the Meta SDK platform, the world-space normal supplied to `ddx`/`ddy` is the SDK's fully composited fragment normal. This already incorporates the SDK's internal normal-map decode, and no additional texture read is therefore required. The derivative and threshold computation itself is identical across all three platforms:

```
float3 dNdx = ddx((float3)worldNormal); // horizontal screen-space derivative
float3 dNdy = ddy((float3)worldNormal); // vertical screen-space derivative
float M = sqrt(dot(dNdx, dNdx) + dot(dNdy, dNdy)); // gradient magnitude
float nMin = _NormalEdgeThreshold - _NormalEdgeSmoothness; // lower threshold
    bound
float nMax = _NormalEdgeThreshold + _NormalEdgeSmoothness; // upper threshold
    bound
float edge = smoothstep(nMin, nMax, M) * _NormalEdgeStrength; // final edge
    weight
```

Listing 5.9: V3 normal gradient and edge weight, Meta SDK.

Table 5.3.: V3 implementation comparison across the three avatar platforms.

	Jody (Mixamo)	**Avaturn**	**Meta SDK**
Normal source	TBN-decoded PBR normal map (baked)	TBN-decoded PBR normal map (glTF)	SDK composited fragment normal
Edge detail	Medium: normal map + mesh curvature	Dense: fine microdetail in normal map	Medium: SDK mesh anatomy
Sample cost	1 (normal map); 0 when disabled	1 (normal map); 0 when disabled	0 extra samples

Figure 4.10 in Chapter 4 presents the V3 output across all three platforms, including how edge thickness depends on viewing distance.

5.4.5. V4 — Sobel Edge Detection

The edge source shifts from signals derived from geometry to the diffuse albedo texture. Each of the nine 3×3 neighbourhood positions is sampled and converted to a scalar luminance using the BT.709 coefficients defined in Equation (3.2). These coefficients match the sRGB primaries used by the rendered textures and the URP colour pipeline, and this prevents colour channel cross-talk from introducing false edges at hue boundaries where RGB channels change in opposite directions but luminance is constant. The Sobel kernels G_x and G_y are applied to the nine luminance samples. Subsequently, the resulting gradient magnitude is compared against a threshold to produce the edge mask.

Mixamo and Avaturn

These platforms read the raw diffuse albedo texture directly. As a result, edge detection responds to surface material boundaries rather than to lighting:

```
float3 L = float3(0.2126, 0.7152, 0.0722); // BT.709 luma weights
float off = _InnerLineBlur * 0.001; // UV-space sample offset

float tl = dot(SAMPLE_TEXTURE2D(_MainTex, sampler_MainTex, uv+float2(-off, off)
    ).rgb, L); // top-left tap
float t  = dot(SAMPLE_TEXTURE2D(_MainTex, sampler_MainTex, uv+float2(0, off)).
    rgb, L); // top tap
// ...the remaining six neighbour samples (tr, l, r, bl, b, br) follow the same
     pattern

float Gx = (tr + 2*r + br) - (tl + 2*l + bl); // horizontal gradient
float Gy = (tl + 2*t + tr) - (bl + 2*b + br); // vertical gradient
float edgeMag = sqrt(Gx*Gx + Gy*Gy); // gradient magnitude
float edge = step(_InnerLineThreshold, edgeMag) * _InnerLineStrength; //
    threshold and scale
color.rgb = lerp(color.rgb, _InnerLineColor.rgb, edge); // composite line
    colour
```

Listing 5.10: Nine-tap luminance sampling and Sobel gradient.

The sampled albedo already excludes lighting. Consequently, the gradient magnitude is compared directly against `_InnerLineThreshold` to produce the edge mask.

Meta Avatars SDK

The Meta SDK hook receives only the composited PBR output rather than a raw albedo texture. The Sobel therefore responds to shadow boundaries and specular gradients in addition to material transitions. This reduces selectivity, but it still adds visible edges along clothing boundaries and broad facial contour lines. The Meta Avatar's stylised, illustrated texture nonetheless contains less high-frequency variation than the photographic albedo used on Avaturn, and this yields a comparatively cleaner edge response, as discussed in Section 4.4.6:

```
float3 L = float3(0.2126, 0.7152, 0.0722); // BT.709 luma weights
float offset = _SobelSampleDist * 0.001; // UV-space sample offset

float tl = dot(tex2D(u_BaseColorSampler, uv+float2(-offset, offset)).rgb, L);
    // top-left tap
float t = dot(tex2D(u_BaseColorSampler, uv+float2(0, offset)).rgb, L); // top
    tap
// ...the remaining six neighbour samples (tr, l, r, bl, b, br) follow the same
     pattern

float Gx = (tr + 2*r + br) - (tl + 2*l + bl); // horizontal gradient
float Gy = (tl + 2*t + tr) - (bl + 2*b + br); // vertical gradient
float edgeMag = sqrt(Gx*Gx + Gy*Gy); // gradient magnitude
```

Listing 5.11: Nine-tap luminance sampling and Sobel gradient, Meta SDK.

Compositing the PBR output reintroduces the UV seam artefacts. This artefact is absent on the other two platforms, where sampling occurs on the pre-lighting albedo instead. The colour difference across a seam registers as a strong edge despite the absence of any real visual boundary. The suppression rejects detections

whose luminance spread across all nine samples exceeds `_SobelSeamLimit`, and a second upper bound `_SobelMax` clips spike magnitudes beyond any plausible structural edge:

```
float lMin = min(min(min(tl,t),min(tr,l)), min(min(r,bl),min(b,br))); // darkest sample
float lMax = max(max(max(tl,t),max(tr,l)), max(max(r,bl),max(b,br))); // brightest sample

float seamMask = step(lMax - lMin, _SobelSeamLimit); // reject seam spikes
float inBand = step(_SobelThreshold, edgeMag)
             * step(edgeMag, _SobelMax); // accept only valid range

float edge = inBand * seamMask * _SobelStrength; // combine gates and scale
color.rgb = lerp(color.rgb, _InnerLineColor.rgb, edge); // composite line colour
```

Listing 5.12: Seam suppression and magnitude band gate, Meta SDK.

Table 5.4.: V4 implementation comparison across the three avatar platforms.

	Mixamo / Avaturn	**Meta SDK**
Sobel input	Raw diffuse albedo texture	Composited PBR colour
Sample count	9 albedo samples	9 samples on PBR output
Threshold gate	Single step threshold	Step band gate + seam gate
Seam suppression	None	Luma range gate + upper magnitude bound
Shadow edges	No	Yes

The V4 output across all three platforms is compared in Figure 4.11 in Chapter 4.

5.4.6. V4.2 — Gaussian Prefiltered Sobel

This method carries over the V4 Sobel gradient computation across the same eight positions, but each luminance sample is now drawn through a prefilter rather than a single point sample. At each call, one centre sample, four axis-aligned cardinal samples, and four diagonal corner samples are drawn at a configurable offset from the given UV position. The three weight groups are normalised to unity before use so that the output is always a valid luminance estimate.

This prefiltering is implemented as a shared helper function, applied identically on all three platforms except for the texture it reads:

```
// samples a 9-tap weighted average of luminance around uv
float SampleLuminanceBlurred(float2 uv, float blurR, float cW, float cardW, float diagW)
{
    float3 L = float3(0.2126, 0.7152, 0.0722); // BT.709 luma weights
    float  v = 0.0; // accumulated luminance
    v += dot(SAMPLE_TEXTURE2D(_MainTex, sampler_MainTex, uv).rgb, L) * cW; // centre tap
    v += dot(SAMPLE_TEXTURE2D(_MainTex, sampler_MainTex, uv + float2(blurR, 0)).rgb, L) * cardW; // cardinal tap
    // ...three more cardinal taps and four diagonal taps follow the same pattern
    return v; // blurred luminance estimate
}
```

```

// Weight normalisation: computed once before the eight Sobel positions are
    sampled
float totalW = _GaussianCenterWeight + 4.0*_GaussianCardinalWeight + 4.0*
    _GaussianDiagonalWeight;
float cW    = _GaussianCenterWeight   / totalW; // centre weight share
float cardW = _GaussianCardinalWeight / totalW; // cardinal weight share
float diagW = _GaussianDiagonalWeight / totalW; // diagonal weight share
```

Listing 5.13: 9-tap Gaussian luminance helper with weight normalisation (unified across platforms).

This helper reads the raw diffuse albedo texture on Mixamo and Avaturn, and the composited PBR output on the Meta SDK, with the rest of the logic unchanged. It is called once at each of the eight Sobel neighbourhood positions, and the resulting values feed the same gradient magnitude computation used in V4.

Additionally, a configurable threshold band, applied identically on all three platforms, gates this gradient magnitude. A lower bound rejects weak gradients as noise, and an upper bound suppresses detections with an unusually large magnitude, such as UV seam spikes. This band gate replaces the luma range seam gate used in V4.

A series of four progressive smoothstep passes then sharpens the thresholded magnitude into a crisp edge mask, shown here for the Meta SDK, where all four passes are always evaluated and driven by a single tightness scalar rather than the individually toggleable flags used on Mixamo and Avaturn:

```
float hw1 = lerp(0.5, 0.03, _GSobelTightness); // pass 1 half-width
edge = smoothstep(0.5 - hw1, 0.5 + hw1, edge); // sharpen pass 1
float hw2 = lerp(0.5, 0.15, _GSobelTightness); // pass 2 half-width
edge = smoothstep(0.5 - hw2, 0.5 + hw2, edge); // sharpen pass 2
// ...two further passes repeat this pattern with hw3 = 0.25 and hw4 = 0.35
edge = pow(edge, _GSobelPowerCurve); // contrast curve
edge *= _GSobelStrength; // final edge weight
```

Listing 5.14: 4-pass progressive smoothstep sharpening and power curve, Meta SDK.

At tightness = 0 every half-width equals 0.5, and each pass reduces to a linear identity. At tightness = 1 the widths shrink to 0.03, 0.15, 0.25, and 0.35, and the passes together compose a steep sigmoid. A power curve above 1 suppresses soft halos, while values below 1 amplify thin lines instead.

Furthermore, three preset configurations that are Light, Moderate, and Aggressive are offered on Mixamo and Avaturn. These bundle the parameters into predefined combinations for rapid setup across avatars, and this removes the need to tune each parameter individually. Table 5.5 lists the parameter values assigned by each preset. The default parameter set on all three platforms corresponds to the Light configuration, and this is the configuration shown in every later V4.2 comparison figure in this thesis. V4.2 itself, however, was not carried forward into the final study composite once texture preprocessing made the basic Sobel operator sufficient on its own (Section 4.5).

Table 5.5.: V4.2 preset parameter values (Mixamo and Avaturn).

Parameter	Light	Moderate	Aggressive
Blur radius multiplier (`_BlurRadiusMultiplier`)	0.8	1.0	1.2
Centre weight (`_GaussianCenterWeight`)	0.25	0.30	0.25
Cardinal weight (`_GaussianCardinalWeight`)	0.125	0.12	0.125
Diagonal weight (`_GaussianDiagonalWeight`)	0.0625	0.05	0.0625
Threshold min multiplier (`_ThresholdMinMultiplier`)	0.2	0.4	0.85
Threshold max multiplier (`_ThresholdMaxMultiplier`)	2.0	1.6	1.15
Smoothstep tightness (`_SmoothstepTightness`)	0.2	0.3	1.0
Power curve (`_PowerCurve`)	1.5	2.0	3.0

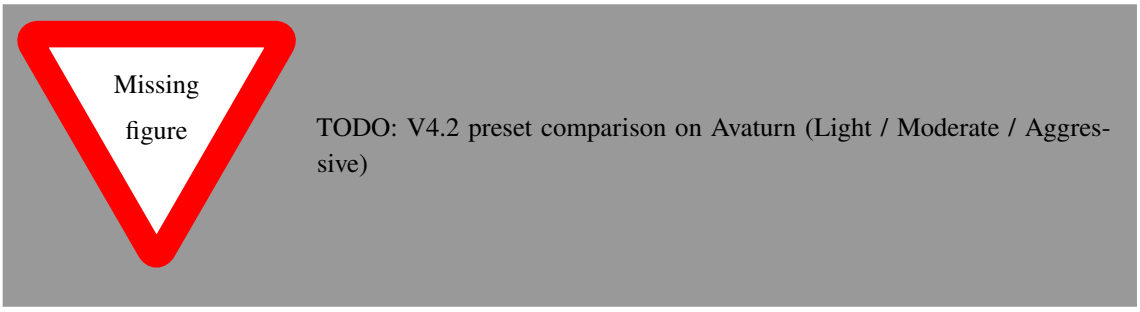


Figure 5.1.: Light, Moderate, and Aggressive preset configurations compared on the Avaturn avatar.

Table 5.6.: V4.2 implementation comparison across the three avatar platforms.

	Mixamo / Avaturn	Meta SDK
Gaussian input	Raw diffuse albedo	Composited PBR output
Smoothstep passes	4, individually toggleable per pass	4, always on via `_GSobelTightness`
Preset configurations	Light / Moderate / Aggressive	N/A

Figure 4.13 in Chapter 4 compares the V4.2 output across all three platforms.

5.4.7. V5 — Hierarchical Edge Detection

V5 combines three independent edge cues into a unified boundary representation. Each cue is thresholded into a line signal, and the three signals are fused by weighted max-pooling rather than addition. Thus, the strongest cue at a fragment dominates rather than being diluted by the others:

```
float edge = max(depthLine * _DepthWeight, // depth cue contribution
             max(normalLine * _NormalWeight, // normal cue contribution
                 colorLine * _ColorWeight)); // colour cue contribution
```

Listing 5.15: Weighted max-pooling fusion (shared formulation across platforms).

The three input layers, however, are computed differently on each platform, since Mixamo and Avaturn access the depth, normal, and colour buffers directly, while the Meta SDK is restricted to the per-fragment hook.

Mixamo and Avaturn

Mixamo and Avaturn run V5 as a URP renderer feature: a post-processing pass that samples Unity's scene depth, normal, and colour buffers directly. Each layer applies the Roberts Cross operator to a 2×2 diagonal sample pattern, as defined in Section 3.2.1:

```
// diagonal difference operator on a 2x2 sample quad
float RobertsCross(float tl, float tr, float bl, float br)
{
    return abs(tl - br) + abs(tr - bl); // sum of diagonal differences
}

float d_tl = GetLinearEyeDepth(uv + float2(-off.x,  off.y)); // top-left depth
    tap
// ...remaining three depth taps (tr, bl, br) follow the same pattern
float depthEdge = RobertsCross(d_tl, d_tr, d_bl, d_br)
                  / max(GetLinearEyeDepth(uv), 0.01) * _DepthScale; //
    normalised, scaled edge
float depthLine = smoothstep(_DepthThreshold - 0.01, _DepthThreshold + 0.01,
    depthEdge); // to line signal
```

Listing 5.16: Roberts Cross depth layer (Mixamo and Avaturn).

The normal layer applies the same Roberts Cross pattern to the screen-space normal buffer. The colour layer applies it to the rendered colour buffer's luminance, and an added chromatic term also responds to boundaries defined by colour alone, with no luminance difference. An adaptive factor derived from local brightness reduces the depth and colour layers' sensitivity in dark image regions, where noise would otherwise dominate the response. The normal layer receives only half this reduction, as its response depends less on scene brightness. Since this post-processing pass operates on the entire screen, background geometry outside the avatar would also register spurious edges without correction. A dedicated avatar mask therefore restricts the composite to the avatar's screen-space silhouette. This mask is captured in a separate render pass, filtered to a Unity rendering layer assigned exclusively to the avatar, so that only avatar fragments are drawn into it. The resulting mask value is then multiplied into the edge blend factor, so that background pixels retain the original shaded colour unmodified.

Meta Avatars SDK

The Meta SDK's single fragment hook has no access to a dedicated depth or normal buffer. The depth layer therefore estimates a distance gradient computed from the fragment's world-space distance to the camera. This gradient is normalised for perspective:

```
float depth = length(worldPos - _WorldSpaceCameraPos); // fragment-to-camera
    distance
float dDepthX = ddx(depth); // horizontal depth derivative
float dDepthY = ddy(depth); // vertical depth derivative
float depthGrad = sqrt(dDepthX*dDepthX + dDepthY*dDepthY) / max(depth, 0.01) *
    _DepthScale; // normalised, scaled gradient
float depthLine = smoothstep(_DepthThreshold - 0.005,
    _DepthThreshold + 0.005, depthGrad); // threshold to line signal
```

Listing 5.17: Depth layer, Meta SDK.

This proxy detects distance discontinuities within the avatar mesh. However, it cannot separate the avatar from the background. The background lies entirely outside the per-fragment computation. The normal layer reuses the world-space normal derivative computation from V3 (Section 5.4.4), and an AHEAD-style adaptive gain then amplifies this layer specifically in darker image regions [51]:

```
float luma = dot(color.rgb, float3(0.2126, 0.7152, 0.0722)); // perceptual
    luminance
float adapt = min(1.0 / (3.0 * luma + 0.1), 4.0); // inverse-linear gain,
    capped at 4x
normalLine *= lerp(1.0, adapt, _AdaptiveStrength); // blend gain by strength
```

Listing 5.18: AHEAD adaptive gain on the normal layer, Meta SDK.

The composited PBR output also contains baked lighting gradients that produce false Roberts Cross responses on this platform. The colour layer therefore replaces Roberts Cross with XDoG: the difference between two Gaussian-blurred luminance samples at different radii, sharpened with a tanh curve:

```
float g1 = GaussianLuma(uv, sigma1); // fine blur
float g2 = GaussianLuma(uv, sigma1 * _XDoGK); // coarse blur
float D = g1 - _XDoGTau * g2; // difference of Gaussians
float colorLine = D < -_ColorThreshold
    ? saturate(-tanh(_XDoGPhi * (D + _ColorThreshold))) : 0.0; // sharpened
    line signal
```

Listing 5.19: XDoG colour layer, Meta SDK.

Subtracting the coarser blur from the finer one cancels out broad, slowly varying illumination gradients. The sharper transitions that mark genuine material boundaries remain unaffected. This directly targets the baked lighting artefact described above, without requiring any change to the underlying composited input.

After the three cues are fused, the Meta SDK applies an additional smoothstep step to sharpen the combined edge value. The Mixamo and Avaturn implementation instead composites the fused value directly, without this extra sharpening step.

Table 5.7.: V5 implementation comparison across the three avatar platforms.

	Mixamo / Avaturn	**Meta SDK**
Architecture	URP renderer feature (post-processing pass)	Per-fragment hook
Depth layer	Roberts Cross on real scene depth texture	`ddx`/`ddy` camera-distance proxy
Normal layer	Roberts Cross on real scene normal texture	`ddx`/`ddy` normal derivative, reused from V3
Colour layer	Roberts Cross on luminance, plus a chromatic term	XDoG (difference of Gaussians)
Adaptive gain	Depth and colour layers, normal layer at half strength	Normal layer only, AHEAD-style
Avatar isolation	Dedicated avatar mask	Not required

Figure 4.15 in Chapter 4 compares the V5 output across all three platforms.

5.4.8. V6 — Kuwahara Painterly Filter

All three platforms use the anisotropic Kuwahara filter of Kyprianidis et al. [13]. This filter relies on a structure tensor to estimate the local feature orientation and anisotropy, and the sampling neighbourhood then becomes an ellipse aligned with that orientation rather than a fixed square.

Mixamo and Avaturn

The structure tensor is computed as a Sobel gradient over luminance in a dedicated post-processing pass on these platforms. A second pass subsequently smooths the tensor with a 5-tap separable Gaussian blur before the main filter pass reads it:

```
float gx = -tl - 2.0*l - bl + tr + 2.0*r + br; // Sobel horizontal
float gy = -tl - 2.0*t - tr + bl + 2.0*b + br; // Sobel vertical
return float4(gx*gx, gx*gy, gy*gy, 1.0); // pack tensor components (E, F, G)

// Second pass: 5-tap separable Gaussian blur over the packed tensor
float4 result = SAMPLE_TEXTURE2D_X_LOD(_BlitTexture, sampler_LinearClamp, uv,
    0) * 0.375; // centre tap
// ...four more taps at +/-offset and +/-2*offset, weighted 0.25 and 0.0625,
    complete the blur
```

Listing 5.20: Structure tensor pass and Gaussian blur pass (Mixamo and Avaturn).

Meta Avatars SDK

The Meta SDK cannot perform a blur pass inside its single fragment hook. The tensor is therefore estimated directly from ddx/ddy on the already-shaded luminance, and this produces a noisier, unblurred orientation estimate:

```
float lum = dot(color.rgb, float3(0.299, 0.587, 0.114)); // shaded luminance
float gx = ddx(lum); // horizontal luminance derivative
float gy = ddy(lum); // vertical luminance derivative
float E = gx * gx; // tensor component E
float F = gx * gy; // tensor component F
float G = gy * gy; // tensor component G
```

Listing 5.21: Structure tensor, Meta SDK (unblurred).

All three platforms then derive the dominant orientation angle and the anisotropy ratio from the same eigenvalue decomposition of this tensor, given by

$$\lambda_{1,2} = \frac{E+G}{2} \pm \sqrt{\left(\frac{E-G}{2}\right)^2 + F^2}, \tag{5.2}$$

$$\phi = \frac{1}{2}\arctan\left(\frac{2F}{E-G}\right), \tag{5.3}$$

$$A = \frac{\lambda_1 - \lambda_2}{\lambda_1 + \lambda_2}, \tag{5.4}$$

where λ_1 and λ_2 are the tensor eigenvalues, ϕ is the dominant orientation, and A is the anisotropy ratio:

```
float disc = sqrt(max(0.0, (E-G)*(E-G) + 4.0*F*F)); // eigenvalue discriminant
float lambda1 = 0.5 * (E + G + disc); // larger eigenvalue
float lambda2 = 0.5 * (E + G - disc); // smaller eigenvalue
float phi = 0.5 * atan2(2.0 * F, E - G); // dominant orientation
float A = (lambda1 - lambda2) / (lambda1 + lambda2); // anisotropy
```

Listing 5.22: Eigenanalysis: orientation and anisotropy (unified across platforms).

The anisotropy A scales the ellipse's major and minor axes, elongating the sampling neighbourhood along the feature direction and narrowing it across. This axis scaling is a clamped linear blend between the isotropic and fully anisotropic extremes on Mixamo and Avaturn. The Meta SDK instead applies the ratio formula

from Kyprianidis et al. directly, driven by an additional eccentricity parameter with no equivalent on the other platforms.

Eight overlapping elliptical sectors are then sampled around this ellipse rather than four fixed quadrants, though the sampling strategy itself differs by platform. Mixamo and Avaturn scan every pixel within the kernel radius and accumulate it into whichever sector it falls into, so the sample count grows with the radius. The Meta SDK instead takes a fixed three samples per sector at set radii, plus one sample shared across all sectors, for a constant total regardless of radius. Each sector accumulates a mean colour and a variance. These sectors are subsequently blended by an inverse-variance weight, so that the smoothest sector dominates the result. The exact weighting formula differs by platform, since each implementation follows a different variant of the same principle from Kyprianidis et al. [13]:

```
float w_i = 1.0 / pow(max(tau, sigma_i), q); // inverse-variance sector weight
result += mean_i * w_i; // accumulate weighted colour
totalWeight += w_i; // accumulate normalisation factor
```

Listing 5.23: Inverse-variance sector weight, Meta SDK.

On Mixamo and Avaturn, the weight instead grows as an inverse power of the sector variance directly:

```
float w_i = 1.0 / (1.0 + pow(variance_i * 1000.0, 0.5 * _Sharpness)); //
    inverse-variance sector weight
result += mean_i * w_i; // accumulate weighted colour
totalWeight += w_i; // accumulate normalisation factor
```

Listing 5.24: Inverse-variance sector weight (Mixamo and Avaturn).

The final colour on both platforms is obtained by normalising the accumulated weighted colour by the total weight. Mixamo and Avaturn then composite this result over the original shaded colour through the same avatar mask pass used in V5 (Section 5.4.7), since V6 also runs as a post-processing effect on these platforms. The Meta SDK instead blends by a strength parameter.

Table 5.8.: V6 implementation comparison across the three avatar platforms.

	Mixamo / Avaturn	**Meta SDK**
Structure tensor source	Sobel gradient over luminance, dedicated pass	`ddx`/`ddy` on already-shaded luminance
Tensor smoothing	5-tap separable Gaussian blur pass	None, unblurred
Final compositing	Masked composite via dedicated avatar mask pass	Strength-parameter blend

Figure 4.16 in Chapter 4 compares the V6 output across all three platforms.

5.5. Runtime Tooling

Two runtime utilities were developed to address constraints that cannot be resolved at shader compilation time: the first enables interactive parameter exploration within the VR headset during the visual assessment phase, and the second freezes the avatar's pose for evaluation screenshots.

5.5.1. In-VR Parameter Tuning Interface

Shader parameters can also be adjusted through a custom world-space UI panel during live headset sessions, reproducing the same controls available in the Unity Inspector. This allowed values to be explored and compared while the avatar remained visible in the headset at correct VR scale. This panel was central to the visual assessment process used to select the technique for the user study.

5.5.2. Pose Freeze for Evaluation

A dedicated script freezes the avatar's body-tracking pose at a chosen instant, preserving a specific expression and posture for evaluation screenshots. Since the Meta Avatars SDK exposes no public API for pausing tracking, the freeze is implemented through reflection, substituting a snapshot provider that returns a fixed pose state in place of the SDK's live input provider.

5.6. Animation and Recording Pipeline for the User Study

Facial animation for the Avaturn avatar is driven using Apple ARKit: the Live Link Face application streams all 52 ARKit blendshape coefficients and head rotation over UDP from an iPhone to Unity, where a custom component maps each coefficient to the corresponding Avaturn blendshape target introduced in Section 5.2.3. This stream drives the corresponding blendshape weights at runtime, so the avatar's expression follows the iPhone performer directly with no intermediate animation clip.

For the C2 and C3 stimuli, this blendshape stream was captured simultaneously while filming the C1 real person video, matching the synchronized recording session described in Section 4.8.2. The same captured stream was played back for both the photorealistic and NPR renders, so any visible difference between C2 and C3 in the resulting footage originates from the shader rather than from the underlying animation data.

The camera in the recording scene uses a reduced field of view of 20°, compared with the 60° default used in the general platform comparison scenes described in Section 5.2.4. A wide angle at the close camera distance required for a talking-head framing exaggerates the size of facial features nearest the camera, most visibly the nose, through perspective distortion. Narrowing this angle removes the artefact while preserving an equivalent framing of the avatar.

6. Evaluation

This chapter evaluates the NPR avatar system developed in this thesis. The first part documents the visual quality assessment of the eight NPR techniques across the three avatar platforms, establishing which technique was selected for the user study and the reasoning behind that selection. The second part presents the user study that tests whether NPR stylisation affects how participants perceive trust in a speaking avatar, and interprets each result in relation to the study hypotheses as it is reported.

6.1. Visual Quality Assessment

6.1.1. Assessment Methodology

Each NPR technique was evaluated qualitatively through a structured visual assessment conducted inside the Unity Editor and on the Meta Quest 3 headset using the world-space UI panel described in Section 5.5.1. The avatar was placed in the shared studio environment described in Section 5.2.4 and animated with a repeating idle loop. It was observed from several distances, including the close conversational framing of approximately 1 m used to record the user study videos. Parameters were adjusted interactively through this panel until the visual output satisfied four criteria:

1. **Facial readability.** Facial features (eyes, mouth boundary, nose bridge, and jaw line) must remain individually legible under stylisation. A technique that obscures or merges these features cannot serve as a plausible avatar representation in the study.
2. **Temporal stability.** Edges and stylisation effects must remain stable during animation. Screen-space techniques are particularly susceptible to flickering when the avatar translates across the screen, because the pixel-space neighbourhood shifts while the underlying geometry does not.
3. **Artefact suppression.** Visible UV seam lines, rectangular outlines around hair cards, depth-fighting at the outline hull, and false edge detections at skin-tone boundaries must be suppressed to a level that is not noticeable under normal observation.
4. **Expressive abstraction.** The stylisation should be visually legible as an intentional artistic choice, not as a rendering error. A technique that produces only isolated noise marks rather than coherent contours would not read as deliberate stylisation.

Evaluation proceeded iteratively for each technique. Parameters were first tuned on the Mixamo avatar, where unrestricted shader access made per-pass debugging straightforward, and then verified on Avaturn and the Meta SDK avatar. Techniques that satisfied all four criteria on at least two of the three platforms were considered viable candidates for the user study.

6.1.2. Per-Technique Assessment

V1 — Inverted Hull Outline. As the shared silhouette pass underlying every other technique (Section 4.4.2), V1 was also assessed on its own as the minimal abstraction reference in the study materials. Temporal stability is satisfied without additional tuning, since the outline geometry is generated in the vertex stage rather than from a screen-space pixel neighbourhood. Artefact suppression required the alpha cutout and depth bias described in Section 4.4.2 to remove rectangular borders around hair cards and silhouette z-fighting respectively, and no artefacts remained visible once these were applied. Facial readability remains satisfied in isolation, since the technique does not alter or obscure the underlying texture or geometry. On the contrary, expressive abstraction is not. The method marks only the external silhouette, and creases, clothing seams, and

facial contours consequently remain undelineated, consistent with the interior boundary limitation established in Section 4.4.2. V1 was therefore retained as the foundational outline pass and as the lowest abstraction reference in the study materials, but not as a standalone candidate for the composite assembled for the user study.

V2 — Toon Shading. The quantised diffuse term produces a consistent, low-frequency tonal structure that satisfies both facial readability and temporal stability without flicker. Clothing seams and facial features carry no dedicated edge cue beyond the local tonal contrast that the shading bands happen to produce. These accordingly remain undelineated at the 1 m framing used in the study. This is consistent with the interior line limitation established in Section 4.4.3. Nevertheless, unlike V1, the coherent tonal bands still read as a deliberate stylistic choice rather than an incomplete or erroneous render. Expressive abstraction therefore remains satisfied despite the missing interior detail.

V2.2 — X-Toon Shading. The abstraction level axis described in Section 4.4.4 is driven by camera distance or surface curvature. At the wider framing used for general platform comparison, these two modes showed only minor differences between them, with Avaturn showing slightly greater variation than the other platforms. Neither mode, however, produces a meaningful signal at the 1 m conversational framing used in the study. The depth range across the avatar's torso and face spans only a few centimetres, too narrow relative to the configured depth window. Correspondingly, the screen-space normal derivatives produced by curvature mode across the smooth head and body geometry remain close to zero, too faint to register as a meaningful signal. Neither dynamic mode therefore produces a strongly different outcome at this fixed conversational distance. This finding is why manual mode was selected as the driving mode for the composite assembled for the user study, as Section 4.5 explains. Because manual mode holds a fixed constant rather than recomputing a value per fragment, it carries none of the flicker risk that affects the screen-space techniques assessed elsewhere in this section. Facial readability and temporal stability are therefore both satisfied without qualification. Expressive abstraction is likewise satisfied. The fixed detail bias visibly compresses the tonal bands towards the ramp midpoint, even though the parameter adjustment study found no significant preference across the tested bias values (Section 4.7).

V3 — Screen-Space Normal Edge Detection. The camera distance stays fixed throughout each study recording. This is why the distance-dependent edge thickening identified in Section 4.4.5 remains mild here rather than growing as it would under a varying viewing distance. Facial readability is nonetheless not satisfied. Smooth skin produces minimal normal discontinuities, so the eyebrow, lip, and eyelid boundaries central to this criterion remain undetected. This is the primary reason V3 was not selected for the composite (Section 4.5). Despite this exclusion, V3 requires zero additional texture samples. This makes it the least costly technique after V2.

V4 — Sobel Edge Detection. Baked-illumination artefacts, such as the nose bridge artefact identified in Section 4.4.6, are suppressed on the Avaturn avatar through the Blender texture preprocessing described in Section 4.6 and a raised threshold on the head material. The skin-to-clothing boundary produces the most visually significant lines. Eyebrow and lip boundaries emerge at moderate threshold settings. Since the step function used for thresholding on Mixamo and Avaturn (Section 5.4.5) can toggle a pixel between frames as lighting varies slightly during animation, temporal stability required the same threshold tuning to keep lines at material boundaries from flickering.

V4.2 — Gaussian Prefiltered Sobel. The facial segment recovery shortfall on Avaturn and the resulting shift to texture preprocessing are described in Section 4.4.7. This came at a cost of up to 72 texture samples per fragment. The preprocessing pipeline made the basic Sobel operator sufficient on its own. The added sampling cost of Gaussian prefiltering was therefore no longer justified for the composite assembled for the user study.

V5 — Hierarchical Multi-Cue Edge Detection. V5 was a candidate addition to the C3 composite, assessed together with the Kuwahara filter through the parameter adjustment study (Section 4.7). The interior contours do not produce clear boundaries between adjacent facial segments, as Section 4.4.8 describes. This leaves

facial readability unsatisfied. The same limitation extends to artefact suppression. The colour layer's response to illumination gradients around the nose bridge is left unaddressed on Avaturn, unlike on the Meta Avatars SDK. This candidate addition was subsequently excluded from the final composite following the parameter adjustment study (Section 6.2.1).

V6 — Kuwahara Painterly Filter. Consistent with the trade-off between abstraction and legibility identified in Section 4.4.9, V6 does not satisfy facial readability as a standalone condition at radius values large enough to read as clearly painterly. This trade-off motivated testing it as an addition to the composite candidate for C3, and the parameter adjustment study thereafter excluded it from the final configuration (Section 6.2.1).

6.1.3. Technique Comparison Summary

Table 6.1.: Summary of visual quality assessment results across the eight NPR techniques on the Avaturn platform. Criteria are rated qualitatively: ✓= satisfactory without tuning, (✓) = satisfactory after tuning, × = limitation observed.

Technique	Facial readability	Temporal stability	Artefact suppression	Expressive abstraction
V1 — Outline	✓	✓	(✓) alpha cutout, depth bias	× not in isolation
V2 — Toon	✓	✓	✓	(✓) no interior lines
V2.2 — X-Toon	✓	✓	✓	(✓) manual mode chosen for flexibility
V3 — Normal edge	× facial landmarks undetected	✓	✓	✓
V4 — Sobel	(✓) moderate threshold	(✓)	(✓) preprocessing	✓
V4.2 — Gaussian Sobel	× insufficient for facial segments	(✓)	✓	✓
V5 — Hierarchical	× segment boundaries unclear	✓	× nose bridge artefact	✓
V6 — Kuwahara	× at large r	✓	✓	✓

6.1.4. Technique Selection for the User Study

The five candidates were narrowed from the eight techniques assessed. Basic toon shading, screen-space normal edge detection, and Gaussian prefiltered Sobel were set aside before the parameter adjustment study began. This selection was guided by limitations in facial segment readability and in each technique's contribution to the resulting image in comparison with the alternative method. Toon shading was superseded by the finer abstraction control of X-Toon. Screen-space normal edge detection could not recover facial boundaries defined by colour or texture. These boundaries determine facial readability. Gaussian prefiltered Sobel no longer offered a measurable benefit once texture preprocessing resolved the noise problem it was designed to address.

Five candidate components were assembled for the user study condition C3: an inverted hull outline, X-Toon tonal shading, Sobel edge detection, and two optional post-processing passes, a Kuwahara painterly filter and a hierarchical depth edge layer. A parameter adjustment study was conducted to determine which of these components to retain and how to configure them. As a result, only the first three were carried forward into the final composite, and Section 6.2.1 reports that process in detail. The combination uses the submesh material split established for the Avaturn avatar (Section 5.3), and this split assigns the head and clothing regions to independent materials. That independence is what allows the texture preprocessed in Blender to be applied to the head material alone, and this preprocessing suppresses the baked-illumination artefacts while facial features remain individually legible. Raising the threshold alone was insufficient, since it also removed genuine facial contour lines. The combination also reads clearly as an intentional stylisation rather

than a rendering artefact. The manipulation check embedded in the study questionnaire (Part I) depends on this clarity. A technique that produces only subtle modifications to the photorealistic baseline could fail the check, and this would reduce usable data.

6.2. User Study Results

The user study was designed and conducted following the protocol described in Section 4.8. This section reports the findings. Full descriptive statistics, raw scale scores, and counterbalancing verification tables are provided in Appendix A.

6.2.1. Parameter Adjustment Study Results

The parameter adjustment study described in Section 4.7 fixed the parameter values used in the C3 composite and produced the empirical basis for excluding two candidate components. Full per-round rank distributions, Borda scores, and statistical test results are reported in Section 4.7 and Chapter B. This section interprets what the outcome means for the composite carried forward into the main study.

Two of the five rounds, R4 (Kuwahara painterly fill) and R5 (hierarchical depth edge layer), produced both a significant Friedman omnibus result and a significant pairwise difference against the unmodified baseline. In each case, participants ranked the composite without the additional layer above every tested variant that included it. The remaining three rounds, R1, R2, and R3, adjusted line threshold, outline geometry, and shadow strength within that same baseline composite, and none produced a significant preference among the tested values. This pattern suggests that perceived trustworthiness in this ranking paradigm was more sensitive to whether an additional abstraction layer was present at all than to fine calibration within a layer already established. It is a pattern worth revisiting when interpreting the H1 and H2 results in Section 6.2.4.

The high response validity rate, 40 to 44 of 45 submitted ballots retained per round, indicates that participants understood the ranking task despite the requirement to assign a unique rank to every image. This calibration exercise fixed the exact C3 stimulus used in the main study, providing the empirical basis for the technique selection already reported in Section 6.1.4.

6.2.2. Participants

Table 6.2.: Participant demographics and study version distribution.

Parameter	**Value**
Total enrolled	*[N]*
Excluded (comprehension check)	*[n]*
Final analytic sample	*[N_{final}]*
Age: mean (SD)	*[M] ([SD])* years
Gender	*[breakdown]*
Study version (A–C)	*[n per version]*

Participants were recruited via [recruitment channel]. Inclusion criteria required age 18 or older, sufficient proficiency in the study language to understand the vignette scenarios, and access to the study URL. No specific domain expertise was required. Participants who selected the wrong option on the comprehension check (Part A) for any block were excluded from the analytic sample. The closing suspicion-check question, asking participants what they believed the study was investigating, was reviewed for explicit mentions of avatar realism or stylisation as the manipulation; *[insert number of participants flagged and whether a sensitivity analysis excluding them changes the pattern of results]*.

6.2.3. Manipulation Check

Part I of the questionnaire asked participants to rate how visually stylised each advisor appeared on a single seven-point scale ranging from completely realistic (1) to very stylised (7), administered once after all three video blocks. This single-item manipulation check was designed to verify that participants perceived the rendering difference between conditions without priming stylisation awareness during the rating blocks. A successful manipulation requires that C3 (NPR) scores significantly higher than C2 (photorealistic), while C1 (real person) scores lowest.

[Insert Part I descriptive statistics and one-way repeated-measures ANOVA results here. Expected pattern: C1 < C2 < C3 on the stylisation scale. Report F, df, p, η_p^2 and pairwise comparisons with Bonferroni correction. If the C2 vs C3 comparison is not significant, the manipulation has failed and trust results must be interpreted with caution.]

6.2.4. Hypothesis Tests

The primary analysis was a one-way repeated-measures MANOVA with condition (C1, C2, C3) as the within-subjects factor and all six subscale composite scores (B_score, C_score, D_score, E_score, F_score, G_score) as dependent variables. Significant omnibus effects were followed up with univariate repeated-measures ANOVAs per subscale and pairwise comparisons with Bonferroni correction ($\alpha_{\text{corrected}} = .05/3 = .017$). Effect sizes are reported as partial η^2.

H1 — NPR Reduces Eeriness

H1 was motivated by the uncanny valley account: a photorealistic avatar occupies the near-human fidelity region of the affinity curve, where negative affect peaks [6]. The NPR composite applied in C3 reduces photorealism while preserving facial feature legibility, as established in Section 6.1.4, and the prediction follows that this reduction should lower scores on the Ho & MacDorman eeriness subscale (D_score) relative to C2 (photorealistic avatar).

[Insert ANOVA results for D_score across C1, C2, C3. Report F, df, p, η_p^2. Report C2 vs C3 pairwise comparison with Bonferroni-corrected p and effect size. If D_score is significantly lower for C3 than for C2, H1 is confirmed. The effect size provides an estimate of how far the C3 composite displaces the Avaturn avatar along the affinity curve from its photorealistic counterpart. The C1 baseline anchors the lower bound of the eeriness distribution: if C3 does not differ significantly from C1, the stylised avatar has reached the same range of the affinity curve as a real human face under the conditions of this study. If H1 is not supported, two explanations deserve consideration. The Avaturn photorealistic baseline may already score below the midpoint of the eeriness scale at the 1 m framing, leaving little room for further reduction. Alternatively, the C3 stylisation may not introduce enough perceptual distance from the photorealistic baseline under video presentation conditions, despite being qualitatively legible as a distinct style in direct rendering.]

H2 — NPR Does Not Reduce Trust

H2 was a null hypothesis: NPR stylisation was not expected to reduce scores on the trust subscales relative to the photorealistic baseline. The prediction was grounded in results from Canales et al., who found that trust ratings were broadly equivalent across three levels of stylisation in a scripted advisor paradigm [7], and in evidence from Gao et al. that realistic and stylised agents attract comparable levels of confidence [24]. The current study mirrors the Canales et al. paradigm closely enough that a null finding across the Competence (E_score), Benevolence (F_score), and Integrity (G_score) subscales is the most plausible outcome for the C2 versus C3 comparison.

[Insert ANOVA results for E_score, F_score, G_score. For each: F, df, p, η_p^2. Report C2 vs C3 Bonferroni comparison. A non-significant pairwise result alone is insufficient evidence for H2, since H2 predicts

equivalence rather than merely an absence of a detected difference. Insert the TOST (two one-sided tests) equivalence test applied to the C2 versus C3 trust composite, with the pre-specified bound of Cohen's $d = 0.3$ established in Chapter 4: report both one-sided test statistics, their p-values, and the 90% confidence interval for the effect size relative to the equivalence bound. H2 is confirmed only if this TOST result falls within the equivalence bound, providing positive evidence of negligible difference, in addition to non-significant pairwise comparisons across E_score, F_score, and G_score. Together, these would replicate the Canales et al. null finding in the context of an NPR-shaded avatar. If any trust subscale does differ significantly, the identity of the subscale is diagnostically informative: a competence difference would indicate that NPR stylisation interferes with perceptions of advisor expertise, whereas a benevolence difference would locate the effect in perceived goodwill rather than professional credibility. An integrity difference would be the least expected outcome, since integrity judgments typically rest on behavioural cues such as argumentative consistency and explicit acknowledgment of trade-offs, both held constant across conditions by the script design. Should C1 score significantly higher than both avatar conditions on one or more subscales, this is consistent with established findings on the real-person advantage in trust [1], [3], though it remains an incidental observation, since the critical comparison for this thesis is C2 versus C3.]

H3 — Eeriness Predicts Trust

H3 examined whether eeriness and trust are related at the individual level. The uncanny valley account predicts that they should be: an avatar that provokes stronger affective discomfort should also attract lower confidence in the advisor role. A person-centred multilevel regression was specified for each trust subscale composite (E_score, F_score, G_score) in turn, with D_score as the within-participant predictor and a random intercept per participant. Person-centring removes each participant's own mean from both variables before the analysis, and this isolates the within-participant association between eeriness and trust from variance driven by condition or by stable between-person differences.

[Insert the fixed-effect slope b, its 95% confidence interval, and the intra-class correlation (ICC) for each trust subscale regressed on D_score. Report whether the eeriness-trust association is consistent across subscales. If D_score negatively predicts each of the three trust subscales, H3 is confirmed. This result would be consistent with the systematic review of Alipour et al., who found a direct negative relationship between eeriness and trust in the majority of studies that examined this association [22]. An asymmetric pattern across subscales, where eeriness predicts benevolence more strongly than competence, would be informative: benevolence reflects perceived warmth and goodwill and is plausibly more sensitive to affective discomfort than competence, which rests partly on the propositional content of the recommendation. If H3 is not supported, this would suggest that eeriness and trust are separable at the within-participant level, even where they may appear related when compared across conditions.]

The relationship between eeriness and trust, if confirmed, also provides a theoretical mechanism for the H1 prediction: any reduction in D_score for C3 relative to C2 should propagate into a corresponding improvement in trust ratings via this pathway. The degree to which H3 and H1 together account for the observed trust pattern is therefore an important secondary question for the analysis.

6.2.5. Post-Study Rankings

After completing all three video blocks and questionnaires, participants ranked the three advisors on three separate questions: from most to least preferred to consult (P1), from most to least trustworthy overall, and from most to least unsettling overall. A rank order was used rather than the strict forced-choice employed by Canales et al. [7], since both avatar conditions are expected to score below C1. A forced binary choice would concentrate responses on C1 and leave insufficient variance to separate C2 from C3 in the behavioural measure, and a ranking instead preserves the C2 versus C3 signal regardless of how many participants place C1 first.

P1 is the primary behavioural measure. Canales et al. observed a dissociation between explicit and behavioural measures in their related study: scores on the trust scales were equivalent across stylisation levels, yet the avatar at intermediate realism attracted the highest selection rate [7]. This dissociation motivates the inclusion of P1 alongside the Likert trust battery, and the trustworthiness and unsettlingness rankings extend the same logic to the two dimensions central to H1 and H2, serving as behavioural analogues to the eeriness subscale and the trust battery respectively and allowing each explicit rating to be compared against an implicit forced ordering of the same three advisors.

Mean ranks per condition (1 = most preferred, 3 = least preferred) were compared using a Friedman test for each of the three questions, followed by pairwise Wilcoxon signed-rank tests with Bonferroni correction ($\alpha_{\text{corrected}} = .017$). The theoretically critical comparison is C2 vs C3 in each case.

[For each of the three ranking questions (P1, trustworthiness, unsettlingness): insert Friedman result χ^2_F, df = 2, p; insert mean rank ± SD per condition; insert Wilcoxon results for C1 vs C2, C1 vs C3, and C2 vs C3 with corrected p-values. If C2 vs C3 Wilcoxon is significant, report direction (which condition received the lower, more preferred mean rank). If not significant on P1 or the trustworthiness ranking, interpret as consistent with H2: NPR stylisation does not reduce implicit preference or perceived trustworthiness relative to the photorealistic avatar. If the unsettlingness ranking places C3 significantly below C2 (i.e. less unsettling), this provides behavioural corroboration of H1 alongside the eeriness subscale result. If the mean rank assigned to C3 is lower than that assigned to C2 on P1 despite equivalent trust subscale scores, this pattern replicates the Canales et al. finding and supports the interpretation that NPR stylisation carries a net advantage at the behavioural level that explicit Likert ratings do not capture, meaning that the design strategy of applying NPR stylisation to a virtual advisor is viable and may even be preferable in contexts where advisor preference, rather than explicit endorsement, is the outcome of interest. If C3 receives a higher mean rank than C2 on P1, indicating weaker preference, this would suggest that the departure from photorealism is perceived as a disadvantage in an implicit preference judgment. If C1 is ranked first by a large majority of participants but trust subscale scores do not separate C1 from the avatar conditions, the behavioural measure is more sensitive to the real-person advantage than the Likert scales, and this discrepancy would itself be a noteworthy finding. Compare the direction of the trustworthiness ranking against the Likert trust subscales (H2) and the unsettlingness ranking against the eeriness subscale (H1) to assess convergence between explicit and behavioural measures for each dimension: if the two agree on the direction of the C2 versus C3 comparison, this strengthens confidence that the explicit trust battery reflects a genuine attitude rather than an artefact of the Likert format, whereas a divergence would parallel the pattern Canales et al. observed for overall preference and would suggest that forced ranking surfaces a preference structure that graded self-report scales attenuate.]

6.2.6. Secondary Measures

Part H of the questionnaire measured agreement with the advisor's recommendation on a seven-point Likert item. This item was included as a secondary exploratory measure, distinct from the pre- to post-video choice repeat used to compute preference shift (Section 6.2.7).

[Insert descriptive statistics for Part H recommendation agreement across the three conditions and a repeated-measures ANOVA or Friedman test as appropriate for the scale. Note whether agreement with the recommendation itself, independent of the choice made, differs by rendering condition.]

The propensity-to-trust covariate (Part Q9) was examined as an individual difference moderator. Participants with high baseline propensity to trust may show smaller condition-driven differences; low-propensity participants may be more sensitive to appearance-based cues.

[Insert moderation analysis or descriptive breakdown of trust scores by propensity quartile if sample size permits.]

6.2.7. Pre-Video Baseline Shift

Before each video, participants recorded their preference between the two scenario options (the exploratory pre-video baseline item). Part H repeats the same binary choice after the video has played, and comparing the two provides a measure of preference shift that isolates the persuasive effect of the advisor's recommendation from the participant's prior inclination. Preference shift was computed per participant per block as the binary change from the pre-video choice to the post-video choice.

[Insert preference shift rates per condition. If participants are more likely to shift their preference towards the advisor's recommendation in one condition, this suggests a condition-level difference in persuasive credibility that is independent of explicit trust self-report.]

Limitations specific to this study are discussed together with the system-level limitations of the thesis in Section 7.2.

7. Conclusion

7.1. Summary of Contributions

This thesis developed and evaluated a system for applying non-photorealistic rendering to real-time VR avatars across three distinct avatar platforms, with the aim of understanding both the technical feasibility of NPR integration and the perceptual consequences of visual abstraction for avatar trust. The work was structured around three research questions: whether NPR can be integrated into platforms with different rendering architectures, which technique achieves the best trade-off between visual quality and performance, and how visual abstraction affects the perceived trustworthiness of a speaking avatar.

The first contribution is an extensible multi-platform NPR rendering system implemented in Unity 6 with URP, supporting three avatar platforms — Mixamo, Avaturn, and the Meta Avatars SDK — that differ substantially in their shader accessibility and runtime management. Eight stylisation techniques were designed and implemented across this system, each motivated by a specific limitation of its predecessor. The sequence progresses from a geometry-based inverted hull outline through toon shading and its X-Toon extension, screen-space derivative edge detection, Sobel and Gaussian Sobel colour-boundary detection, hierarchical multi-cue edge fusion, and ends with the anisotropic Kuwahara painterly filter. For the Meta SDK platform, all effects were implemented inside the `AppSpecificPostManipulation` hook exposed by the SDK's shader pipeline, which is the only integration point that does not break the SDK's skinning, LOD management, or PBR material system.

The second contribution is the adaptation of the X-Toon extended toon shading technique [10] to a production avatar context, including a Normal Field Abstraction mechanism on the Mixamo and Avaturn platforms that uses a simplified normal map and TBN-space interpolation to progressively flatten surface shape towards its low-frequency form, and an abstraction compression formula that widens shading bands as the detail axis increases. On the Meta platform, where intermediate lighting terms are not accessible within the post-processing hook, a luminance-based proxy replaces the direct Lambertian dot product, and the light direction is sourced from the URP per-frame global `_MainLightPosition.xyz`, enabling a correct Blinn-Phong specular response.

The third contribution is a runtime tooling infrastructure supporting iterative visual assessment in VR. The fully procedural world-space parameter panel (`NPREdgeDetectionUI.cs`) exposes over sixty tunable parameters across all technique groups via controller-driven sliders and supports real-time technique switching without shader recompilation. A preset save and recall system allows parameter configurations to be stored to persistent storage and reloaded across sessions, supporting systematic exploration of the parameter space. A material management script (`AvatarShaderSwapper.cs`) resolves the Meta SDK's LOD-driven material reversion problem through a polling coroutine, ensuring that custom NPR assignments persist through all SDK LOD transitions.

7.2. Limitations

Several limitations qualify the contributions of this thesis, spanning both the rendering system and the user study.

System and implementation limitations.

- **Meta SDK hook constraint.** The Meta SDK integration is constrained to the `AppSpecificPostManipulation` hook, which fires after the PBR lighting stage. Techniques that require access to intermediate lighting

terms, per-fragment NdotL, or the tangent frame are therefore not directly realisable on the Meta platform. This required separate implementations for the X-Toon and toon shading techniques, introducing a divergence in the parameter space across platforms that complicates direct visual comparison.

- **Unsmoothed structure tensor estimation.** The anisotropic Kuwahara formulation of Kyprianidis et al. [13] is implemented on all three platforms. The structure tensor, however, requires a Gaussian blur pass for a stable orientation estimate, and this blur is not feasible within the single-pass Meta SDK hook. The Meta SDK therefore estimates the tensor directly from unblurred screen-space derivatives. This noisier estimate degrades the inverse-variance weighting, and the resulting brushstrokes often fail to follow feature orientation on this platform, as Section 4.4.9 describes.
- **Unbenchmarked Quest 3 performance.** Performance on the Meta Quest 3 target hardware was not formally benchmarked in this thesis. The sample counts for the higher-complexity techniques, Gaussian Sobel and the composite Kuwahara pipeline, scale with kernel radius, and the worst-case fragment cost was not profiled across the full range of avatar mesh complexity and viewing distances. This leaves open whether the most sample-intensive techniques run within the Quest 3's frame budget under the conditions of the user study.

User study limitations.

- **Single NPR condition, scenario set, and avatar platform.** The evaluation uses a single NPR condition, the C3 composite, so results cannot be generalised to NPR stylisation as a category. A different technique or abstraction strength might produce different trust outcomes. The three recommendation scenarios are similarly fixed rather than varied, and the results reflect trust responses to this specific set of decision contexts. Because Avaturn was the study platform, the results characterise NPR applied to a photorealistic avatar reconstructed from photographs specifically. This characterisation may not extend to less realistic avatar types, including the Mixamo model used during development.
- **Video-based modality.** The study is conducted online via video stimuli. The absence of spatial presence, interactivity, and stereoscopic depth means that the results reflect trust responses to a speaking head in a video interface, not to an embodied avatar in VR.
- **Single avatar identity and recruitment channel.** The Avaturn avatar is a single identity constructed for this study, and individual characteristics including perceived age, gender, and attractiveness may interact with the NPR effect in ways the current design cannot decompose. The study recruits participants through [channel], which may oversample populations with higher digital literacy and more prior exposure to AI-generated content, potentially attenuating the uncanny valley response relative to a general population sample.
- **Real-person baseline confound.** The C1 stimulus features the thesis researcher as the speaking advisor, meaning participants who knew the researcher personally may have rated C1 with elevated trust scores reflecting social familiarity rather than advisor credibility. Because a recognition check was not included in the questionnaire, it is not possible to identify and separately analyse these participants. C1 ratings should therefore be interpreted as an approximate rather than clean real-person baseline. The theoretically critical comparison between C2 and C3 is unaffected by this confound, since neither condition features a recognisable person.
- **Head texture asymmetry between C2 and C3.** The Blender preprocessing pass applied to suppress baked-illumination edge artefacts (Section 4.6) was applied to the texture used in C3 but not to the texture retained for C2 (Section 6.1.4). The modification does not alter facial geometry, proportions, or identity. The lip and skin tone adjustments are visible in the unstylised preprocessed texture, though they become much less noticeable once the NPR pipeline is applied. This equivalence was not independently verified through a perceptual matching study, and a residual difference between C2 and C3 beyond rendering style alone therefore cannot be fully excluded.

- **Trust subscale item count.** The trust subscales use three items per dimension, adapted from the single-item measures of Canales et al. [7] and grounded in the McKnight & Chervany [21] typology. While this exceeds the minimum for internal consistency assessment, three items per subscale represents a lower bound for robust reliability estimation, and a fully validated multi-item instrument would provide a stronger psychometric foundation for the trust outcome measures.
- **Untested order effects.** The Latin Square counterbalancing scheme assumes the absence of systematic order effects, and this assumption was not formally tested. If fatigue or carry-over effects are present, condition-by-position interactions could inflate or attenuate observed condition differences. Future work should include an explicit order effect test, particularly for the eeriness subscale, which may be more susceptible to contrast effects when a photorealistic condition follows a stylised one or vice versa.
- **Parameter adjustment study coverage.** Concerns about participant burden and study length meant the parameter adjustment study (Section 4.7) tested only the parameter groups expected to have the greatest visual impact. Parameters with smaller expected effects were instead fixed through direct visual inspection during development rather than through participant ranking. The final composite may therefore reflect a locally optimal configuration among the tested parameters rather than a global optimum across the full parameter space.

7.3. Future Work

Several directions extend naturally from this work. The most direct extension is a parametric study of abstraction degree: rather than comparing a single NPR condition to a photorealistic baseline, future work could vary the `_LightingStrength` or `_XToonLightingStrength` blend parameter continuously to measure how trust, uncanny valley ratings, and social presence co-vary with abstraction level. This would produce a more complete mapping of the trust-abstraction curve than the single-point comparison of this thesis.

On the technical side, the Meta SDK hook constraint could potentially be relaxed in future SDK versions that expose earlier pipeline stages; access to per-fragment NdotL and the tangent frame would enable Normal Field Abstraction and direct NdotL toon shading on the Meta platform, eliminating the current cross-platform divergence. Alternatively, a deferred-injection approach using Unity's Render Graph could apply NPR effects to the Meta avatar in a separate pass outside the SDK hook, at the cost of a second depth sample.

The temporal stability of screen-space edge detection under head motion in VR is a known challenge that was addressed during development through threshold calibration but not through explicit temporal filtering. A moving-average filter on the edge magnitude signal, or a reprojection-based approach that warps edge maps between frames, could reduce the flickering that screen-space derivatives produce at seam boundaries when the avatar is in motion.

Finally, the avatar identity used in the user study was designed to be neutral and gender-unspecified; future work could investigate whether the trust effect of NPR stylisation interacts with avatar identity characteristics such as perceived gender, age, or species, extending the study design towards the broader question of how the form of an avatar mediates social responses to its behaviour.

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A. Study Materials

Declaration of Authorship

I hereby declare, on my honour, that this Master's thesis was written by me independently and that no sources or aids other than those cited have been used. All passages drawn verbatim or in paraphrase from published or unpublished sources are identified as such. This work has not been submitted to any other examination authority, in whole or in part, in the same or in a substantially similar form, nor has it been published.

The thesis was prepared under the supervision of Prof. Dr. Oliver Deussen (first reviewer) and Prof. Dr. Tiare Feuchtner (second reviewer) at the University of Konstanz, Department of Computer and Information Science, within the Master of Science programme in Computer Science.

I further declare that in preparing this thesis I used generative artificial intelligence tools to assist with written expression. Specifically, Anthropic Claude was used to refine sentence formulation, improve clarity, and check adherence to academic style conventions. All intellectual content, including the research design, technical development, data collection and analysis, interpretation of results, and conclusions, is exclusively my own work. No AI tool was used to generate research content, arguments, or findings. All AI-assisted passages were reviewed and revised by me, and I bear full academic responsibility for the entire content of this thesis.

Konstanz, August 2026

_____

Nidanur Günay

Parameter Adjustment Study Questionnaire

The following pages reproduce the parameter adjustment study questionnaire administered before the main study. The resulting ranking data and aggregated scores are reported in Chapter B.

Avatar NPR Study: User Preference Evaluation

Thank you for participating in this avatar appearance study. You will see images of stylized 3D avatars across several rounds and rank them based on your preferences.

For the best experience, please complete this study on a desktop or laptop computer rather than a mobile phone. Images are easier to compare on a larger screen, and the ranking questions are more comfortable to use with a mouse or trackpad.

Each round takes approximately 1–2 minutes. The full study takes around 10 minutes. There are no right or wrong answers. Please rank based on your personal impression.

* Indicates required question

Round 1

View the 5 avatar images below, then rank them.

A

B

C

D

E

Overview — All Avatars in This Round

1. Imagine these avatars are virtual advisors presenting you with advice on an important personal decision. Rank them from most to least trustworthy. *
   Assign each number to only one avatar. Each rank can be used once.

   Mark only one oval per row.

	A	B	C	D	E
1st (Best)	◯	◯	◯	◯	◯
2nd	◯	◯	◯	◯	◯
3rd	◯	◯	◯	◯	◯
4th	◯	◯	◯	◯	◯
5th (Worst)	◯	◯	◯	◯	◯

2. *What influenced your ranking most in this round? (optional)*

Round 2

View the 6 avatar images below, then rank them.

A

B

C

D

E

F

Overview — All Avatars in This Round

3. Imagine these avatars are virtual advisors presenting you with advice on an important personal decision. Rank them from most to least trustworthy. *
   Assign each number to only one avatar. Each rank can be used once.

   Mark only one oval per row.

	A	B	C	D	E	F
1st (Best)	◯	◯	◯	◯	◯	◯
2nd	◯	◯	◯	◯	◯	◯
3d	◯	◯	◯	◯	◯	◯
4th	◯	◯	◯	◯	◯	◯
5th	◯	◯	◯	◯	◯	◯
6th (Worst)	◯	◯	◯	◯	◯	◯

4. *What influenced your ranking most in this round? (optional)*

Round 3

View the 5 avatar images below, then rank them.

A

B

C

D

E

Overview — All Avatars in This Round

5. Imagine these avatars are virtual advisors presenting you with advice on an important personal decision. Rank them from most to least trustworthy. *
   Assign each number to only one avatar. Each rank can be used once.

   Mark only one oval per row.

	A	B	C	D	E
1st (Best)	◯	◯	◯	◯	◯
2nd	◯	◯	◯	◯	◯
3rd	◯	◯	◯	◯	◯
4th	◯	◯	◯	◯	◯
5th (Worst)	◯	◯	◯	◯	◯

6. *What influenced your ranking most in this round? (optional)*

Round 4

View the 4 avatar images below, then rank them.

A

B

C

D

Overview — All Avatars in This Round

7. Imagine these avatars are virtual advisors presenting you with advice on an important personal decision. Rank them from most to least trustworthy. *
Assign each number to only one avatar. Each rank can be used once.

Mark only one oval per row.

	A	B	C	D
1st (Best)	◯	◯	◯	◯
2nd	◯	◯	◯	◯
3rd	◯	◯	◯	◯
4th (Worst)	◯	◯	◯	◯

8. *What influenced your ranking most in this round? (optional)*

Round 5

View the 6 avatar images below, then rank them.

A

B

C

D

E

F

Overview — All Avatars in This Round

9. Imagine these avatars are virtual advisors presenting you with advice on an important personal decision. Rank them from most to least trustworthy. *
Assign each number to only one avatar. Each rank can be used once.

Mark only one oval per row.

	A	B	C	D	E	F
1st (Best)	◯	◯	◯	◯	◯	◯
2nd	◯	◯	◯	◯	◯	◯
3rd	◯	◯	◯	◯	◯	◯
4th	◯	◯	◯	◯	◯	◯
5th	◯	◯	◯	◯	◯	◯
6th (Worst)	◯	◯	◯	◯	◯	◯

10. *What influenced your ranking most in this round? (optional)*

Overall Impression

The images above show all unique avatar configurations from this study. R4 A, and R5 A are the same base avatar as R1 E and are not shown again.

R stands for Round — for example, R2 C means Round 2, image C.

11. *Looking back at all the avatars you have seen across this study, write the round and label of the single one you found most visually appealing (for example: R2 C). Note that R4 A, and R5 A are the same base avatar as R1 E. You can scroll back to remind yourself.*

Thank you for completing this evaluation

Your responses will help select the visual parameters for a follow-up study on avatar perception. If you have any questions, please contact: nidanur.guenay@uni-konstanz.de

This evaluation is part of Master's thesis research at the University of Konstanz.

This content is neither created nor endorsed by Google.

Google Forms

Online Questionnaire

The following pages reproduce the main study questionnaire as administered to participants via Google Forms.

Evaluation of Video Advisors

Welcome, and thank you for taking part.

In this study you will watch three short videos of advisors giving their opinion on everyday decisions. After each video you will answer brief questions about your impression of the advisor. At the end you will be asked to compare all three advisors overall, so please watch each video carefully and pay attention to each advisor.

Please be aware that the visual presentation of the advisors in this study varies across videos. Some advisors appear more realistic, while others are more stylised or abstracted. This variation is a deliberate aspect of the study design. You are asked to focus on your overall impression of each advisor rather than on the realism of the animation.

The study takes approximately 20 minutes — please complete it in one sitting with sound on. A desktop or laptop is strongly recommended.

Before you begin:

- Participation is voluntary and you may stop at any time.
- No personally identifying information is collected.
- Your anonymised data will be used solely for academic research at the University of Konstanz.
- Questions? Contact: nidanur.guenay@uni-konstanz.de

* Indicates required question

1. I have read the information above, I am 18 years or older, * and I agree to participate in this study.

Check all that apply.

- [ ] I consent

Demographics

2. Please indicate the gender you identify with *

Mark only one oval.

- ◯ Man
- ◯ Woman
- ◯ Non-binary
- ◯ Prefer not to say
- ◯ Other: ____

3. Please indicate your age *

_____

4. How often do you interact with virtual avatars or computer-generated characters * (in games, VR, video calls, social media, etc.)?

Mark only one oval.

	1	2	3	4	5	
Nevı	◯	◯	◯	◯	◯	Very often

5. In general, I tend to trust people I meet for the first time. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

Round 1

Read the scenario below carefully, then answer one quick question before watching the video.

Situation

You recently completed your master's degree and have received two job offers. Both start in one month. You have five days to decide.

JOB A — Generalist Role (Small Company)

Job A is a generalist role at a smaller company. You would define your own work area, your contributions would be directly visible to the team, and the skills you build would transfer across industries. The starting salary is below the industry average for your field.

JOB B — Specialist Role (Large Company)

Job B is a specialist role at a large, well-known company. The employer name carries weight on a CV and the role is in a high-demand field. However, the position involves deep specialisation within a large team, with limited individual visibility. The starting salary is above the industry average for your field.

6. Round 1: Before watching, which job would you choose? *

Mark only one oval.

- ◯ Job A
- ◯ Job B

Video 1

Please watch the video below in full, then continue to the next page.
Make sure your sound is on.

Advisor 1

http://youtube.com/watch?v=68sg8fSFH1E

The advisor said
"I'd go with Job A. Yes, the salary is lower right now, but the generalist role gives you a range of experience that is genuinely hard to get later. Salary gaps can be closed in the future. But once you specialise, switching back is costly. Take the pivot while you can."

7. I have watched the full video. *

Check all that apply.

- [ ] I have watched the video

Round 1 Questionaire

8. Which job did the advisor recommend? *

Mark only one oval.

- ◯ Job A
- ◯ Job B
- ◯ I am not sure

Warmth and Appeal

Below you will see pairs of opposite words. Select the number that best matches your impression of this advisor. A lower number means you lean toward the word on the left. A higher number means you lean toward the word on the right. If you feel neither or somewhere in between, select 3.

9. *

Mark only one oval.

	1	2	3	4	5	
Disli	◯	◯	◯	◯	◯	Like

10. *

Mark only one oval.

	1	2	3	4	5	
Unfr	◯	◯	◯	◯	◯	Friendly

11. *

Mark only one oval.

	1	2	3	4	5	
Unki	◯	◯	◯	◯	◯	Kind

12. *

Mark only one oval.

	1	2	3	4	5	
Unp	◯	◯	◯	◯	◯	Pleasant

13. *

Mark only one oval.

	1	2	3	4	5	
Awfi	◯	◯	◯	◯	◯	Nice

Impression

Please rate your impression of the advisor on the following scales. 1 = left word, 5 = right word.

14. *"Eerie" means strangely unsettling in a way that is hard to explain.* *

Mark only one oval.

	1	2	3	4	5	
Rea:	◯	◯	◯	◯	◯	Eerie

15. *"Freaky"* means very strange or disturbing in an unexpected way. *

Mark only one oval.

	1	2	3	4	5	
Nurr	◯	◯	◯	◯	◯	Freaky

16. *"Supernatural"* means beyond what feels natural or normal. *

Mark only one oval.

	1	2	3	4	5	
Ordi	◯	◯	◯	◯	◯	Supernatural

17. *"Uncanny" means something that feels strangely unsettling, as if almost right but slightly off.* *

Mark only one oval.

	1	2	3	4	5	
Blar	◯	◯	◯	◯	◯	Uncanny

18. *"Hair-raising" means causing a strong feeling of unease or alarm.* *

Mark only one oval.

	1	2	3	4	5	
Une	◯	◯	◯	◯	◯	Hair-raising

19. *"Spine-tingling" means producing a strong feeling of discomfort or dread.* *

Mark only one oval.

	1	2	3	4	5	
Unir	◯	◯	◯	◯	◯	Spine-tingling

20. *"Shocking"* means surprising you in an unpleasant or startling way. *

Mark only one oval.

	1	2	3	4	5	
Bori	◯	◯	◯	◯	◯	Shocking

21. *"Thrilling"* here means a strong, uncomfortable physical sensation in your body. *

Mark only one oval.

	1	2	3	4	5	
Prec	◯	◯	◯	◯	◯	Thrilling

Trust

The following questions use a different scale: 1 to 7, where 1 = Strongly Disagree and 7 = Strongly Agree.

22. This advisor is capable and competent. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

23. This advisor seems knowledgeable. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

24. This advisor seems to have the expertise needed to give sound advice on this kind of decision. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

25. This advisor will act in a benevolent (well meaning) manner. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

26. This advisor would act in my best interest. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

27. This advisor appears to have good intentions *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

28. This advisor adheres to a set of principles that I find acceptable. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

29. This advisor appears to be honest. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

30. This advisor seems sincere *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

Response to the Recommendation & Visual Style

31. How much do you agree with the recommendation this advisor gave? *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

32. After watching, which job would you choose? *

Mark only one oval.

- ◯ Job A
- ◯ Job B

33. Please select option 2 regardless of anything else.

Mark only one oval.

- ◯ 1
- ◯ 2
- ◯ 3
- ◯ 4
- ◯ 5

Round 2

Read the scenario below carefully, then answer one quick question before watching the video.

Situation
You have just arrived by train in a city you have never visited. You need to reach your hotel, where colleagues are expecting you in one hour.

ROUTE A
Route A is a direct route on a single line with no transfers. It is straightforward to navigate in an unfamiliar city. If the journey runs on time, you would arrive approximately 10 minutes before your meeting begins.

ROUTE B
Route B involves one transfer at a mid-size station you have not used before. If the connection runs on time, you would arrive approximately 40 minutes before your meeting begins. However, the transfer station is unfamiliar and a missed connection could leave you with very little time.

34. Round 2: Before watching, which route would you choose? *

Mark only one oval.

- ◯ Route A
- ◯ Route B

Video 2

Please watch the video below in full, then continue to the next page.
Make sure your sound is on.

Advisor 2

http://youtube.com/watch?v=3vfPChpASas

The advisor said:
"I'd take Route B. Yes, it has a transfer at a station you have never used. But it gets you there forty minutes early. If something goes wrong, you still have enough time to recover. Route A is direct but leaves only ten minutes. That may not be enough. Take the buffer."

35. I have watched the full video.

Check all that apply.

- ☐ I have watched the video

Round 2 Questionaire

36. Which route did the advisor recommend? *

Mark only one oval.

- ◯ Route A
- ◯ Route B
- ◯ I am not sure

Warmth and Appeal

Below you will see pairs of opposite words. Select the number that best matches your impression of this advisor. A lower number means you lean toward the word on the left. A higher number means you lean toward the word on the right. If you feel neither or somewhere in between, select 3.

37. *

Mark only one oval.

	1	2	3	4	5	
Disli	◯	◯	◯	◯	◯	Like

38. *

Mark only one oval.

	1	2	3	4	5	
Unfr	◯	◯	◯	◯	◯	Friendly

39. *

Mark only one oval.

	1	2	3	4	5	
Unki	◯	◯	◯	◯	◯	Kind

40. *

Mark only one oval.

	1	2	3	4	5	
Unp	◯	◯	◯	◯	◯	Pleasant

41. *

Mark only one oval.

	1	2	3	4	5	
Awfi	◯	◯	◯	◯	◯	Nice

Impression

Please rate your impression of the advisor on the following scales. 1 = left word, 5 = right word.

42. *"Eerie" means strangely unsettling in a way that is hard to explain.* *

Mark only one oval.

	1	2	3	4	5	
Rea:	◯	◯	◯	◯	◯	Eerie

43. *"Freaky"* means very strange or disturbing in an unexpected way. *

Mark only one oval.

	1	2	3	4	5	
Nurr	◯	◯	◯	◯	◯	Freaky

44. *"Supernatural"* means beyond what feels natural or normal. *

Mark only one oval.

	1	2	3	4	5	
Ordi	◯	◯	◯	◯	◯	Supernatural

45. *"Uncanny" means something that feels strangely unsettling, as if almost right but slightly off.* *

Mark only one oval.

	1	2	3	4	5	
Blar	◯	◯	◯	◯	◯	Uncanny

46. *"Hair-raising" means causing a strong feeling of unease or alarm.* *

Mark only one oval.

	1	2	3	4	5	
Une	◯	◯	◯	◯	◯	Hair-raising

47. *"Spine-tingling" means producing a strong feeling of discomfort or dread.* *

Mark only one oval.

	1	2	3	4	5	
Unir	◯	◯	◯	◯	◯	Spine-tingling

48. *"Shocking"* means surprising you in an unpleasant or startling way. *

Mark only one oval.

	1	2	3	4	5	
Bori	◯	◯	◯	◯	◯	Shocking

49. *"Thrilling"* here means a strong, uncomfortable physical sensation in your body. *

Mark only one oval.

	1	2	3	4	5	
Prec	◯	◯	◯	◯	◯	Thrilling

Trust

The following questions use a different scale: 1 to 7, where 1 = Strongly Disagree and 7 = Strongly Agree.

50. This advisor is capable and competent. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

51. This advisor seems knowledgeable. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

52. This advisor seems to have the expertise needed to give sound advice on this kind of decision. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

53. This advisor will act in a benevolent (well meaning) manner. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

54. This advisor would act in my best interest. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

55. This advisor appears to have good intentions *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

56. This advisor adheres to a set of principles that I find acceptable. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

57. This advisor appears to be honest. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

58. This advisor seems sincere *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

Response to the Recommendation & Visual Style

59. How much do you agree with the recommendation this advisor gave? *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

60. After watching, which route would you choose? *

Mark only one oval.

- ◯ Route A
- ◯ Route B

Round 3

Read the scenario below carefully, then answer one quick question before watching the video.

Situation

You want to develop a professional skill that would strengthen your CV and improve your career prospects. You have a few weeks available and you are a student with a limited budget.

OPTION A - Paid Online Course
Option A is a structured online course from a well-known learning platform. It follows a clear curriculum and awards a recognised certificate you can add to your CV. It costs €350.

OPTION B - Free Tutorial Series
Option B is a video tutorial series by a practitioner with a strong reputation. It covers the same material at no cost and you set your own pace. However, there is no certificate and no structured curriculum.

61. Round 3: Before watching, which option would you choose?

Mark only one oval.

- ◯ Option A
- ◯ Option B

Video 3

Please watch the video below in full, then continue to the next page.
Make sure your sound is on.

Advisor 3

http://youtube.com/watch?v=oP5tg9IelYQ

The advisor said
"I would take Option A. Yes, 350 euros is a real cost. But the certificate is what matters in here. It is not just about what you learn, it is about what you can show. A recruiter cannot verify a free tutorial series. That is worth it."

62. I have watched the full video.

Check all that apply.

- ☐ I have watched the video

Round 3 Questionaire

63. Which option did the advisor recommend? *

Mark only one oval.

- ◯ Option A
- ◯ Option B
- ◯ I am not sure

Warmth and Appeal

Below you will see pairs of opposite words. Select the number that best matches your impression of this advisor. A lower number means you lean toward the word on the left. A higher number means you lean toward the word on the right. If you feel neither or somewhere in between, select 3.

64. *

Mark only one oval.

	1	2	3	4	5	
Disli	◯	◯	◯	◯	◯	Like

65. *

Mark only one oval.

	1	2	3	4	5	
Unfr	◯	◯	◯	◯	◯	Friendly

66. *

Mark only one oval.

	1	2	3	4	5	
Unki	◯	◯	◯	◯	◯	Kind

67. *

Mark only one oval.

	1	2	3	4	5	
Unp	◯	◯	◯	◯	◯	Pleasant

68. *

Mark only one oval.

	1	2	3	4	5	
Awf	◯	◯	◯	◯	◯	Nice

Impression

Please rate your impression of the advisor on the following scales. 1 = left word, 5 = right word.

69. *"Eerie" means strangely unsettling in a way that is hard to explain.* *

Mark only one oval.

	1	2	3	4	5	
Rea:	◯	◯	◯	◯	◯	Eerie

70. *"Freaky"* means very strange or disturbing in an unexpected way. *

Mark only one oval.

	1	2	3	4	5	
Nurr	◯	◯	◯	◯	◯	Freaky

71. *"Supernatural"* means beyond what feels natural or normal. *

Mark only one oval.

	1	2	3	4	5	
Ordi	◯	◯	◯	◯	◯	Supernatural

72. *"Uncanny" means something that feels strangely unsettling, as if almost right but slightly off.* *

Mark only one oval.

	1	2	3	4	5	
Blan	◯	◯	◯	◯	◯	Uncanny

73. *"Hair-raising" means causing a strong feeling of unease or alarm.* *

Mark only one oval.

	1	2	3	4	5	
Une:	◯	◯	◯	◯	◯	Hair-raising

74. *"Spine-tingling" means producing a strong feeling of discomfort or dread.* *

Mark only one oval.

	1	2	3	4	5	
Unir	◯	◯	◯	◯	◯	Spine-tingling

75. *"Shocking"* means surprising you in an unpleasant or startling way. *

Mark only one oval.

	1	2	3	4	5	
Bori	◯	◯	◯	◯	◯	Shocking

76. *"Thrilling"* here means a strong, uncomfortable physical sensation in your body. *

Mark only one oval.

	1	2	3	4	5	
Prec	◯	◯	◯	◯	◯	Thrilling

Trust

The following questions use a different scale: 1 to 7, where 1 = Strongly Disagree and 7 = Strongly Agree.

77. This advisor is capable and competent. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

78. This advisor seems knowledgeable. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

79. This advisor seems to have the expertise needed to give sound advice on this kind of decision. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

80. This advisor will act in a benevolent (well meaning) manner. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

81. This advisor would act in my best interest. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

82. This advisor appears to have good intentions *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

83. This advisor adheres to a set of principles that I find acceptable. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

84. This advisor appears to be honest. *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

85. This advisor seems sincere *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

Response to the Recommendation & Visual Style

86. How much do you agree with the recommendation this advisor gave? *

Mark only one oval.

	1	2	3	4	5	6	7	
Stro	◯	◯	◯	◯	◯	◯	◯	Strongly Agree

87. After watching, which option would you choose? *

Mark only one oval.

- ◯ Option A
- ◯ Option B

88. Please select option 4 regardless of anything else.

Mark only one oval.

- ◯ 1
- ◯ 2
- ◯ 3
- ◯ 4
- ◯ 5

Overall Impressions

You have now watched all three advisors.

The following questions ask you to compare them overall. Take a moment to think back on each one before answering.

Advisors in the Videos

89. Imagine each of these advisors could give you advice on any type of decision. * Please rank them from the one you would most want to consult to the one you would least want to consult. Assign each a different rank.

Mark only one oval per row.

	Video 1	Video 2	Video 3
1	◯	◯	◯
2	◯	◯	◯
3	◯	◯	◯

90. Please rank the three advisors from most to least trustworthy overall. *

Mark only one oval per row.

	Video 1	Video 2	Video 3
1	◯	◯	◯
2	◯	◯	◯
3	◯	◯	◯

91. Please rank the three advisors from most to least unsettling overall. *

Mark only one oval per row.

	Video 1	Video 2	Video 3
1	◯	◯	◯
2	◯	◯	◯
3	◯	◯	◯

92. How visually stylised did each advisor appear? *
(1 = Completely realistic, 7 = Very stylised)

Mark only one oval per row.

	1	2	3	4	5	6	7
Video 1	◯	◯	◯	◯	◯	◯	◯
Video 2	◯	◯	◯	◯	◯	◯	◯
Video 3	◯	◯	◯	◯	◯	◯	◯

93. Looking back at all three advisors, what was the most important factor in deciding which one you trusted the most?

Please describe in a few sentences.

94. What do you think this study was investigating?

Thank you for completing this study!

If you have any questions, please contact: nidanur.guenay@uni-konstanz.de

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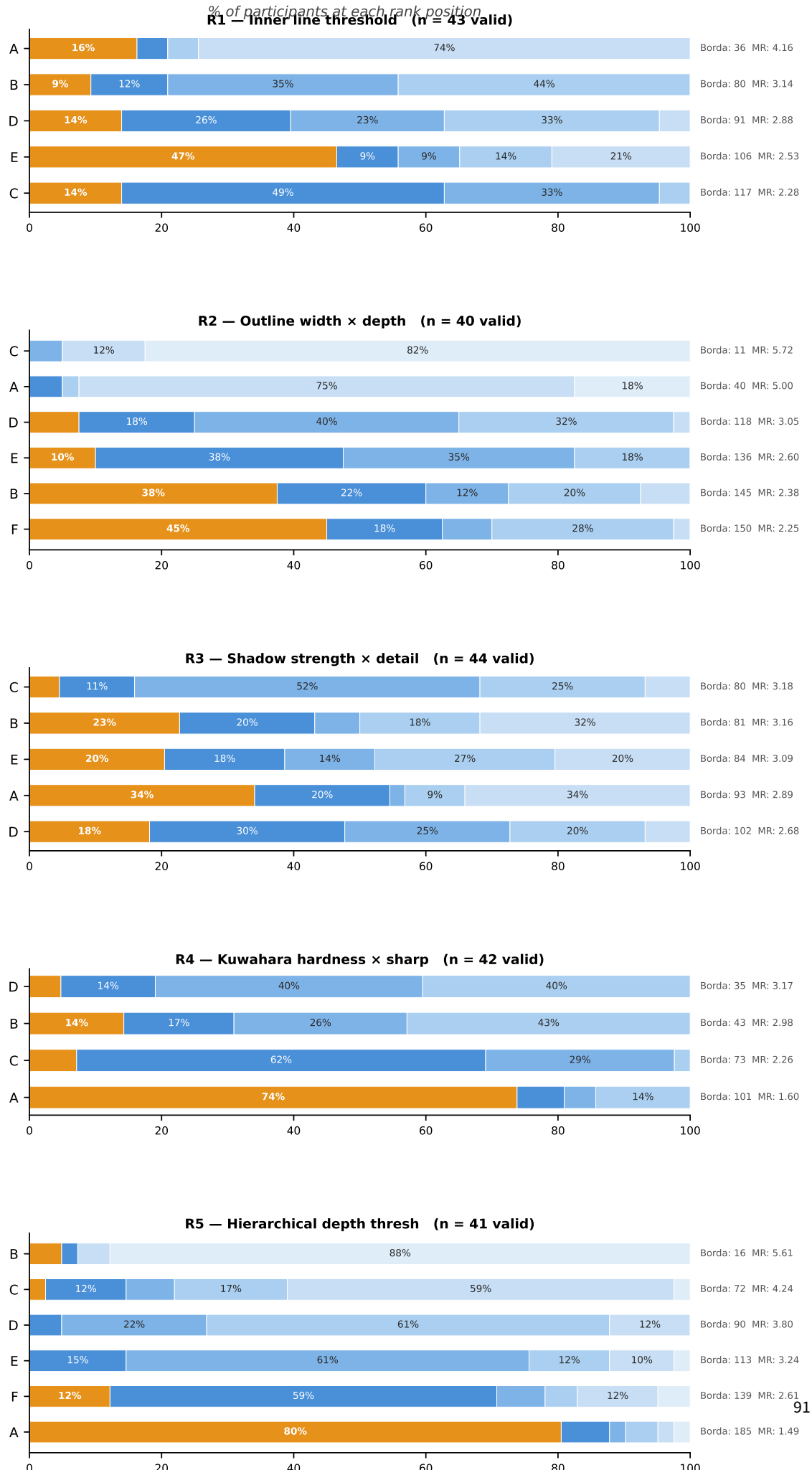
Google Forms

B. Parameter Adjustment Study Results

The figures and tables below present the full ranking data from the five-round parameter adjustment study described in Section 4.7. Each bar shows the percentage of participants who assigned a given option to that rank position. Rows are sorted by Borda score. The Borda score for each option is the sum of $(N - r)$ across all valid responses, where N is the number of options in that round and r the rank position assigned by a participant. A higher Borda score indicates a more consistently preferred option across the full distribution. The mean rank (MR) is reported alongside; a lower value indicates higher preference.


% of participants at each rank position
R1 — Inner line threshold (n = 43 valid)
A
B
D
E
C
16%
74%
9%
12%
35%
44%
14%
26%
23%
33%
47%
9%
9%
14%
21%
14%
49%
33%
Borda: 36 MR: 4.16
Borda: 80 MR: 3.14
Borda: 91 MR: 2.88
Borda: 106 MR: 2.53
Borda: 117 MR: 2.28
0
20
40
60
80
100
R2 — Outline width × depth (n = 40 valid)
C
A
D
E
B
F
12%
82%
75%
18%
18%
40%
32%
10%
38%
35%
18%
38%
22%
12%
20%
45%
18%
28%
Borda: 11 MR: 5.72
Borda: 40 MR: 5.00
Borda: 118 MR: 3.05
Borda: 136 MR: 2.60
Borda: 145 MR: 2.38
Borda: 150 MR: 2.25
0
20
40
60
80
100
R3 — Shadow strength × detail (n = 44 valid)
C
B
E
A
D
11%
52%
25%
23%
20%
18%
32%
20%
18%
14%
27%
20%
34%
20%
9%
34%
18%
30%
25%
20%
Borda: 80 MR: 3.18
Borda: 81 MR: 3.16
Borda: 84 MR: 3.09
Borda: 93 MR: 2.89
Borda: 102 MR: 2.68
0
20
40
60
80
100
R4 — Kuwahara hardness × sharp (n = 42 valid)
D
B
C
A
14%
40%
40%
14%
17%
26%
43%
62%
29%
74%
14%
Borda: 35 MR: 3.17
Borda: 43 MR: 2.98
Borda: 73 MR: 2.26
Borda: 101 MR: 1.60
0
20
40
60
80
100
R5 — Hierarchical depth thresh (n = 41 valid)
B
C
D
E
F
A
88%
12%
17%
59%
22%
61%
12%
15%
61%
12%
10%
12%
59%
12%
80%
Borda: 16 MR: 5.61
Borda: 72 MR: 4.24
Borda: 90 MR: 3.80
Borda: 113 MR: 3.24
Borda: 139 MR: 2.61
Borda: 185 MR: 1.49
0
20
40
60
80
100

The tables below list the computed Borda scores, maximum attainable score, mean rank, and percentage ranked first for each option in each round. The winner by Borda score is typeset in bold.

Round 1 — Inner line threshold (n = 43 valid responses)

Option	Borda	Max	Mean rank	% ranked 1st
C	117	172	2.28	14.0%
E	106	172	2.53	46.5%
D	91	172	2.88	14.0%
B	80	172	3.14	9.3%
A	36	172	4.16	16.3%

Round 2 — Outline width × depth offset (n = 40 valid responses)

Option	Borda	Max	Mean rank	% ranked 1st
F	150	200	2.25	45.0%
B	145	200	2.38	37.5%
E	136	200	2.60	10.0%
D	118	200	3.05	7.5%
A	40	200	5.00	0.0%
C	11	200	5.72	0.0%

Round 3 — Shadow strength × detail bias (n = 44 valid responses)

Option	Borda	Max	Mean rank	% ranked 1st
D	102	176	2.68	18.2%
A	93	176	2.89	34.1%
E	84	176	3.09	20.5%
B	81	176	3.16	22.7%
C	80	176	3.18	4.5%

Round 4 — Kuwahara hardness × sharpness (n = 42 valid responses)

Option	Borda	Max	Mean rank	% ranked 1st
A	101	126	1.60	73.8%
C	73	126	2.26	7.1%
B	43	126	2.98	14.3%
D	35	126	3.17	4.8%

Round 5 — Hierarchical depth threshold (n = 41 valid responses)

Option	Borda	Max	Mean rank	% ranked 1st
A	185	205	1.49	80.5%
F	139	205	2.61	12.2%
E	113	205	3.24	0.0%
D	90	205	3.80	0.0%
C	72	205	4.24	2.4%
B	16	205	5.61	4.9%

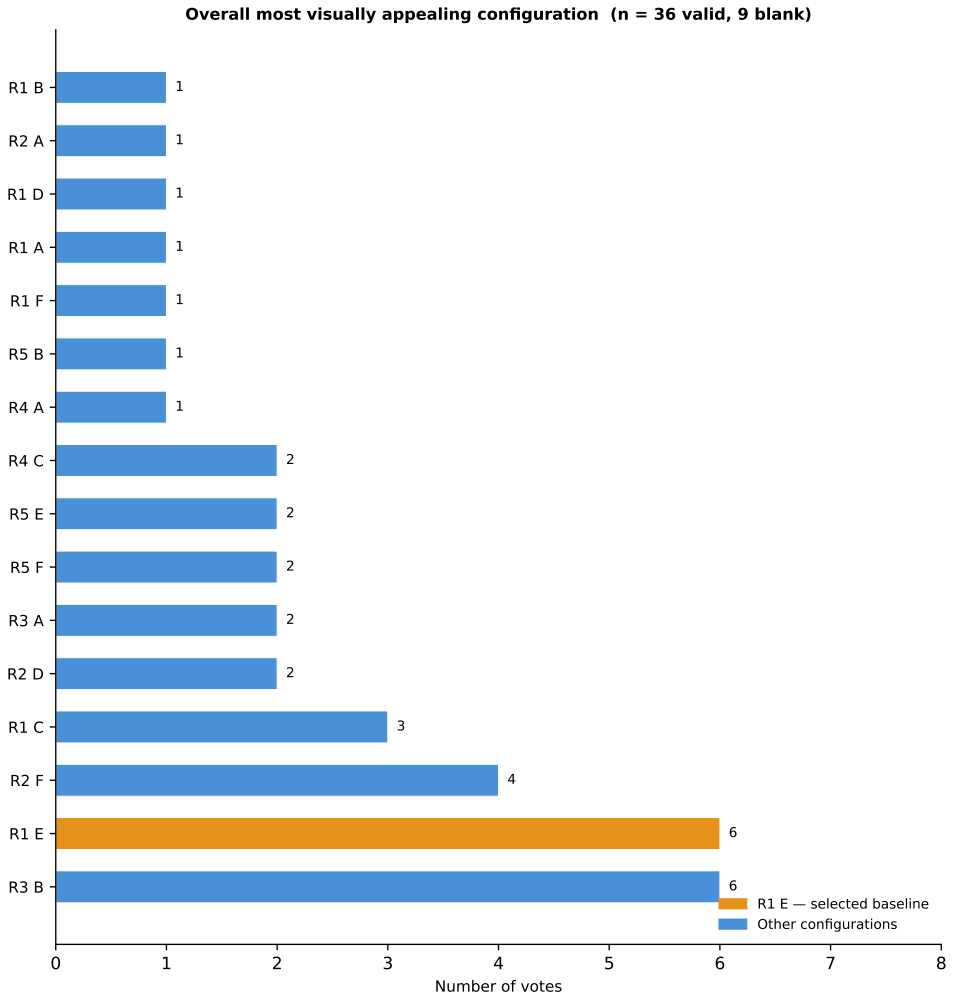


Figure B.2.: Votes for the overall most visually appealing configuration from the final question of the parameter adjustment study (n = 36 valid, 9 blank). The selected study baseline, R1 E, is highlighted in amber.

Overall visual appeal (n = 36 valid responses, 9 blank)

Configuration	Votes
R3 B	6
R1 E	6
R2 F	4
R1 C	3
R2 D	2
R3 A	2
R5 F	2
R5 E	2
R4 C	2
R4 A	1
R5 B	1
R1 F	1
R1 A	1
R1 D	1
R2 A	1
R1 B	1
Blank	9